Algebraic Number Fields

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Contents

1 Arithmetic of Number Fields

Throughout, all rings will be understood to be commutative and with unity.

1.1 Integrality

Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains. We say that $\beta \in \mathcal{O}$ is **integral** over $\mathcal{o}$ if β is the root of a monic polynomial $f(x) \in \mathcal{o}[x]$. In other words, β satisfies an equation of the form

$$\beta^n + \alpha_{n-1}\beta^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. We say that $\mathcal{O}$ is **integral** over $\mathcal{o}$ if every element of $\mathcal{O}$ is integral over $\mathcal{o}$. As we may expect, the set of integral elements form a ring.

Proposition 1.1. *Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains. Then $\beta_1, \ldots, \beta_n \in \mathcal{O}$ are all integral over $\mathcal{o}$ if and only if $\mathcal{o}[\beta_1, \ldots, \beta_n]$ is a finitely generated $\mathcal{o}$-module. In particular, the elements of $\mathcal{O}$ that are integral over $\mathcal{o}$ form a ring.*

Proof. We will prove the forward implication by induction. First suppose $\beta \in \mathcal{O}$ is integral over $\mathcal{o}$. Then there exists a monic polynomial $f(x) \in \mathcal{o}[x]$ of say degree n such that $f(\beta) = 0$. Let $g(x) \in \mathcal{o}[x]$ and write

$$g(x) = q(x)f(x) + r(x),$$

with $q(x), r(x) \in \mathcal{o}[x]$ where the degree of $r(x)$ is strictly less than n. Then $g(\beta) = r(\beta)$. Letting

$$r(x) = \alpha_{n-1}x^{n-1} + \cdots + \alpha_1 x + a_0,$$

with $\alpha_i \in \mathcal{o}$, it follows that

$$g(\beta) = \alpha_{n-1}\beta^{n-1} + \cdots + \alpha_1\beta + \alpha_0.$$

As $g(x)$ was arbitrary, we see that $1, \beta, \ldots, \beta^{n-1}$ generate $\mathcal{o}[\beta]$ as an $\mathcal{o}$-module. Now assume by induction that $\mathcal{o}[\beta_1, \ldots, \beta_{n-1}]$ is a finitely generated $\mathcal{o}$-module. Then $\mathcal{o}[\beta_1, \ldots, \beta_n]$ is a finitely generated $\mathcal{o}$-module. This proves the forward implication of the first statement. For the reverse implication, suppose $\mathcal{o}[\beta_1, \ldots, \beta_n]$ is a finitely generated $\mathcal{o}$-module and let $\omega_1, \ldots, \omega_r$ be generators. Then for any $\beta \in \mathcal{o}[\beta_1, \ldots, \beta_n]$, we have

$$\beta\omega_i = \sum_j \alpha_{i,j}\omega_j,$$

with $\alpha_{i,j} \in \mathcal{o}$. These r equations are equivalent to the identity

$$\begin{pmatrix} \beta - \alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & \beta - \alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & \beta - \alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

The determinant of the matrix on the left-hand side must be zero. This shows that β is the root of the characteristic polynomial $\det(xI - (\alpha_{i,j}))$ which is a monic polynomial with coefficients in $\mathcal{o}$. Hence β is integral over $\mathcal{o}$. As β was arbitrary, the elements $\beta_1, \ldots, \beta_n$ are all integral over $\mathcal{o}$ and that the sum and product of elements that are integral over $\mathcal{o}$ are also integral over $\mathcal{o}$. This proves the reverse implication and the second statement. □

Integrality is also transitive.

Proposition 1.2. *Let $\widetilde{\mathcal{O}}/\mathcal{O}/\mathcal{o}$ be a tower of integral domains. If $\widetilde{\mathcal{O}}$ is integral over $\mathcal{O}$ and $\mathcal{O}$ is integral over $\mathcal{o}$ then $\widetilde{\mathcal{O}}$ is integral over $\mathcal{o}$.*

Proof. Let $\gamma \in \widetilde{\mathcal{O}}$. Since $\widetilde{\mathcal{O}}$ is integral over $\mathcal{O}$, we have

$$\gamma^n + \beta_{n-1}\gamma^{n-1} + \cdots + \beta_0 = 0,$$

for some positive integer n and $\beta_i \in \mathcal{O}$. Set $R = \mathcal{o}[\beta_0, \ldots, \beta_{n-1}]$. Then $R[\gamma]$ is a finitely generated R-module. As $\mathcal{O}$ is integral over $\mathcal{o}$, Proposition 1.1 implies that $R[\gamma]$ is also a finitely generated $\mathcal{o}$-module whence γ is integral over $\mathcal{o}$. As γ was arbitrary, this proves $\widetilde{\mathcal{O}}$ is integral over $\mathcal{o}$. □

In light of this result, we define the **integral closure** $\overline{\mathcal{o}}$ of $\mathcal{o}$ in $\mathcal{O}$ by

$$\overline{\mathcal{o}} = \{\beta \in \mathcal{O} : \beta \text{ is integral over } \mathcal{o}\}.$$

Then $\overline{\mathcal{o}}$ is a subring of $\mathcal{O}$ by Proposition 1.1. Clearly $\mathcal{o} \subseteq \overline{\mathcal{o}}$. Moreover, we say that $\mathcal{o}$ is **integrally closed** in $\mathcal{O}$ if $\mathcal{o} = \overline{\mathcal{o}}$. As $\mathcal{o} \subseteq \overline{\mathcal{o}} \subseteq \overline{\overline{\mathcal{o}}}$, Proposition 1.2 implies that $\overline{\mathcal{o}}$ is automatically integrally closed in $\mathcal{O}$. It will often be more fruitful to work with the integral closure rather than a generic ring.

There is a particular situation of integrally closed rings which is deserving of additional interest. Suppose $\mathcal{o}$ is an integral domain with field of fractions K. We call the integral closure $\overline{\mathcal{o}}$ of $\mathcal{o}$ in K the **normalization** of $\mathcal{o}$ and simply say that $\mathcal{o}$ is an **integrally closed domain** if $\mathcal{o}$ is equal to its normalization. It turns out that every unique factorization domain is an integrally closed domain.

Lemma 1.3. *Let $\mathcal{o}$ be a unique factorization domain with field of fractions K. Then $\mathcal{o}$ is an integrally closed domain. In particular, every principal ideal domain is an integrally closed domain.*

Proof. Let $\kappa \in K$ be such that

$$\kappa^n + \alpha_{n-1}\kappa^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Write $\kappa = \frac{\alpha}{\beta}$ for $\alpha, \beta \in \mathcal{o}$ with β nonzero and $(\alpha, \beta) = 1$. Multiplying by β^n and isolating the leading term shows

$$\alpha^n = -(\alpha_{n-1}\beta\alpha^{n-1} + \cdots + \alpha_0\beta^n).$$

As β divides the right-hand side it divides the left-hand side as well. But then β is a unit in $\mathcal{o}$ since $(\alpha, \beta) = 1$. This means $\kappa \in \mathcal{o}$ whence $\mathcal{o}$ is an integrally closed domain. This proves the first statement. The second statement is immediate since every principal ideal domain is a unique factorization domain. $\square$

We will often consider the more refined setting where $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K is a finite separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. In this setting, L is the field of fractions of $\mathcal{O}$ and the elements of L which are integral over $\mathcal{o}$ have a simple description.

Proposition 1.4. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then every $\lambda \in L$ is of the form*

$$\lambda = \frac{\beta}{\alpha},$$

for some $\beta \in \mathcal{O}$ and nonzero $\alpha \in \mathcal{o}$. In particular, $\mathcal{O}$ is an integrally closed domain with field of fractions L. Moreover, $\lambda \in L$ is integral over $\mathcal{o}$ if and only if the minimal polynomial $m_\lambda(x)$ of λ over K has coefficients in $\mathcal{o}$.

Proof. As L/K is finite, it is necessarily algebraic so that any $\lambda \in L$ satisfies

$$\alpha\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha, \alpha_i \in \mathcal{o}$ with α nonzero. We claim that $\alpha\lambda$ is integral over $\mathcal{o}$. Indeed, multiplying the previous identity by α^{n-1} yields

$$(\alpha\lambda)^n + \alpha'_{n-1}(\alpha\lambda)^{n-1} + \cdots + \alpha'_0 = 0,$$

where $\alpha'_i = \alpha_i\alpha^{n-1-i}$. Whence $\alpha\lambda$ is integral over $\mathcal{o}$ and thus $\alpha\lambda \in \mathcal{O}$. Then $\alpha\lambda = \beta$ for some $\beta \in \mathcal{O}$ which is equivalent to $\lambda = \frac{\beta}{\alpha}$. Therefore L is the field of fractions of $\mathcal{O}$. Now $\mathcal{O}$ is an integral domain as it is a subring of a field. Being the integral closure of $\mathcal{o}$ in L, $\mathcal{O}$ is an integrally closed domain with field of fractions L. It remains to prove the last statement. For the forward implication, suppose λ is integral over $\mathcal{o}$. Then λ is a root of a monic polynomial $f(x) \in \mathcal{o}[x]$. As $m_\lambda(x)$ divides $f(x)$ in $\mathcal{o}[x]$, all of the roots of $m_\lambda(x)$ are integral over $\mathcal{o}$ too. By Vieta's formulas, the coefficients of $m_\lambda(x)$ integral over $\mathcal{o}$ as well whence $m_\lambda(x) \in \mathcal{o}[x]$. For the reverse implication, if the minimal polynomial $m_\lambda(x)$ of λ over K has coefficients in $\mathcal{o}$ then λ is automatically integral over $\mathcal{o}$. □

The previous result can also be generalized to towers in light of the fact that integral closure is transitive.

Proposition 1.5. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, let $M/L/K$ be a finite separable tower, and let $\mathcal{O}$ and $\widetilde{\mathcal{O}}$ be the integral closures of $\mathcal{o}$ in L and M respectively. Then $\mathcal{O}$ and $\widetilde{\mathcal{O}}$ are integrally closed domains. In particular, $\widetilde{\mathcal{O}}$ is the integral closure of $\mathcal{O}$ in L.*

Proof. The first statement follows from two applications of Proposition 1.4. To prove the second statement, we must show $\widetilde{\mathcal{O}} = \overline{\mathcal{O}}$. The forward inclusion is obvious as $\mathcal{o}$ is a subset of $\mathcal{O}$. For the reverse inclusion, Proposition 1.2 implies that $\overline{\mathcal{O}}$ is integral over $\mathcal{o}$. As $\widetilde{\mathcal{O}}$ is the integral closure of $\mathcal{o}$ in M, the reverse inclusion follows. □

Number Fields

Number fields are a particular instance of the aforementioned setting. A **number field** K is a field extension of $\mathbb{Q}$ of finite degree. That is, K is a finite dimensional $\mathbb{Q}$-vector space. In particular, $K/\mathbb{Q}$ is a finite separable extension since $\mathbb{Q}$ is perfect whence the primitive element theorem applies. Moreover, $K/\mathbb{Q}$ is Galois if and only if it is normal. We say that the **degree** of K is $[K : \mathbb{Q}]$ which is simply the degree of K as a $\mathbb{Q}$-vector space. In the cases of small degrees we will often use the latin derived names **quadratic**, **cubic**, etc. Any $\kappa \in K$ is called an **algebraic number**. We define the **ring of integers** $\mathcal{O}_K$ of K to be the integral closure of $\mathbb{Z}$ in K. In other words,

$$\mathcal{O}_K = \{\kappa \in K : \kappa \text{ is integral over } \mathbb{Z}\}.$$

Any $\alpha \in \mathcal{O}_K$ is called an **algebraic integer**. Note that α is an algebraic integer if and only if it is the root of a monic polynomial $f(x) \in \mathbb{Z}[x]$. It follows from Proposition 1.4 that every $\kappa \in K$ is of the form

$$\kappa = \frac{\alpha}{a},$$

for some algebraic integer α and nonzero integer $\mathcal{o}$. In particular, K is the field of fractions of $\mathcal{O}_K$ and $\mathcal{O}_K$ is an integrally closed domain. Moreover, κ is an algebraic integer if and only if the minimal polynomial $m_\kappa(x)$ of κ over $\mathbb{Q}$ has coefficients in $\mathbb{Z}$.

Remark 1.6. By Lemma 1.3, $\mathbb{Z}$ is an integrally closed domain and therefore the ring of integers of $\mathbb{Q}$ is exactly $\mathbb{Z}$.

1.2 Traces and Norms

Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. Recall that by fixing a basis for $\mathcal{O}/\mathcal{o}$, there is an isomorphism

$$\mathrm{End}_{\mathcal{O}/\mathcal{o}} \cong \mathrm{Mat}_n(\mathcal{o})$$

This permits us to view $\mathcal{o}$-linear operators from $\mathcal{O}$ to itself as $n \times n$ matrices with coefficients in $\mathcal{o}$. To any $\beta \in \mathcal{O}$, we define the $\mathcal{o}$-linear operator T_β on $\mathcal{O}$ by

$$T_\beta : \mathcal{O} \to \mathcal{O} \qquad x \mapsto \beta x.$$

This is simply multiplication by β. The trace and determinant of T_β are deserving of special interest. For instance, let $f_\beta(x)$ denote the characteristic polynomial of T_β so that

$$f_\beta(x) = \det(xI - T_\beta) = x^n - \alpha_{n-1}x^{n-1} + \cdots + (-1)^n\alpha_0,$$

with $\alpha_i \in \mathcal{o}$. Then the trace and determinant have the alternative representations

$$\mathrm{trace}(T_\beta) = \alpha_{n-1} \quad \text{and} \quad \det(T_\beta) = \alpha_0,$$

We define the **trace** and **norm** of $\mathcal{O}/\mathcal{o}$, denoted $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}$ and $\mathrm{N}_{\mathcal{O}/\mathcal{o}}$, by

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}} : \mathcal{O} \to \mathcal{o} \qquad \beta \mapsto \mathrm{trace}(T_\beta) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}} : \mathcal{O} \to \mathcal{o} \qquad \beta \mapsto \det(T_\beta).$$

Our primary aim will be to develop a sufficient understanding of the trace and norm. The trace is additive while the norm is multiplicative because they correspond to the trace and determinant of matrices. Whence

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta + \beta') = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) + \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta') \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta\beta') = \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta)\,\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta'),$$

for all $\beta, \beta' \in \mathcal{O}$ making the trace and norm into homomorphisms. In fact, we have the additional relations

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\alpha\beta) = \alpha\,\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\alpha\beta) = \alpha^n\,\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta),$$

for all $\alpha \in \mathcal{o}$ by scalar multiplication of matrices. The trace and norm also behave well with respect to direct sums. If $\mathcal{O} = \mathcal{O}_1 \oplus \mathcal{O}_2$ and we have a decomposition $\beta = \beta_1 + \beta_2$ with $\beta_1 \in \mathcal{O}_1$ and $\beta_2 \in \mathcal{O}_2$ then

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) = \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_1) + \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_2) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta) = \mathrm{N}_{\mathcal{O}_1/\mathcal{o}}(\beta_1)\,\mathrm{N}_{\mathcal{O}_2/\mathcal{o}}(\beta_2),$$

because the corresponding matrix for β is block diagonal.

In the case of a degree n field extension L/K, we call $\mathrm{Tr}_{L/K}$ and $\mathrm{N}_{L/K}$ the **field trace** and **field norm** of L/K. For any $\lambda \in L$, the **field trace** and **field norm** of λ are $\mathrm{Tr}_{L/K}(\lambda)$ and $\mathrm{N}_{L/K}(\lambda)$. In this case, $\mathrm{N}_{L/K}(\lambda) = 0$ if and only if $\lambda = 0$ because otherwise T_λ has inverse $T_{\lambda^{-1}}$ and hence a nonzero determinant. Therefore we obtain homomorphisms

$$\mathrm{Tr}_{L/K} : L \to K \quad \text{and} \quad \mathrm{N}_{L/K} : L^* \to K^*.$$

When L/K is also separable, we can derive alternative descriptions of the field trace and field norm of L/K. This additional assumption is weak because we are mostly interested in field extensions of $\mathbb{Q}$ and $\mathbb{F}_p$ of finite degree which are always separable as these fields are perfect. In any case, to do this we need to work in the algebraic closure $\overline{K}$ of K. As L/K is a degree n separable extension, there are exactly n distinct K-embeddings $\sigma_1, \dots, \sigma_n$ of L into $\overline{K}$. That is, there are n elements of $\mathrm{Hom}_K(L, \overline{K})$. These K-embeddings are constructed by letting θ be a primitive element for L/K and sending θ to one of its conjugate roots in the minimal polynomial $m_\theta(x)$ of θ over K. These K-embeddings are deeply connected to the field trace and field norm.

Proposition 1.7. *Let L/K be a degree n separable extension and let σ run over the elements of $\mathrm{Hom}_K(L, \overline{K})$. For any $\lambda \in L$, the characteristic polynomial $f_\lambda(x)$ of T_λ over K is a power of the minimal polynomial $m_\lambda(x)$ of λ over K and satisfies*

$$f_\lambda(x) = \prod_\sigma (x - \sigma(\lambda)).$$

We also have

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_\sigma \sigma(\lambda) \quad \textit{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_\sigma \sigma(\lambda).$$

Moreover, if L/K is Galois and $\lambda_1, \dots, \lambda_n$ are the conjugates of λ then

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_i \lambda_i \quad \textit{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_i \lambda_i.$$

Proof. Let m and d be the degrees of $K(\lambda)/K$ and $L/K(\lambda)$ respectively. Write

$$m_\lambda(x) = x^m + \kappa_{m-1}x^{m-1} + \cdots + \kappa_0,$$

with $\kappa_i \in K$. We claim

$$f_\lambda(x) = m_\lambda(x)^d.$$

Indeed, recall that $1, \lambda, \dots, \lambda^{m-1}$ is a basis of $K(\lambda)/K$. If $\alpha_1, \dots, \alpha_d$ is a basis for $L/K(\lambda)$ then

$$\alpha_1, \alpha_1\lambda, \dots, \alpha_1\lambda^{m-1}, \dots, \alpha_d, \alpha_d\lambda, \dots, \alpha_d\lambda^{m-1},$$

is a basis for L/K. The matrix of T_λ with respect to this basis is block diagonal with d blocks each of the form

$$\begin{pmatrix} & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\kappa_0 & -\kappa_1 & \cdots & -\kappa_{m-1} \end{pmatrix}.$$

This matrix is the companion matrix to $m_\lambda(x)$ and hence the characteristic polynomial is $m_\lambda(x)$ as well. Our claim follows since the characteristic polynomial of a block diagonal matrix is the product of the characteristic polynomials of the blocks. Since λ is algebraic over K of degree m, $K(\lambda)$ is the splitting field of $m_\lambda(x)$ and there are m elements of $\mathrm{Hom}_K(K(\lambda), \overline{K})$. Then the elements of $\mathrm{Hom}_K(L, \overline{K})$ are partitioned into m many equivalence classes each of size d where two K-embeddings are equivalent if and only if they take the same value at λ. If τ runs over the elements of $\mathrm{Hom}_K(K(\lambda), \overline{K})$ then this a complete set of representatives. As

$$m_\lambda(x) = \prod_\tau (x - \tau(\lambda))$$

it follows that

$$f_\lambda(x) = \prod_\sigma (x - \sigma(\lambda)).$$

The formulas for the field trace and field norm follow from Vieta's formulas applied to this product for $f_\lambda(x)$. Now suppose L/K is Galois. Then $\mathrm{Hom}_K(L, \overline{K}) = \mathrm{Gal}(L/K)$. Therefore the conjugates of λ are exactly the images of λ under these K-embeddings and the last statement follows. $\square$

As an application of this result, we can show how the trace and norm act when $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K is a finite separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L.

Proposition 1.8. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\beta \in \mathcal{O}$ then $\mathrm{Tr}_{L/K}(\beta) \in \mathcal{o}$ and $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$.*

Proof. By Proposition 1.4, the minimal polynomial $m_\beta(x)$ of β over K has coefficients in $\mathcal{o}$. From Proposition 1.7, the characteristic polynomial $f_\beta(x)$ is a power of $m_\beta(x)$ and hence $f_\beta(x)$ has coefficients in $\mathcal{o}$ too. Whence $\mathrm{Tr}_{L/K}(\beta) \in \mathcal{o}$ and $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ as they are coefficients of $f_\beta(x)$ up to sign. $\square$

In this setting it is also easy to classify the units of $\mathcal{O}$ in terms of the units of $\mathcal{o}$.

Proposition 1.9. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\beta \in \mathcal{O}$ is a unit if and only if $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ is a unit.*

Proof. For the forward implication, if $\beta \in \mathcal{O}$ is a unit then $\frac{1}{\beta} \in \mathcal{O}$. Whence

$$\mathrm{N}_{L/K}(\beta)\, \mathrm{N}_{L/K}\left(\frac{1}{\beta}\right) = 1.$$

By Proposition 1.8, both of the field norms on the left-hand side are in $\mathcal{o}$. This proves $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ is a unit. For the reverse implication, we know by assumption that β is nonzero. Also, Proposition 1.4 implies that the minimal polynomial $m_\beta(x)$ of β over K has coefficients in $\mathcal{o}$. The constant term is a unit since it is $\mathrm{N}_{L/K}(\beta)$ up to sign. Letting the degree of $m_\beta(x)$ be m, we have shown that

$$m_\beta(x) = x^m + \alpha_{m-1}x^{m-1} + \cdots + \alpha,$$

with $\alpha, \alpha_i \in \mathcal{o}$ where α is a unit. Dividing this relation by $\alpha\beta^m$, we find that $\frac{1}{\beta}$ is a root of the monic polynomial

$$f(x) = x^m + \frac{\alpha_1}{\alpha}x^{m-1} + \cdots + \frac{1}{\alpha},$$

whose coefficients are in $\mathcal{o}$. Hence $\frac{1}{\beta} \in \mathcal{O}$ and thus β is a unit. $\square$

Having introduced traces and norms, we turn to discussing discriminants of free modules. Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. If $\beta_1, \ldots, \beta_n$ is a basis for $\mathcal{O}/\mathcal{o}$, we define its **trace matrix** $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$ by

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) = \begin{pmatrix} \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1\beta_1) & \cdots & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1\beta_n) \\ \vdots & & \vdots \\ \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_n\beta_1) & \cdots & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_n\beta_n) \end{pmatrix}.$$

The **discriminant** $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$ of $\beta_1, \ldots, \beta_n$ is defined to be the determinant of the trace matrix. That is,

$$d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) = \det(\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)).$$

In particular, $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) \in \mathcal{o}$ since the entries of the trace matrix belong to $\mathcal{o}$. It is also independent of the choice of basis up to elements of $(\mathcal{o}^*)^2$. For if $\beta_1', \ldots, \beta_n'$ is another basis for $\mathcal{O}/\mathcal{o}$, we have

$$\beta_i' = \sum_j \alpha_{i,j}\beta_j,$$

with $\alpha_{i,j} \in \mathcal{o}$. Then $(\alpha_{i,j})$ is the base change matrix from $\beta_1, \ldots, \beta_n$ to $\beta_1', \ldots, \beta_n'$ and so has nonzero determinant. Thus $\det((\alpha_{i,j})) \in \mathcal{o}^*$. Moreover,

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1', \ldots, \beta_n') = (\alpha_{i,j})\, \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)(\alpha_{i,j})^t,$$

which, upon taking the determinant, shows

$$d_{\mathcal{O}/\mathcal{o}}(\beta_1', \ldots, \beta_n') = \det((\alpha_{i,j}))^2 d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n). \tag{1}$$

Accordingly, we define the **discriminant** $d_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ to be the coset in $\mathcal{o}/(\mathcal{o}^*)^2$ represented by any discriminant $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$. In other words,

$$d_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)(\mathcal{o}^*)^2.$$

In particular, $d_{\mathcal{O}/\mathcal{o}} = 0$ is independent of the choice of representative. The discriminant is also multiplicative with respect to direct sums.

Proposition 1.10. *Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. Suppose we have a direct sum decomposition*

$$\mathcal{O} = \mathcal{O}_1 \oplus \mathcal{O}_2,$$

for free $\mathcal{o}$-modules $\mathcal{O}_1$ and $\mathcal{O}_2$ of ranks n_1 and n_2 with bases $\beta_{1,1}, \ldots, \beta_{n_1,1}$ and $\beta_{1,2}, \ldots, \beta_{n_2,2}$ over $\mathcal{o}$ respectively. Then $\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}$ is a basis of $\mathcal{O}/\mathcal{o}$ and

$$d_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}) = d_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) d_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}).$$

Proof. Clearly $\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}$ is a basis for $\mathcal{O}/\mathcal{o}$ and $\beta_{i,1}\beta_{j,2} = 0$. It follows that $d_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2})$ is the determinant of the block diagonal matrix

$$\begin{pmatrix} \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) & \\ & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}) \end{pmatrix}.$$

Recall that

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1) = \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_1) \quad \text{and} \quad \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_2) = \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_2),$$

for any $\beta_1 \in \mathcal{O}_1$ and $\beta_2 \in \mathcal{O}_2$. Whence the block diagonal matrix above is

$$\begin{pmatrix} \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) & \\ & \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}) \end{pmatrix}.$$

The determinant of this matrix is $d_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) d_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2})$ which completes the proof. $\square$

We now specialize to the setting of a degree n separable extension L/K. In this case, it turns out that the discriminant of a basis is nonzero. To see this will require some work. First, we define the **trace form** of L/K to be the map

$$\mathrm{Tr}_{L/K} : L \times L \to K \qquad (\lambda, \eta) \mapsto \mathrm{Tr}_{L/K}(\lambda\eta).$$

In other words, this is just the field trace of L/K considered as a pairing. Clearly this is a symmetric bilinear form. In fact, the trace form is also nondegenerate as Proposition 1.7 implies

$$\mathrm{Tr}_{L/K}(1) = n,$$

and we can pair any $\lambda \in L$ with its inverse. Using the trace form, we can show that the discriminant of any basis for L/K is nonzero.

Proposition 1.11. *Let L/K be a degree n separable extension and let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K. Then $d_{L/K}(\lambda_1, \ldots, \lambda_n) \neq 0$.*

Proof. Suppose by contradiction that $d_{L/K}(\lambda_1, \ldots, \lambda_n) = 0$. Then it follows that the trace matrix $\mathrm{Tr}_{L/K}(\lambda_1, \ldots, \lambda_n)$ is not invertible. Therefore

$$\mathrm{Tr}_{L/K}(\lambda_1, \ldots, \lambda_n) \begin{pmatrix} \kappa_1 \\ \vdots \\ \kappa_n \end{pmatrix} = \mathbf{0},$$

for some $\kappa_i \in K$ not all of which are zero. Then the element

$$\lambda = \sum_j \kappa_j \lambda_j.$$

is nonzero. Moreover, the previous identity is equivalent to the fact that $\mathrm{Tr}_{L/K}(\lambda\lambda_i) = 0$. As $\lambda_1, \dots, \lambda_n$ is a basis for L/K, it follows that the trace form is degenerate which is a contradiction. Hence $d_{L/K}(\lambda_1, \dots, \lambda_n) \neq 0$. $\square$

In addition to the discriminant $d_{L/K}(\lambda_1, \dots, \lambda_n)$ being nonzero, we can also write it in a more useful form. To do this, we define the **embedding matrix** $M(\lambda_1, \dots, \lambda_n)$ of the basis $\lambda_1, \dots, \lambda_n$ by

$$M(\lambda_1, \dots, \lambda_n) = \begin{pmatrix} \sigma_1(\lambda_1) & \cdots & \sigma_1(\lambda_n) \\ \vdots & & \vdots \\ \sigma_n(\lambda_1) & \cdots & \sigma_n(\lambda_n) \end{pmatrix},$$

where $\sigma_1, \dots, \sigma_n$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$. The discriminant turns out to be the square of the determinant of the embedding matrix.

Proposition 1.12. *Let L/K be a degree n separable extension and let $\lambda_1, \dots, \lambda_n$ be a basis for L/K. Then*

$$d_{L/K}(\lambda_1, \dots, \lambda_n) = \det(M(\lambda_1, \dots, \lambda_n))^2.$$

Proof. Recall that the ij-entry of $M(\lambda_1, \dots, \lambda_n)^t M(\lambda_1, \dots, \lambda_n)$ is the dot product of the i-th and j-th columns of $M(\lambda_1, \dots, \lambda_n)$. Then a short computation shows

$$\det(M(\lambda_1, \dots, \lambda_n)^t M(\lambda_1, \dots, \lambda_n)) = \det\left(\left(\sum_{\sigma \in \mathrm{Hom}_K(L, \overline{K})} \sigma(\lambda_i\lambda_j)\right)_{i,j}\right).$$

By Proposition 1.7, this identity is equivalent to the claim. $\square$

In general, it is difficult to compute the discriminant of a basis for L/K. However, if the basis is of the form $1, \lambda, \dots, \lambda^{n-1}$ (take $\lambda = \theta$ for a primitive element θ of L/K) then the discriminant of this basis can be easily computed. Indeed, the embedding matrix becomes

$$M(1, \lambda, \dots, \lambda^{n-1}) = \begin{pmatrix} 1 & \sigma_1(\lambda) & \cdots & \sigma_1(\lambda)^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \sigma_n(\lambda) & \cdots & \sigma_n(\lambda)^{n-1} \end{pmatrix},$$

which is a Vandermonde matrix. By Proposition 1.12, we have

$$d_{L/K}(1, \lambda, \dots, \lambda^{n-1}) = \prod_{i \leq j} (\sigma_i(\lambda) - \sigma_j(\lambda))^2, \tag{2}$$

which is the square of the Vandermonde determinant of $M(1, \lambda, \dots, \lambda^{n-1})$.

Let us now consider the situation when $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. If $\lambda_1, \ldots, \lambda_n$ is a basis for L/K then using Proposition 1.4 we may multiply by a nonzero element of $\mathcal{o}$, if necessary, to ensure that this basis belongs to $\mathcal{O}$. In many instances this will be useful. As a first application, the discriminant of such a basis relates inclusion of $\mathcal{O}$ relative to $\mathcal{o}$.

Lemma 1.13. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\lambda_1, \ldots, \lambda_n$ is a basis for L/K that is contained in $\mathcal{O}$ then*

$$d_{L/K}(\lambda_1, \ldots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

In particular, $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n.

Proof. Let $\beta \in \mathcal{O}$ and write

$$\beta = \sum_j \kappa_j \lambda_j.$$

for some $\kappa_j \in K$. Linearity of the trace implies

$$\mathrm{Tr}_{L/K}(\lambda_i \beta) = \sum_j \kappa_j \, \mathrm{Tr}_{L/K}(\lambda_i \lambda_j),$$

These n equations are equivalent to the identity

$$\begin{pmatrix} \mathrm{Tr}_{L/K}(\beta\lambda_1) \\ \vdots \\ \mathrm{Tr}_{L/K}(\beta\lambda_n) \end{pmatrix} = \mathrm{Tr}_{L/K}(\lambda_1, \ldots, \lambda_n) \begin{pmatrix} \kappa_1 \\ \vdots \\ \kappa_n \end{pmatrix}.$$

By Proposition 1.8, the column vector on the left-hand side and the trace matrix have entries in $\mathcal{o}$. Cramer's rule implies $d_{L/K}(\lambda_1, \ldots, \lambda_n)\kappa_i \in \mathcal{o}$. As β was arbitrary, this means

$$d_{L/K}(\lambda_1, \ldots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n,$$

proving the first statement. The second statement is immediate. $\square$

In this situation, we say that $\beta_1, \ldots, \beta_n$ is an **integral basis** for $\mathcal{O}/\mathcal{o}$ if

$$\mathcal{O} = \mathcal{o}\beta_1 + \cdots + \mathcal{o}\beta_n.$$

Equivalently, $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. An integral basis is necessarily a basis for L/K by Proposition 1.4. However, an integral basis need not always exist. For if $\lambda_1, \ldots, \lambda_n$ is a basis of L/K, Proposition 1.4 implies that we can multiply by a nonzero element of $\mathcal{o}$ to ensure that this basis is contained in $\mathcal{O}$. However, $\lambda_1, \ldots, \lambda_n$ need not also be a basis of $\mathcal{O}/\mathcal{o}$. Nevertheless, if $\mathcal{o}$ is a principal ideal domain then we can ensure the existence of an integral basis.

Theorem 1.14. *Let $\mathcal{o}$ be a principal ideal domain with field of fractions K, let L/K be a degree n separable extension, and let $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\mathcal{O}/\mathcal{o}$ admits an integral basis. Moreover, every finitely generated nonzero $\mathcal{O}$-submodule M of L is a free $\mathcal{o}$-module of rank n.*

Proof. Let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$. Then Lemma 1.13 implies

$$d_{L/K}(\lambda_1, \ldots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

It follows from the structure theorem for finitely generated modules over principal ideal domains that $\mathcal{O}$ is also a free $\mathcal{o}$-module of rank at most n. But any basis for $\mathcal{O}/\mathcal{o}$ must also be a basis for L/K by Proposition 1.4. Hence the rank is exactly n and $\mathcal{O}$ admits an integral basis over $\mathcal{o}$ which proves the first statement.

Now let M is a finitely generated nonzero $\mathcal{O}$-submodule of L with generators $\omega_1, \ldots, \omega_r$. By Proposition 1.4, there exists a nonzero $\alpha \in \mathcal{o}$ such that $\alpha\omega_i \in \mathcal{O}$. Whence $\alpha M \subseteq \mathcal{O}$. But then from our previous work, we have

$$\alpha d_{L/K}(\lambda_1, \ldots, \lambda_n)M \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

By the structure theorem for finitely generated modules over principal ideal domains again, M is a free $\mathcal{o}$-module of rank at most n. To see that the rank is at least n, let $\lambda \in M$ be nonzero. Then $\lambda\lambda_1, \ldots, \lambda\lambda_n$ is a basis for L/K contained in M. Thus the rank of M is at least n, and in particular it must be n. This proves the second statement. □

Recall that if L_1/K and L_2/K are finite separable extensions then the composite L of L_1 and L_2 is such that L/K is also a finite separable extension. Integral bases behave well with respect to composite fields provided the fields are linearly disjoint.

Proposition 1.15. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L_1/K and L_2/K be degree n_1 and n_2 separable extensions, and $\mathcal{O}_1$ and $\mathcal{O}_2$ be the integral closures of $\mathcal{o}$ in L_1 and L_2 respectively. Suppose L_1 and L_2 are linearly disjoint over K in $\overline{K}$ and that $\mathcal{O}_1/\mathcal{o}$ and $\mathcal{O}_2/\mathcal{o}$ admit integral bases $\beta_{1,1}, \ldots, \beta_{n_1,1}$ and $\beta_{1,2}, \ldots, \beta_{n_2,2}$ with*

$$\alpha_1 d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1}) + \alpha_2 d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2}) = 1,$$

for some $\alpha_1, \alpha_2 \in \mathcal{o}$. Let L be the composite of L_1 and L_2 and let $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$ and

$$\mathcal{O} = \mathcal{O}_1\mathcal{O}_2.$$

In particular,

$$d_{L/K}(\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}) = d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})^{n_2} d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})^{n_1}.$$

Proof. We will start by proving $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. Since L_1 and L_2 are linearly disjoint over K in $\overline{K}$, it must be that $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is a basis for L/K. Therefore any $\beta \in \mathcal{O}$ is of the form

$$\beta = \sum_{i,j} \kappa_{i,j}\beta_{i,1}\beta_{j,2},$$

for some $\kappa_{i,j} \in K$. We need to show that $\kappa_{i,j} \in \mathcal{o}$. To this end, let

$$\alpha_{i,2} = \sum_j \kappa_{i,j}\beta_{j,2} \quad \text{and} \quad \alpha_{j,1} = \sum_i \kappa_{i,j}\beta_{i,1},$$

so that

$$\beta = \sum_i \alpha_{i,2}\beta_{i,1} \quad \text{and} \quad \beta = \sum_j \alpha_{j,1}\beta_{j,2}.$$

In particular, $\alpha_{i,2} \in L_2$ and $\alpha_{j,1} \in L_1$. By linear disjointness, we have

$$\mathrm{Hom}_K(L, \overline{K}) \cong \mathrm{Hom}_K(L_1, \overline{K}) \times \mathrm{Hom}_K(L_2, \overline{K}).$$

We will now let $\sigma_{1,1}, \ldots, \sigma_{n_1,1}$ and $\sigma_{1,2}, \ldots, \sigma_{n_2,2}$ denote the elements of $\mathrm{Hom}_K(L_1, \overline{K})$ and $\mathrm{Hom}_K(L_2, \overline{K})$ respectively. Then $\sigma_{1,1}\sigma_{1,2}, \ldots, \sigma_{n_1,1}\sigma_{n_2,2}$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$ and we may view $\sigma_{1,1}, \ldots, \sigma_{n_1,1}$ and $\sigma_{1,2}, \ldots, \sigma_{n_2,2}$ as elements of $\mathrm{Hom}_K(L, \overline{K})$ that act as the identity on L_2 and L_1 respectively. Then the n_1 and n_2 equations

$$\sum_k \sigma_{i,1}(\beta_{k,1})\alpha_{k,2} = \sigma_{i,1}(\beta) \quad \text{and} \quad \sum_\ell \sigma_{j,2}(\beta_{\ell,2})\alpha_{\ell,1} = \sigma_{j,2}(\beta),$$

are equivalent to the identities

$$M(\beta_{1,1}, \ldots, \beta_{n_1,1}) \begin{pmatrix} \alpha_{1,2} \\ \vdots \\ \alpha_{n_1,2} \end{pmatrix} = \begin{pmatrix} \sigma_{1,1}(\beta) \\ \vdots \\ \sigma_{n_1,1}(\beta) \end{pmatrix},$$

and

$$M(\beta_{1,2}, \ldots, \beta_{n_2,2}) \begin{pmatrix} \alpha_{1,1} \\ \vdots \\ \alpha_{n_2,1} \end{pmatrix} = \begin{pmatrix} \sigma_{1,2}(\beta) \\ \vdots \\ \sigma_{n_2,2}(\beta) \end{pmatrix},$$

respectively. Observe that the embedding matrices and column vectors on the right-hand sides have entries in $\mathcal{O}$. Then $d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})\alpha_{i,2} \in \mathcal{O}$ and $d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})\alpha_{j,1} \in \mathcal{O}$ by Cramer's rule and Proposition 1.12 . As $\mathcal{O}_1 = L_1 \cap \mathcal{O}$ and $\mathcal{O}_2 = L_2 \cap \mathcal{O}$, we have $\alpha_{i,2} \in \mathcal{O}_2$ and $\alpha_{j,1} \in \mathcal{O}_1$. Expanding $\alpha_{i,2}$ and $\alpha_{j,1}$ show that $d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})\kappa_{i,j} \in \mathcal{o}$ and $d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})\kappa_{i,j} \in \mathcal{o}$. From the identity

$$k_{i,j} = k_{i,j}(\alpha_1 d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1}) + \alpha_2 d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})),$$

we conclude $k_{i,j} \in \mathcal{o}$. This proves $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. The fact that

$$\mathcal{O} = \mathcal{O}_1\mathcal{O}_2,$$

follows at once. It remains to prove the discriminant formula which will be achieved by decomposing the embedding matrix into blocks. Recall that $\sigma_{1,1}\sigma_{1,2}, \ldots, \sigma_{n_1,1}\sigma_{n_2,2}$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$, we have

$$M(\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}) = \begin{pmatrix} \sigma_{1,1}(\beta_{1,1})\sigma_{1,2}(\beta_{1,2}) & \cdots & \sigma_{1,1}(\beta_{n_1,1})\sigma_{1,2}(\beta_{n_2,2}) \\ \vdots & & \vdots \\ \sigma_{n_1,1}(\beta_{1,1})\sigma_{n_2,2}(\beta_{1,2}) & \cdots & \sigma_{n_1,1}(\beta_{n_1,1})\sigma_{n_2,2}(\beta_{n_2,2}) \end{pmatrix}.$$

This left-hand side admits the block factorization

$$\begin{pmatrix} M(\beta_{1,1},\ldots,\beta_{n_1,1}) & & \\ & \ddots & \\ & & M(\beta_{1,1},\ldots,\beta_{n_1,1}) \end{pmatrix} \begin{pmatrix} I\sigma_{1,2}(\beta_{1,2}) & \cdots & I\sigma_{1,2}(\beta_{n_2,2}) \\ \vdots & & \vdots \\ I\sigma_{n_2,2}(\beta_{1,2}) & \cdots & I\sigma_{n_2,2}(\beta_{n_2,2}) \end{pmatrix},$$

where the second matrix is the Kronecker product $M(\beta_{1,2},\ldots,\beta_{n_2,2})\otimes I$. As this Kronecker product can be expressed as $(I\otimes M(\beta_{1,2},\ldots,\beta_{n_2,2}))P$ for some permutation matrix P, it takes the form

$$\begin{pmatrix} M(\beta_{1,2},\ldots,\beta_{n_2,2}) & & \\ & \ddots & \\ & & M(\beta_{1,2},\ldots,\beta_{n_2,2}) \end{pmatrix} P,$$

with $\det(P)=\pm 1$. Putting these decompositions together and applying Proposition 1.12 shows

$$d_{L/K}(\beta_{1,1}\beta_{1,2},\ldots,\beta_{n_1,1}\beta_{n_2,2}) = d_{L_1/K}(\beta_{1,1},\ldots,\beta_{n_1,1})^{n_2} d_{L_2/K}(\beta_{1,2},\ldots,\beta_{n_2,2})^{n_1}.$$

This proves the discriminant formula. $\square$

It is generally a difficult problem to write down an integral basis explicitly. However, there is one instance in which this is possible. We say that $\mathcal{O}/\mathcal{o}$ is **monogenic** if $\mathcal{O}=\mathcal{o}[\beta]$ for some $\beta\in\mathcal{O}$. It follows immediately that $1,\beta,\ldots,\beta^{n-1}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. The discriminant of this basis is then given by Equation (2).

Regardless of if $\mathcal{O}/\mathcal{o}$ is monogenic or not, if an integral basis exists then the field trace $\mathrm{Tr}_{L/K}$ and field norm $\mathrm{N}_{L/K}$ agree with the trace $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}$ and norm $\mathrm{N}_{\mathcal{O}/\mathcal{o}}$ respectively.

Proposition 1.16. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\mathcal{O}$ admits an integral basis over $\mathcal{o}$ then*

$$\mathrm{Tr}_{L/K}(\beta)=\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \textit{and} \quad \mathrm{N}_{L/K}(\beta)=\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta),$$

for all $\beta\in\mathcal{O}$.

Proof. Let $\beta_1,\ldots,\beta_n$ be an integral basis for $\mathcal{O}/\mathcal{o}$. Then $\beta_1,\ldots,\beta_n$ is also a basis for L/K. It follows that the multiplication by β map in L has the same matrix representation as it does in $\mathcal{O}$. Whence

$$\mathrm{Tr}_{L/K}(\beta)=\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \text{and} \quad \mathrm{N}_{L/K}(\beta)=\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta).$$ $\square$

Number Fields

We now turn to the case of a number field K of degree n. We write $\mathrm{Tr}_K=\mathrm{Tr}_{K/\mathbb{Q}}$ and $\mathrm{N}_K=\mathrm{N}_{K/\mathbb{Q}}$ and call these the **field trace** and **field norm** of K. Moreover, for any $\kappa\in K$ we call $\mathrm{Tr}_K(\kappa)$ and $\mathrm{N}_K(\kappa)$ the **field trace** and **field norm** of κ. Observe from Propositions 1.8

and 1.9 that the field trace and field norm of algebraic integers are themselves integers and any $\kappa \in \mathcal{O}_K$ is a unit if and only if $\mathrm{N}_K(\kappa) = \pm 1$. We also call the nondegenerate symmetric bilinear form

$$\mathrm{Tr}_{K/\mathbb{Q}} : K \times K \to \mathbb{Q} \qquad (\kappa, \lambda) \mapsto \mathrm{Tr}_{K/\mathbb{Q}}(\kappa\lambda),$$

the **trace form** of K. Moreover, since $\mathbb{Z}$ is a principal ideal domain Theorem 1.14 implies that $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis. Whence $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n. Accordingly, we say that $\alpha_1, \ldots, \alpha_n$ is an **integral basis** for K if it is an integral basis for $\mathcal{O}_K/\mathbb{Z}$. Accordingly, we define the **discriminant** Δ_K of K to be the discriminant of $\mathcal{O}_K/\mathbb{Z}$. As $(\mathbb{Z}^*)^2 = \{1\}$, Δ_K is an integer and satisfies

$$\Delta_K = d_{L/K}(\alpha_1, \ldots, \alpha_n),$$

for any integral basis $\alpha_1, \ldots, \alpha_n$ for K. Moreover, Δ_K is nonzero since the trace form is nondegenerate and may very well be negative. In light of Proposition 1.12, we also have

$$|\det(M(\alpha_1, \ldots, \alpha_n))| = \sqrt{|\Delta_K|}. \tag{3}$$

Lastly, we say K is **monogenic** if $\mathcal{O}_K$ is monogenic over $\mathbb{Z}$.

1.3 Dedekind Domains

Let $\mathcal{O}$ be an integral domain with field of fractions K. Any nonzero ideal $\mathfrak{a}$ of $\mathcal{O}$ is said to be an **integral ideal** of $\mathcal{O}$. We call any prime integral ideal $\mathfrak{p}$ of $\mathcal{O}$ a **prime** of $\mathcal{O}$. If $\mathfrak{p}$ is principal with $\mathfrak{p} = \alpha\mathcal{O}$ for some nonzero $\alpha \in K$ we will also refer to α as the prime instead of $\mathfrak{p}$. In any case, an integral ideal is just an $\mathcal{O}$-submodule of $\mathcal{O}$. More generally, we say $\mathfrak{f}$ is a **fractional ideal** of $\mathcal{O}$ if it is a nonzero $\mathcal{O}$-submodule of K such that there is a nonzero $\delta \in \mathcal{O}$ with $\delta\mathfrak{f} \in \mathcal{O}$. This simply means $\delta\mathfrak{f}$ is an integral ideal. Conversely, if $\mathfrak{a}$ is an integral ideal and $\delta \in \mathcal{O}$ is nonzero then

$$\mathfrak{f} = \frac{1}{\delta}\mathfrak{a},$$

is a fractional ideal. So every fractional ideal is of this form. Clearly fractional ideals are closed under multiplication. The fractional ideal $\mathfrak{f}$ is said to be **principal** if it is generated by a single element. That is, if $\mathfrak{f} = \kappa\mathcal{O}$ for some nonzero $\kappa \in K$. In particular, all integral ideals are fractional ideals by taking $\delta = 1$ and all principal integral ideals are principal fractional ideals. The **inverse** $\mathfrak{f}^{-1}$ of $\mathfrak{f}$ is defined by

$$\mathfrak{f}^{-1} = \{\kappa \in K : \kappa\mathfrak{f} \subseteq \mathcal{O}\}.$$

Note that $\delta \in \mathfrak{f}^{-1}$ whence $\delta\mathfrak{f}^{-1}$ is an integral ideal so that $\mathfrak{f}^{-1}$ is a fractional ideal. It is the largest fractional ideal such $\mathfrak{f}\mathfrak{f}^{-1} \subseteq \mathcal{O}$. We say that $\mathfrak{f}$ is **invertible** if

$$\mathfrak{f}\mathfrak{f}^{-1} = \mathcal{O}.$$

A priori, a fractional ideal need not be invertible nor should we expect it to be. In order to guarantee invertibility, we need to consider a more restrictive setting than mere integral domains. This will allow for integral ideals to factor uniquely into a product of primes and for the fractional ideals to form an abelian group under multiplication in which every fractional ideal is invertible and $\mathcal{O}$ is the identity. This will essentially permit us to treat integral ideas as we would integers.

Accordingly, $\mathcal{O}$ is said to be a **Dedekind domain** if the following properties are satisfied:

(i) $\mathcal{O}$ is an integrally closed domain.

(ii) $\mathcal{O}$ is noetherian.

(iii) Every prime of $\mathcal{O}$ is maximal.

Observe that $\mathcal{O}$ being noetherian forces every integral ideal $\mathfrak{a}$ to be a finitely generated $\mathcal{O}$-module. In fact, as every fractional ideal $\mathfrak{f}$ is of the form $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$, for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$, $\mathfrak{f}$ is a finitely generated $\mathcal{O}$-submodule of K. Also, the condition that every prime of $\mathcal{O}$ is maximal is equivalent to $\mathcal{O}$ being one-dimensional as the longest chain of strict prime ideals is

$$0 \subsetneq \mathfrak{p} \subsetneq \mathcal{O},$$

for any prime $\mathfrak{p}$, and this is equivalent to $\mathfrak{p}$ being maximal as every maximal ideal is necessarily prime. In particular, a Dedekind domain is just a one-dimensional noetherian domain that is also an integrally closed domain. One-dimensional noetherian domains will also play a prominent role alongside Dedekind domains.

Remark 1.17. As $\mathbb{Z}$ is a principal ideal domain which is an integrally closed domain by Remark 1.6, $\mathbb{Z}$ is a Dedekind domain. The primes of $\mathbb{Z}$ are exactly the primes p.

In view of principal ideal domains being integrally closed domains by Lemma 1.3, they are Dedekind domains. In fact, Dedekind domains should be thought of as generalizations of principal ideal domains. As $\mathbb{Z}$ is a prototypical example of a principal ideal domain, a Dedekind domain $\mathcal{O}$ should be thought of as a generalization of $\mathbb{Z}$. In fact, we will show that being a principal ideal domain is equivalent to being a unique factorization domain for $\mathcal{O}$. So if $\mathcal{O}$ is not a principal ideal domain then unique factorization fails. However, it will be possible to remedy most of this as one of our primary aims is to prove that integral ideals factor uniquely into a product of primes. This clearly generalizes unique factorization in $\mathbb{Z}$ which will be enough. To this end, we first show inclusion in one direction for integral ideals. Interesting, this only requires that $\mathcal{O}$ be a noetherian domain.

Lemma 1.18. *Let $\mathcal{O}$ be a noetherian domain. For every integral ideal $\mathfrak{a}$ of $\mathcal{O}$, there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ of $\mathcal{O}$ such that*

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{a}.$$

Proof. Let $\mathcal{S}$ be the set of integral ideals which do not contain a product of primes. It suffices to show $\mathcal{S}$ is empty. Assuming otherwise and ordering $\mathcal{S}$ by inclusion, there exists a maximal element $\mathfrak{a} \in \mathcal{S}$ as $\mathcal{O}$ is noetherian. Moreover $\mathfrak{a}$ cannot be prime. Then there exist $\alpha_1, \alpha_2 \in \mathcal{O}$ with $\alpha_1\alpha_2 \in \mathfrak{a}$ and such that $\alpha_1, \alpha_2 \notin \mathfrak{a}$. Define integral ideals

$$\mathfrak{b}_1 = \mathfrak{a} + \alpha_1\mathcal{O} \quad \text{and} \quad \mathfrak{b}_2 = \mathfrak{a} + \alpha_2\mathcal{O}.$$

Note that $\mathfrak{b}_1$ and $\mathfrak{b}_2$ strictly contain $\mathfrak{a}$. Moreover, $\mathfrak{b}_1\mathfrak{b}_2 \subseteq \mathfrak{a}$ because

$$\mathfrak{b}_1\mathfrak{b}_2 = \mathfrak{a}^2 + \alpha_1\mathfrak{a} + \alpha_2\mathfrak{a} + \alpha_1\alpha_2\mathcal{O}.$$

Maximality of $\mathfrak{a}$ implies that there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \ldots, \mathfrak{q}_\ell$ such that

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{b}_1 \quad \text{and} \quad \mathfrak{q}_1 \cdots \mathfrak{q}_\ell \subseteq \mathfrak{b}_2.$$

But then

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \mathfrak{q}_1 \cdots \mathfrak{q}_\ell \subseteq \mathfrak{a},$$

which contradicts the fact that $\mathfrak{a} \in \mathcal{S}$. Hence $\mathcal{S}$ is empty. □

More work will be necessary obtain reverse inclusion and this requires the full strength of Dedekind domains. We being by showing that every prime $\mathfrak{p}$ is invertible. Note that unlike integral ideals, $1 \in \mathfrak{p}^{-1}$ so that $\mathfrak{p}^{-1}$ contains units.

Lemma 1.19. *Let $\mathcal{O}$ be a Dedekind domain and $\mathfrak{p}$ be a prime of $\mathcal{O}$. Then*

$$\mathcal{O} \subset \mathfrak{p}^{-1} \quad \textit{and} \quad \mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}.$$

Proof. We will first prove the inclusion. As $\mathcal{O} \subseteq \mathfrak{p}^{-1}$, it suffices to show $\mathfrak{p}^{-1} - \mathcal{O}$ is nonempty. To this end, let $\alpha \in \mathfrak{p}$ be nonzero. By Lemma 1.18 let k be the smallest positive integer such that there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ with

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \alpha\mathcal{O}.$$

As $\alpha\mathcal{O} \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is prime, there must be some $\mathfrak{p}_i$ such that $\mathfrak{p}_i \subseteq \mathfrak{p}$. Without loss of generality, we may assume $\mathfrak{p}_1 \subseteq \mathfrak{p}$. As primes are maximal in $\mathcal{O}$, we conclude $\mathfrak{p}_1 = \mathfrak{p}$. Minimality of k forces

$$\mathfrak{p}_2 \cdots \mathfrak{p}_k \not\subseteq \alpha\mathcal{O}.$$

Hence there exists $\beta \in \mathfrak{p}_2 \cdots \mathfrak{p}_k$ with $\beta \notin \alpha\mathcal{O}$. We claim $\beta\alpha^{-1} \in \mathfrak{p}^{-1} - \mathcal{O}$. Since $\mathfrak{p}_1 = \mathfrak{p}$, what we have previously shown implies $\beta\mathfrak{p} \subseteq \alpha\mathcal{O}$ whence $\beta\alpha^{-1}\mathfrak{p} \in \mathcal{O}$ which means $\beta\alpha^{-1} \in \mathfrak{p}^{-1}$. But as $\beta \notin \alpha\mathcal{O}$, we also have $\beta\alpha^{-1} \notin \mathcal{O}$. Hence $\beta\alpha^{-1} \in \mathfrak{p}^{-1} - \mathcal{O}$ as desired.

Now let us prove the identity. Observe that

$$\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq \mathcal{O}.$$

Since $\mathfrak{p}$ is maximal, it follows that $\mathfrak{p}\mathfrak{p}^{-1}$ is either $\mathfrak{p}$ or $\mathcal{O}$. So it suffices to show that the first case cannot hold. Assume by contradiction that $\mathfrak{p}\mathfrak{p}^{-1} = \mathfrak{p}$. Let $\omega_1, \ldots, \omega_r$ be generators of $\mathfrak{p}$ as an $\mathcal{O}$-module and let $\alpha \in \mathfrak{p}^{-1} - \mathcal{O}$. Then $\alpha\omega_i \in \mathfrak{p}\mathfrak{p}^{-1}$ whence $\alpha\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1}$ and $\alpha\mathfrak{p} \subseteq \mathfrak{p}$. But then

$$\alpha\omega_i = \sum_j \alpha_{i,j}\omega_j,$$

with $\alpha_{i,j} \in \mathcal{O}$. These r equations are equivalent to the identity

$$\begin{pmatrix} \alpha - \alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & \alpha - \alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & \alpha - \alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

Thus the determinant of the matrix on the left-hand side must be zero. But this means α is a root of the characteristic polynomial $\det(xI - (\alpha_{i,j}))$ which is a monic polynomial with coefficients $\mathcal{O}$. As $\mathcal{O}$ is an integrally closed domain, $\alpha \in \mathcal{O}$ which is a contradiction. Thus $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}$. $\square$

We can now show that every integral ideal factors uniquely into a product of primes.

Theorem 1.20. *Let $\mathcal{O}$ be a Dedekind domain. Then for every integral ideal $\mathfrak{a}$ of $\mathcal{O}$ there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ of $\mathcal{O}$ such that $\mathfrak{a}$ factors as*

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_k.$$

Moreover, this factorization is unique up to reordering of the factors.

Proof. We first prove existence and then uniqueness. For existence, let $\mathcal{S}$ be the set of integral ideals that are not a product of primes. We will show that $\mathcal{S}$ is empty. Assuming otherwise and ordering $\mathcal{S}$ by inclusion, there exists a maximal element $\mathfrak{a} \in \mathcal{S}$ as $\mathcal{O}$ is noetherian. Necessarily $\mathfrak{a}$ is not prime. Since primes are maximal in $\mathcal{O}$, there is some prime $\mathfrak{p}_1$ for which $\mathfrak{a} \subset \mathfrak{p}_1$. Then Lemma 1.19 implies

$$\mathfrak{a} \subset \mathfrak{a}\mathfrak{p}_1^{-1} \subset \mathcal{O}.$$

In particular, $\mathfrak{a}\mathfrak{p}_1^{-1}$ is an integral ideal. By maximality of $\mathfrak{a}$, $\mathfrak{a}\mathfrak{p}_1^{-1}$ factors into a product of primes. That is, there exist primes $\mathfrak{p}_2, \ldots, \mathfrak{p}_k$ such that

$$\mathfrak{a}\mathfrak{p}_1^{-1} = \mathfrak{p}_2, \ldots, \mathfrak{p}_k.$$

Hence

$$\mathfrak{a} = \mathfrak{p}_1, \ldots, \mathfrak{p}_k,$$

which is a contradiction. Therefore $\mathcal{S}$ is empty thus proving the existence of such a factorization.

Now we prove uniqueness. Suppose $\mathfrak{a}$ admits factorizations

$$\mathfrak{a} = \mathfrak{p}_1, \ldots, \mathfrak{p}_k \quad \text{and} \quad \mathfrak{a} = \mathfrak{q}_1, \ldots, \mathfrak{q}_\ell,$$

for primes $\mathfrak{p}_i$ and $\mathfrak{q}_j$. Since $\mathfrak{p}_1$ is prime, there is some j for which $\mathfrak{q}_j \subseteq \mathfrak{p}_1$. Without loss of generality, we may assume $\mathfrak{q}_1 \subseteq \mathfrak{p}_1$ and since primes are maximal in $\mathcal{O}$ we have $\mathfrak{q}_1 = \mathfrak{p}_1$. Then

$$\mathfrak{p}_2, \ldots, \mathfrak{p}_k = \mathfrak{q}_2, \ldots, \mathfrak{q}_\ell.$$

Repeating this process, we see that the factorizations are the same. This proves uniqueness of the factorization. $\square$

As a near immediate corollary, fractional ideal admits analogous factorizations which implies that they are invertible.

Corollary 1.21. *Let $\mathcal{O}$ be a Dedekind domain. Then for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}$ there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \ldots, \mathfrak{q}_\ell$ of $\mathcal{O}$ such that $\mathfrak{f}$ factors as*

$$\mathfrak{f} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \mathfrak{q}_1^{-1}, \ldots, \mathfrak{q}_\ell^{-1}.$$

Moreover, this factorization is unique up to reordering of the factors.

Proof. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. By Theorem 1.20, $\mathfrak{a}$ and $\delta\mathcal{O}$ admit unique factorizations

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \quad \text{and} \quad \delta\mathcal{O} = \mathfrak{q}_1, \dots, \mathfrak{q}_\ell,$$

for some primes $\mathfrak{p}_1, \dots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \dots, \mathfrak{q}_\ell$ up to reordering of the factors. Hence

$$\mathfrak{f} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \mathfrak{q}_1^{-1}, \dots, \mathfrak{q}_\ell^{-1}.$$

□

So for any fractional ideal $\mathfrak{f}$ there exist distinct prime $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ such that $\mathfrak{f}$ admits a factorization

$$\mathfrak{f} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

for some nonzero integers e_i. This is called the **prime factorization** of $\mathfrak{f}$ with **prime factors** $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. In particular, the prime factorization of an integral ideal $\mathfrak{a}$ is of the form

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

for some positive integers e_i. Accordingly, for any two integral ideal $\mathfrak{a}$ and $\mathfrak{b}$ we say that $\mathfrak{b}$ **divides** $\mathfrak{a}$ and write $\mathfrak{b} \mid \mathfrak{a}$ if $\mathfrak{a} \subseteq \mathfrak{b}$. Sometimes this condition is expressed as *to divide is to contain*. By prime factorization, this is equivalent to the fact that every prime power factor of $\mathfrak{b}$ appears in the prime factorization of $\mathfrak{a}$. In the case of a prime $\mathfrak{p}$, we also say that $\mathfrak{p}^e$ **exactly divides** $\mathfrak{a}$ and write $\mathfrak{p}^e \mid\mid \mathfrak{a}$ if $\mathfrak{p}^e$ divides $\mathfrak{a}$ but no higher power of $\mathfrak{p}$ divides $\mathfrak{a}$. This is equivalent to the fact that $\mathfrak{p}^e$ is the power of $\mathfrak{p}$ appearing in the prime factorization of $\mathfrak{a}$. Moreover, if $\mathfrak{a} \mid \mathfrak{p}$ then $\mathfrak{a} = \mathfrak{p}$. The **greatest common divisor** $(\mathfrak{a}, \mathfrak{b})$ of $\mathfrak{a}$ and $\mathfrak{b}$ is defined to be the integral ideal that all other common integral ideal divisors divide. Since to divide is to contain, $(\mathfrak{a}, \mathfrak{b})$ is the smallest ideal that contains both $\mathfrak{a}$ and $\mathfrak{b}$. This is $\mathfrak{a} + \mathfrak{b}$ and so $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b}$. The **least common multiple** $[\mathfrak{a}, \mathfrak{b}]$ of $\mathfrak{a}$ and $\mathfrak{b}$ is defined to be the integral ideal that divides all other common multiples. Since to divide is to contain, $[\mathfrak{a}, \mathfrak{b}]$ is the largest integral ideal that is contained in both $\mathfrak{a}$ and $\mathfrak{b}$. This is $\mathfrak{a} \cap \mathfrak{b}$ and so $[\mathfrak{a}, \mathfrak{b}] = \mathfrak{a} \cap \mathfrak{b}$. Lastly, we say that $\mathfrak{a}$ and $\mathfrak{b}$ are **relatively prime** if $(\mathfrak{a}, \mathfrak{b}) = \mathcal{O}$. In other words, $\mathfrak{a}$ and $\mathfrak{b}$ are comaximal which is to say

$$\mathfrak{a} + \mathfrak{b} = \mathcal{O}.$$

This is equivalent to the prime factorizations of $\mathfrak{a}$ and $\mathfrak{b}$ containing distinct primes. In particular, distinct primes and their powers are relatively prime. As these integral ideals are comaximal, we also have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$. In terms of the least common multiple, this means $[\mathfrak{a}, \mathfrak{b}] = \mathfrak{a}\mathfrak{b}$.

Just as it is common to suppress the fundamental theorem of arithmetic and just state the prime factorization of an integer, we suppress referencing Theorem 1.20 and simply state the prime factorization of a fractional ideal.

With the prime factorization in hand, we will discuss the group structure of the fractional ideals of $\mathcal{O}$. Let $I(\mathcal{O})$ denote the set of fractional ideals of $\mathcal{O}$. This is indeed an abelian group.

Theorem 1.22. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K. Then $I(\mathcal{O})$ is an abelian group with identity $\mathcal{O}$.*

Proof. Associativity and commutativity of $I(\mathcal{O})$ are obvious. The identity is $\mathcal{O}$ because every fractional ideal is a finitely generated $\mathcal{O}$-submodule of K. It remains to show that every fractional ideal $\mathfrak{f}$ is invertible. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. Then we must show $\mathfrak{a}$ is invertible. In view of Lemma 1.19, if

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

is the prime factorization of $\mathfrak{a}$ then let

$$\mathfrak{b} = \mathfrak{p}_1^{-e_1} \cdots \mathfrak{p}_r^{-e_r}.$$

We must show $\mathfrak{b} = \mathfrak{a}^{-1}$. For the forward inclusion, $\mathfrak{a}\mathfrak{b} = \mathcal{O}$ implies $\mathfrak{b} \subseteq \mathfrak{a}^{-1}$ by the definition of $\mathfrak{a}^{-1}$. For the reverse inclusion, suppose $\kappa \in \mathfrak{a}^{-1}$. Then $\kappa\mathfrak{a} \subseteq \mathcal{O}$ and multiplying by $\mathfrak{b}$ shows $\kappa\mathcal{O} \subseteq \mathfrak{b}$ whence $\kappa \in \mathfrak{b}$. □

It follows from this result that every fractional ideal is invertible. We will use this fact so often that we suppress it for legibility.

Let $P(\mathcal{O})$ denote the subgroup of principal fractional ideals of $I(\mathcal{O})$. Since $I(\mathcal{O})$ is abelian by the previous result, $P(\mathcal{O})$ is normal. The **ideal class group** $\mathrm{Cl}(\mathcal{O})$ of $\mathcal{O}$ is defined to be the quotient

$$\mathrm{Cl}(\mathcal{O}) = I(\mathcal{O})/P(\mathcal{O}).$$

A class of $\mathrm{Cl}(\mathcal{O})$ is called an **ideal class** of $\mathcal{O}$. As every fractional ideal $\mathfrak{f}$ can be expressed as $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta in \mathcal{O}$ and integral ideal $\mathfrak{a}$, we have $\delta\mathfrak{f} = \mathfrak{a}$ and it follows that every ideal class can be represented by an integral ideal. The **class number** $h_{\mathcal{O}}$ of $\mathcal{O}$ is defined by

$$h_{\mathcal{O}} = |\,\mathrm{Cl}(\mathcal{O})|.$$

The ideal class group encodes how much $\mathcal{O}$ fails to be a principal ideal domain and the class number is a measure of the degree of failure. Indeed, $\mathcal{O}$ is a principal ideal domain if and only if $h_{\mathcal{O}} = 1$.

Remark 1.23. The class number $h_{\mathcal{O}}$ need not be finite for a general Dedekind domain $\mathcal{O}$.

The **unit group** of $\mathcal{O}$ is defined to be $\mathcal{O}^*$. That is, the unit group is the group of units in $\mathcal{O}$. The ideal class and unit groups of $\mathcal{O}$ fit into a short exact sequence.

Proposition 1.24. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K. Then there is a short exact sequence*

$$1 \longrightarrow K^*/\mathcal{O}^* \longrightarrow I(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the second map is induced by $\kappa \mapsto \kappa\mathcal{O}$.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $K^*/\mathcal{O}^*$ and $\mathrm{Cl}(\mathcal{O})$. So it remains to prove exactness at $I(\mathcal{O})$. This follows because the principal fractional ideals represent the identity class of $\mathrm{Cl}(\mathcal{O})$ and these are generated by a complete set of representatives for $K^*/\mathcal{O}^*$. □

Thinking of the third map in this exact sequence as passing from numbers in K^* to fractional ideals in $I(\mathcal{O})$, exactness means that unit group is measuring the contraction (how many numbers are annihilated) taking place during this process while the ideal class group is measuring the expansion (how many fractional ideals are created).

Remark 1.25. The class number $h_{\mathcal{O}}$ and unit group $\mathcal{O}^*$ are two of the most difficult pieces of algebraic data of $\mathcal{O}$ to compute.

We will now turn to proving some interesting results about Dedekind domains. As a first result, being a principal ideal domain is equivalent to being a unique factorization domain for Dedekind domains.

Proposition 1.26. *Let $\mathcal{O}$ be a Dedekind domain. Then $\mathcal{O}$ is a principal ideal domain if and only if it is a unique factorization domain.*

Proof. The forward implication is immediate as every principal ideal domain is a unique factorization domain. For the reverse implication, suppose $\mathcal{O}$ is a unique factorization domain. As every integral ideal factors into a product of primes by Theorem 1.20 and the product of principal integral ideals is principal, it suffices to show that primes are principal. Let $\mathfrak{p}$ be a prime. Then there exists nonzero $\alpha \in \mathfrak{p}$ and α is not a unit since $\mathfrak{p}$ a proper ideal. Since $\mathcal{O}$ is a unique factorization domain, let $\alpha = \varepsilon\pi_1^{e_1}\cdots\pi_r^{e_r}$ be the prime factorization of α. Since $\mathfrak{p}$ is prime, it follows that there is some $\pi_i \in \mathfrak{p}$. Without loss of generality, we may assume $\pi_1 \in \mathfrak{p}$. Then the integral ideal $\pi_1\mathcal{O}$ satisfies $\pi_1\mathcal{O} \subseteq \mathfrak{p}$. As π_1 is prime, $\pi_1\mathcal{O}$ is a prime and hence maximal since prime ideals are maximal in $\mathcal{O}$. Whence $\pi_1\mathcal{O} = \mathfrak{p}$ proving $\mathfrak{p}$ is principal. □

The Chinese remainder theorem also applies to Dedekind domains which will allow us to deduce some more results. Suppose $\mathfrak{a}$ is an integral ideal with prime factorization

$$\mathfrak{a} = \mathfrak{p}_1^{e_1}\cdots\mathfrak{p}_r^{e_r}.$$

As powers of distinct primes are relatively prime, the integral ideals $\mathfrak{p}_1^{e_1}, \ldots, \mathfrak{p}_r^{e_r}$ are pairwise relatively prime. Whence they are pairwise comaximal so that the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_i \mathcal{O}/\mathfrak{p}_i^{e_i},$$

induced by natural projection. In particular, for any choice of $\alpha_i \in \mathcal{O}$ there exists a $\alpha \in \mathcal{O}$ such that

$$\alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i}}.$$

We will use this fact to prove a few useful lemmas about Dedekind domains. Our first lemma shows that given two integral ideals, we can multiply by a relatively prime integral ideal to produce a principal integral ideal.

Lemma 1.27. *Let $\mathcal{O}$ be a Dedekind domain and $\mathfrak{a}$ and $\mathfrak{b}$ be integral ideals of $\mathcal{O}$. Then there exists an integral ideal $\mathfrak{c}$ of $\mathcal{O}$ relatively prime to $\mathfrak{b}$ such that $\mathfrak{ac}$ is principal.*

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the prime factors of both $\mathfrak{a}$ and $\mathfrak{b}$ so that

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r} \quad \text{and} \quad \mathfrak{b} = \mathfrak{p}_1^{f_1} \cdots \mathfrak{p}_r^{f_r},$$

for some nonnegative integers e_i and f_i. By prime factorization, there exists $\alpha_i \in \mathfrak{p}_i^{e_i} - \mathfrak{p}_i^{e_i+1}$. As $\mathfrak{p}_1^{e_1+1}, \ldots, \mathfrak{p}_r^{e_r+1}$ are pairwise relatively prime, the Chinese remainder theorem implies the existence of an $\alpha \in \mathcal{O}$ such that $\alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i+1}}$. But then

$$\alpha \equiv 0 \pmod{\mathfrak{p}_i^{e_i}} \quad \text{and} \quad \alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i+1}}.$$

Whence $\mathfrak{p}_i^{e_i} \mid\mid \alpha\mathcal{O}$. It follows that $(\alpha\mathcal{O}, \mathfrak{a}\mathfrak{b}) = \mathfrak{a}$. Then there exists an integral ideal $\mathfrak{c}$ such that $\alpha\mathcal{O} = \mathfrak{a}\mathfrak{c}$ and necessarily $(\mathfrak{a}\mathfrak{c}, \mathfrak{a}\mathfrak{b}) = \mathfrak{a}$. Thus $(\mathfrak{c}, \mathfrak{b}) = \mathcal{O}$ which is to say that $\mathfrak{c}$ must be relatively prime to $\mathfrak{b}$. $\square$

Our second lemma shows that multiplying by fractional ideals does not affect quotients.

Lemma 1.28. *Let $\mathcal{O}$ be a Dedekind domain where $\mathfrak{f}$, $\mathfrak{g}$, and $\mathfrak{h}$ be fractional ideals of $\mathcal{O}$ with $\mathfrak{g} \subseteq \mathfrak{f}$. Then we have an isomorphism*

$$\mathfrak{f}/\mathfrak{g} \cong \mathfrak{f}\mathfrak{h}/\mathfrak{g}\mathfrak{h},$$

induced by multiplication and natural projection. In particular, there is an isomorphism

$$\mathfrak{a}^{-1}/\mathcal{O} \cong \mathcal{O}/\mathfrak{a},$$

induced by multiplication and natural projection for any integral ideal $\mathfrak{a}$ of $\mathcal{O}$.

Proof. Write $\mathfrak{f} = \frac{1}{\alpha}\mathfrak{a}$, $\mathfrak{g} = \frac{1}{\beta}\mathfrak{b}$, and $\mathfrak{h} = \frac{1}{\gamma}\mathfrak{c}$ for some nonzero $\alpha, \beta, \gamma \in \mathcal{O}$ and integral ideals $\mathfrak{a}$, $\mathfrak{b}$, and $\mathfrak{c}$ with $\mathfrak{b} \subseteq \mathfrak{a}$. In view of the isomorphisms

$$\mathfrak{f}/\mathfrak{g} \cong \beta\mathfrak{a}/\alpha\mathfrak{b} \quad \text{and} \quad \mathfrak{f}\mathfrak{h}/\mathfrak{g}\mathfrak{h} \cong \beta\mathfrak{a}\mathfrak{c}/\alpha\mathfrak{b}\mathfrak{c},$$

it suffices to show

$$\mathfrak{a}/\mathfrak{b} \cong \mathfrak{a}\mathfrak{c}/\mathfrak{b}\mathfrak{c}.$$

To this end, as $\mathfrak{a} \mid \mathfrak{b}$ we conclude $\mathfrak{b}\mathfrak{a}^{-1}$ is an integral ideal. Then Lemma 1.27 implies the existence of an integral ideal $\mathfrak{d}$ relatively prime to $\mathfrak{b}\mathfrak{a}^{-1}$ such that $\mathfrak{c}\mathfrak{d} = \delta\mathcal{O}$ for some nonzero $\delta \in \mathcal{O}$. Whence

$$\mathfrak{d} + \mathfrak{b}\mathfrak{a}^{-1} = \mathcal{O} \quad \text{and} \quad \mathfrak{d} \cap \mathfrak{b}\mathfrak{a}^{-1} = \mathfrak{d}\mathfrak{b}\mathfrak{a}^{-1}.$$

Writing $\mathfrak{d} = \delta\mathfrak{c}^{-1}$, short computations show that these identities imply

$$\delta\mathfrak{a} + \mathfrak{b}\mathfrak{c} = \mathfrak{a}\mathfrak{c} \quad \text{and} \quad \delta^{-1}\mathfrak{b}\mathfrak{c} \cap \mathfrak{a} = \mathfrak{b}.$$

Consider the homomorphism

$$\phi : \mathfrak{a} \to \mathfrak{a}\mathfrak{c}/\mathfrak{b}\mathfrak{c} \qquad \alpha \mapsto \delta\alpha + \mathfrak{b}\mathfrak{c},$$

induced by multiplication and natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker\phi = \mathfrak{b}$. Surjectivity follows from the first identity above. As $\ker\phi = \delta^{-1}\mathfrak{b}\mathfrak{c} \cap \mathfrak{a}$, the second identity above implies $\ker\phi = \mathfrak{b}$. This proves the first statement. For the second statement, take $\mathfrak{f} = \mathfrak{a}^{-1}$, $\mathfrak{g} = \mathcal{O}$, and $\mathfrak{h} = \mathfrak{a}$. $\square$

If $\mathcal{O}$ is a Dedekind domain and $\mathfrak{p}$ is a prime of $\mathcal{O}$, we call $\mathbb{F}_{\mathfrak{p}} = \mathcal{O}/\mathfrak{p}$ the **residue class field** of $\mathcal{O}$ by $\mathfrak{p}$. This is field as mfp is maximal. Our next lemma gives an isomorphism between the residue class field and quotients by powers of a prime.

Lemma 1.29. *Let $\mathcal{O}$ be a Dedekind domain. Then for any prime $\mathfrak{p}$ of $\mathcal{O}$ and nonnegative integer n, we have an isomorphism*

$$\mathbb{F}_{\mathfrak{p}} \cong \mathfrak{p}^n/\mathfrak{p}^{n+1},$$

induced by multiplication and natural projection.

Proof. By uniqueness of prime factorizations of fractional ideals, there exists $\beta \in \mathfrak{p}^n - \mathfrak{p}^{n+1}$. Consider the homomorphism

$$\phi : \mathcal{O} \to \mathfrak{p}^n/\mathfrak{p}^{n+1} \qquad \alpha \mapsto \beta\alpha + \mathfrak{p}^{n+1},$$

induced by multiplication and natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker \phi = \mathfrak{p}$. By our choice of β, we find that $\beta\mathfrak{p}^{-n}$ is an integral ideal relatively prime to $\mathfrak{p}$. Whence

$$\beta\mathfrak{p}^{-n} + \mathfrak{p} = \mathcal{O} \quad \text{and} \quad \beta\mathfrak{p}^{-n} \cap \mathfrak{p} = \beta\mathfrak{p}^{1-n}.$$

Short computations show that these identities imply

$$\beta\mathcal{O} + \mathfrak{p}^{n+1} = \mathfrak{p}^n \quad \text{and} \quad \beta^{-1}\mathfrak{p}^{n+1} \cap \mathcal{O} = \mathfrak{p}.$$

Surjectivity follows from the first identity. As $\ker \phi = \beta^{-1}\mathfrak{p}^{n+1} \cap \mathcal{O}$, the second identity implies $\ker \phi = \mathfrak{p}$. □

We now state a few additional interesting propositions about Dedekind domains. The first is that any Dedekind domain with only finitely many primes is a principal ideal domain.

Proposition 1.30. *Let $\mathcal{O}$ be a Dedekind domain. If there are only finitely many primes of $\mathcal{O}$ then $\mathcal{O}$ is a principal ideal domain.*

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the primes of $\mathcal{O}$. Then for any integral ideal $\mathfrak{a}$, Lemma 1.27 implies the existence of an integral ideal $\mathfrak{b}$ relatively prime to $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ such that $\mathfrak{a}\mathfrak{b} = \alpha\mathcal{O}$ for some nonzero $\alpha \in \mathcal{O}$. As $\mathfrak{b}$ is relatively prime to all of the primes of $\mathcal{O}$ we must have $\mathfrak{b} = \mathcal{O}$. Then $\mathfrak{a} = \alpha\mathcal{O}$. As $\mathfrak{a}$ was arbitrary, $\mathcal{O}$ is a principal ideal domain. □

The second is that any Dedekind domain is not far off from being a principal ideal domain in terms of the number of generators needed for an integral ideal. In fact, any fractional ideal is generated by at most two elements.

Proposition 1.31. *Let $\mathcal{O}$ be a Dedekind domain. Then every fractional ideal $\mathfrak{f}$ of $\mathcal{O}$ is generated by at most two elements.*

Proof. Writing $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{o}$ and integral ideal $\mathfrak{a}$, it suffices to prove the claim $\mathfrak{a}$. Let $\alpha \in \mathfrak{a}$ be nonzero. By Lemma 1.27, there exists an integral ideal $\mathfrak{b}$ relatively prime to $\alpha\mathcal{o}$ such that $\mathfrak{a}\mathfrak{b} = \beta\mathcal{o}$ for some nonzero $\beta \in \mathcal{o}$. Whence

$$\alpha\mathcal{o} + \mathfrak{b} = \mathcal{o}.$$

Upon multiplying by $\mathfrak{a}$, a short computation shows

$$\alpha\mathcal{o} + \beta\mathcal{o} = \mathfrak{a}.$$

Therefore $\mathfrak{a}$ is generated by at most two elements. $\square$

For our last result, let $\mathcal{o}$ be a Dedekind domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then it turns out that $\mathcal{O}$ is also a Dedekind domain.

Proposition 1.32. *Let $\mathcal{o}$ be a Dedekind domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\mathcal{O}$ is a Dedekind domain with field of fractions L.*

Proof. By Proposition 1.4, $\mathcal{O}$ is an integrally closed domain with field of fractions L. To show $\mathcal{O}$ is noetherian, let the degree of L/K be n and $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$. Then Lemma 1.13 implies $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. As $\mathcal{o}$ is noetherian, every integral ideal of $\mathcal{O}$ is also a finitely generated $\mathcal{o}$-module and therefore also a finitely generated $\mathcal{O}$-module. Whence $\mathcal{O}$ is noetherian. It remains to show that every prime $\mathfrak{P}$ of $\mathcal{O}$ is maximal. This is equivalent to showing $\mathcal{O}/\mathfrak{P}$ is a field. To this end, consider the homomorphism

$$\phi : \mathcal{o} \to \mathcal{O}/\mathfrak{P} \qquad \alpha \mapsto \alpha + \mathfrak{P},$$

induced by natural projection. Then $\ker\phi = \mathfrak{P} \cap \mathcal{o}$ and we claim $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$. It is clearly an ideal of $\mathcal{o}$ and is prime because $\mathfrak{P}$ is. To see that it is not the zero ideal, let $\lambda \in \mathfrak{P}$ be nonzero. As λ is integral over $\mathcal{o}$, we have

$$\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Taking n minimal ensures $\alpha_0 \neq 0$ and isolating α_0 shows $\alpha_0 \in \mathfrak{P}$ whence $\alpha_0 \in \mathfrak{P} \cap \mathcal{o}$. Therefore $\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}$ for some prime $\mathfrak{p}$. This means $\ker\phi = \mathfrak{p}$. By the first isomorphism theorem, ϕ induces an embedding $\mathbb{F}_{\mathfrak{p}} \to \mathcal{O}/\mathfrak{P}$. Therefore $\mathcal{O}/\mathfrak{P}$ is a finite dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. By primality of $\mathfrak{P}$, $\mathcal{O}/\mathfrak{P}$ is also an integral domain and thus must be a field. $\square$

Accordingly, we say $\mathcal{O}/\mathcal{o}$ is a **Dedekind extension** of a finite separable extension L/K if $\mathcal{O}$ and $\mathcal{o}$ are Dedekind domains whose field of fractions are L and K respectively and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. By the preceding result, it is enough to show that $\mathcal{o}$ is a Dedekind domain and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. These Dedekind domains are related via the identity

$$\mathcal{O} \cap K = \mathcal{o}. \tag{4}$$

From Lemma 1.13 we see that $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. In view of this fact, if $\mathfrak{f}$ is a fractional ideal of $\mathcal{o}$ then $\mathfrak{f}\mathcal{O}$ is a fractional ideal of $\mathcal{O}$. In particular, $\mathfrak{a}\mathcal{O}$ is an integral ideal of $\mathcal{O}$ for any integral ideal $\mathfrak{a}$ of $\mathcal{o}$. Recall that if $\mathfrak{F}$ is a fraction ideal of $\mathcal{O}$ then we may write $\mathfrak{F} = \frac{1}{\delta}\mathfrak{A}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{A}$. Then Proposition 1.4 permits us take $\delta \in \mathcal{o}$. Also, it need not be true that $\mathcal{O}/\mathcal{o}$ admits an integral basis as $\mathcal{O}$ is not guaranteed to be a free $\mathcal{o}$-module of rank n. However, an integral basis will exist if $\mathcal{o}$ has finitely many primes for then it is a principal ideal domain by Proposition 1.30 and we may appeal to Theorem 1.14.

In fact, the previous result motivates the more general discussion of primes in an extension of integral domains $\mathcal{O}/\mathcal{o}$ where $\mathcal{O}$ is integral over $\mathcal{o}$. We start by introducing the notion of primes above other primes. If $\mathfrak{P}$ and $\mathfrak{p}$ are primes of $\mathcal{O}$ and $\mathcal{o}$ respectively, we say that $\mathfrak{P}$ is **above** $\mathfrak{p}$, or equivalently, $\mathfrak{p}$ is **below** $\mathfrak{P}$ if

$$\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}.$$

In fact, every prime of $\mathcal{O}$ is above exactly one prime of $\mathcal{o}$. To see this, it suffices to show $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$. Clearly $\mathfrak{P} \cap \mathcal{o}$ is an ideal of $\mathcal{o}$ and it is prime because $\mathfrak{P}$ is. To see that it is nonzero, let $\lambda \in \mathfrak{P}$ be nonzero. As λ is integral over $\mathcal{o}$, we have

$$\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Taking n minimal ensures $\alpha_0 \neq 0$. Isolating α_0 shows $\alpha_0 \in \mathfrak{P}$ whence $\alpha_0 \in \mathfrak{P} \cap \mathcal{o}$. Therefore $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$.

We can say much more when $\mathcal{O}/\mathcal{o}$ is a Dedekind extension. For if $\mathfrak{P}$ is above $\mathfrak{p}$ then $\mathfrak{P} \mid \mathfrak{p}\mathcal{O}$ because to divide is to contain. This implies that only finitely many primes $\mathfrak{P}$ can lie above a prime $\mathfrak{p}$ and they are exactly the prime factors of $\mathfrak{p}\mathcal{O}$. Moreover, every prime $\mathfrak{p}$ satisfies

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O}.$$

Assume by contradiction that $\mathfrak{p}\mathcal{O} = \mathcal{O}$. By prime factorization, let $\alpha \in \mathfrak{p} - \mathfrak{p}^2$ so that $\alpha\mathcal{o} = \mathfrak{a}\mathfrak{p}$ for some integral ideal $\mathfrak{a}$ relatively prime to $\mathfrak{p}$. Then $\mathfrak{a} + \mathfrak{p} = \mathcal{o}$ and so there exist $\beta \in \mathfrak{a}$ and $\gamma \in \mathfrak{p}$ both nonzero and satisfying $\beta + \gamma = 1$. As $\beta\mathfrak{p} \subseteq \alpha\mathcal{o}$, it follows that $\beta\mathcal{O} \subseteq \alpha\mathcal{O}$ which implies $\beta = \alpha\delta$ for some nonzero $\delta \in \mathcal{O}$. In view of Equation (4), $\delta \in \mathcal{o}$. Whence $\alpha\delta + \gamma = 1$ and thus $1 \in \mathfrak{p}$ contradicting that $\mathfrak{p}$ is proper. Therefore $\mathfrak{p}\mathcal{O}$ is always a integral ideal of $\mathcal{O}$. In fact, we have

$$\mathfrak{p}\mathcal{O} \cap \mathcal{o} = \mathfrak{p}.$$

The reverse inclusion is obvious. For the forward inclusion, as $\mathfrak{p}\mathcal{O} \neq \mathcal{O}$ there exists a prime factor $\mathfrak{P}$ of $\mathfrak{p}\mathcal{O}$ so that $\mathfrak{p}\mathcal{O} \subseteq \mathfrak{P}$. But then $\mathfrak{P}$ is above $\mathfrak{p}$ so that $\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}$ and the forward inclusion follows. Whence the primes above $\mathfrak{p}$ are exactly the prime factors $\mathfrak{P}$ of $\mathfrak{p}\mathcal{O}$. In fact, a more general identity holds. Let $\mathfrak{f}$ be a fractional ideal of $\mathcal{o}$. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{o}$ and integral ideal $\mathfrak{a}$. Expressing $\mathfrak{a}$ as its prime factorization and repeatedly applying the previous identity shows that

$$\mathfrak{f}\mathcal{O} \cap \mathcal{o} = \mathfrak{f}.$$

This is called **contraction** of the fractional ideal $\mathfrak{f}\mathcal{O}$ to $\mathcal{o}$. In practice, it is useful for relating information about the fractional ideal $\mathfrak{f}\mathcal{O}$ of $\mathcal{O}$ to the fraction ideal $\mathfrak{f}$ of $\mathcal{o}$.

Number Fields

We now turn to the case of a number field K for which our developments so far can be refined. By Proposition 1.32, the ring of integers $\mathcal{O}_K$ is a Dedekind domain as $\mathbb{Z}$ is a principal ideal domain. Whence $\mathcal{O}_K/\mathbb{Z}$ is a Dedekind extension. In view of this fact, we simplify some terminology. An **integral ideal** of K is simply an integral ideal of $\mathcal{O}_K$, a **prime** of K is a prime of $\mathcal{O}_K$, and a **fractional ideal** of K is a fractional ideal of $\mathcal{O}_K$. In particular, every fractional ideal $\mathfrak{f}$ is of the form

$$\mathfrak{f} = \frac{1}{d}\mathfrak{a},$$

for some nonzero $d \in \mathbb{Z}$ and integral ideal $\mathfrak{a}$. We will write $I(K)$ and $P(K)$ for the group of fractional ideals of $\mathcal{O}_K$ and the subgroup of principal fractional ideals respectively. The **ideal class group** $\mathrm{Cl}(K)$ of K is the ideal class group of $\mathcal{O}_K$. In particular,

$$\mathrm{Cl}(K) = I(K)/P(K).$$

The **class number** h_K of K is the class number of $\mathcal{O}_K$. The **unit group** of K is the unit group of $\mathcal{O}_K$ and we call any element of $\mathcal{O}_K^*$ a **unit** of K (this is slightly abusive as every element of K^* is invertible). Note that the integral ideals of K admit prime factorizations. One of our core investigations will be to understand how the principal integral ideal $p\mathcal{O}_K$ factors into a product of primes of K for any prime p. We will also leverage geometric ideas to show that the class number is finite and completely describe the unit group.

Now suppose $L/K/\mathbb{Q}$ is an extension of number fields. Then $L/K/\mathbb{Q}$ is a finite separable tower. Moreover, $\mathcal{O}_L/\mathcal{O}_K/\mathbb{Z}$ is a tower of Dedekind extensions of $L/K/\mathbb{Q}$ by Propositions 1.5 and 1.32. In particular, $\mathcal{O}_L/\mathcal{O}_K$ is a Dedekind extension of L/K.

1.4 Localization

Let $\mathcal{O}$ be an integral domain with field of fractions K. A subset D of $\mathcal{O}$ is said to be **multiplicative** if it is closed under multiplication, $1 \in D$, and $0 \notin D$. The set $\mathcal{O} - \{0\}$ is always multiplicative but more interesting examples are when we consider strict subsets. In any case, we define the **localization** $\mathcal{O}D^{-1}$ of $\mathcal{O}$ at D by

$$\mathcal{O}D^{-1} = \left\{ \frac{\eta}{\delta} \in K : \eta \in \mathcal{O} \text{ and } \delta \in D \right\}.$$

Clearly $\mathcal{O}D^{-1}$ is a subring of K which is an integral domain and is obtained from $\mathcal{O}$ by making the elements of D invertible. More generally, for any fractional ideal $\mathfrak{f}$ of $\mathcal{O}$, the **localization** $\mathfrak{f}D^{-1}$ of $\mathfrak{f}$ at D is defined by

$$\mathfrak{f}D^{-1} = \left\{ \frac{\eta}{\delta} \in K : \eta \in \mathfrak{f} \text{ and } \delta \in D \right\}.$$

In particular, $\mathfrak{a}D^{-1}$ is an integral ideal of $\mathcal{O}D^{-1}$ if $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$. Writing $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$, it follows that $\mathfrak{f}D^{-1}$ is a fractional ideal of $\mathcal{O}D^{-1}$. Note that in the case of an integral ideal $\mathfrak{a}$, we will have $\mathfrak{a}D^{-1} = \mathcal{O}D^{-1}$ if $\mathfrak{a}$ contains an element of D.

In the case of primes, we have a bijective correspondence between those of $\mathcal{O}$ disjoint from D and those of $\mathcal{O}D^{-1}$.

Proposition 1.33. *Let $\mathcal{O}$ be an integral domain and D be a multiplicative subset of $\mathcal{O}$. Then the maps*

$$\mathfrak{q} \mapsto \mathfrak{q}D^{-1} \quad \textit{and} \quad \mathfrak{Q} \mapsto \mathfrak{Q} \cap \mathcal{O}.$$

are inverse inclusion-preserving bijections between the primes $\mathfrak{q}$ of $\mathcal{O}$ disjoint from D and the primes $\mathfrak{Q}$ of $\mathcal{O}D^{-1}$.

Proof. First suppose $\mathfrak{q}$ is a prime of $\mathcal{O}$ that is disjoint from D. Then the integral ideal $\mathfrak{q}D^{-1}$ of $\mathcal{O}D^{-1}$ is prime because $\mathfrak{q}$ is. Also, the prime $\mathfrak{q}D^{-1}$ satisfies

$$\mathfrak{q} = \mathfrak{q}D^{-1} \cap \mathcal{O},$$

because $\mathfrak{q}$ is disjoint from D. Now suppose $\mathfrak{Q}$ is a prime of $\mathcal{O}D^{-1}$. Then the integral ideal $\mathfrak{Q} \cap \mathcal{O}$ of $\mathcal{O}$ is prime because $\mathfrak{Q}$ is. Also, $\mathfrak{Q} \cap \mathcal{O}$ is disjoint from D for otherwise it contains a unit which is a contradiction as prime ideals are proper. Moreover, the prime $\mathfrak{Q} \cap \mathcal{O}$ satisfies

$$\mathfrak{Q} = (\mathfrak{Q} \cap \mathcal{O})D^{-1}.$$

All of this together shows that the mappings

$$\mathfrak{q} \mapsto \mathfrak{q}D^{-1} \quad \text{and} \quad \mathfrak{Q} \mapsto \mathfrak{Q} \cap \mathcal{O}.$$

are inverse bijections between the primes $\mathfrak{q}$ of $\mathcal{O}$ disjoint from D and the primes $\mathfrak{Q}$ of $\mathcal{O}D^{-1}$. They are clearly inclusion-preserving. □

It follows from this result that the primes of $\mathcal{O}D^{-1}$ are of the form $\mathfrak{p}D^{-1}$ for primes $\mathfrak{p}$ of $\mathcal{O}$ disjoint from D. Often, D is chosen so that it is the compliment of a prime $\mathfrak{p}$. Indeed, $\mathcal{O} - \mathfrak{p}$ is a multiplicative subset precisely because $\mathfrak{p}$ is prime. In fact, let X be a set of primes of $\mathcal{O}$ and consider

$$\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}.$$

Then $\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}$ is a multiplicative subset because of the identity

$$\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p} = \bigcap_{\mathfrak{p} \in X} (\mathcal{O} - \mathfrak{p}).$$

In any case, we define the **localization** $\mathcal{O}_{\mathfrak{p}}$ of $\mathcal{O}$ at $\mathfrak{p}$ by

$$\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(\mathcal{O} - \mathfrak{p})^{-1}.$$

Localizing at a prime $\mathfrak{p}$ should be thought of as removing all of the algebraic information about $\mathcal{O}$ that has nothing to do with $\mathfrak{p}$. More generally, if $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{f}_{\mathfrak{p}}$ of $\mathfrak{f}$ at $\mathfrak{p}$ is defined to be

$$\mathfrak{f}_{\mathfrak{p}} = \mathfrak{f}(\mathcal{O} - \mathfrak{p})^{-1}.$$

In both of these constructions we have localized at a single prime. It is possible to localize at more than one prime. Let X be a set of primes in $\mathcal{O}$. We define the **localization** $\mathcal{O}(X)$ of $\mathcal{O}$ at X by

$$\mathcal{O}(X) = \mathcal{O}\left(\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

If $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{f}(X)$ of $\mathfrak{f}$ at X is defined to be

$$\mathfrak{f}(X) = \mathfrak{f}\left(\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

If X consists of a single prime $\mathfrak{p}$ then $\mathcal{O}(X) = \mathcal{O}_{\mathfrak{p}}$ and $\mathfrak{f}(X) = \mathfrak{f}_{\mathfrak{p}}$.

The case of localizing at a prime is deserving of special interest. An integral domain $\mathcal{O}$ is said to be a **local** if it has a unique maximal ideal. As may be expected, the resulting integral domain after localizing at a prime will be local. In particular, $\mathcal{O}_{\mathfrak{p}}$ is local where the maximal ideal is $\mathfrak{p}_{\mathfrak{p}}$. Indeed, recall from Proposition 1.33 that the map

$$\mathfrak{q} \to \mathfrak{q}_{\mathfrak{p}},$$

is an inclusion-preserving bijection between the primes of $\mathcal{O}$ contained in $\mathfrak{p}$ the primes of $\mathcal{O}_{\mathfrak{p}}$. As maximal ideals are prime, $\mathfrak{p}_{\mathfrak{p}}$ is the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. As for consequences, we have

$$\mathcal{O}_{\mathfrak{p}}^* = \mathcal{O}_{\mathfrak{p}} - \mathfrak{p}_{\mathfrak{p}} \quad \text{and} \quad \mathcal{O}_{\mathfrak{p}}^* + \mathfrak{p}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}^*.$$

In other words, the units of $\mathcal{O}_{\mathfrak{p}}$ are precisely the elements not in $\mathfrak{p}_{\mathfrak{p}}$ and the sum of a unit of $\mathcal{O}_{\mathfrak{p}}$ and an element of $\mathfrak{p}_{\mathfrak{p}}$ is again a unit of $\mathcal{O}_{\mathfrak{p}}$.

In order for localization to be a useful tool in algebraic investigations, it must behave well with respect to what we have already developed. To this end, we will collect some useful properties about localization. Most importantly, intersections of localizations behave well with respect to fractional ideals and units and recover the original structure before localization.

Proposition 1.34. *Let $\mathcal{O}$ be an integral domain with field of fractions K. Then*

$$\mathcal{O} = \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}} \quad \textit{and} \quad \mathcal{O}^* = \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^*.$$

In particular, for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}$, we have

$$\mathfrak{f} = \bigcap_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}.$$

Proof. For the first identity, the forward inclusion is obvious. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}$ and set

$$\mathfrak{a} = \{\alpha \in \mathcal{O} : \alpha\eta \in \delta\mathcal{O}\}.$$

Then $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$ containing δ. As δ is not contained in any prime of $\mathcal{O}$, it follows that $\mathfrak{a}$ cannot be contained in any prime of $\mathcal{O}$. But every proper ideal is contained in a maximal ideal which is necessarily prime. Whence $\mathfrak{a}$ is not proper and so $\mathfrak{a} = \mathcal{O}$. Therefore $\eta \in \delta\mathcal{O}$ which implies $\frac{\alpha}{\beta} \in \mathcal{O}$ proving the reverse inclusion. As for the second identity, the forward inclusion is clear. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^*$. We have already shown $\frac{\eta}{\delta} \in \mathcal{O}$ so it suffices to show $\frac{\eta}{\delta}$ is a unit. As δ is not contained in any prime of $\mathcal{O}$, it cannot be contained in any maximal ideal as maximal ideals are prime. This means δ is a unit. Interchanging the roles of η and δ shows that η is a unit as well. Hence $\frac{\eta}{\delta}$ is a unit and the reverse inclusion follows. The second statement is immediate from the first upon multiplying by $\mathfrak{f}$. $\square$

It also turns out that the localization of the inverse is the inverse of the localization for any fractional ideal of $\mathcal{o}$.

Proposition 1.35. *Let $\mathcal{o}$ be an integral domain with field of fractions K and D be a multiplicative subset of $\mathcal{o}$. Then for any fractional ideal $\mathfrak{f}$ of $\mathcal{o}$, we have*

$$(\mathfrak{f}D^{-1})^{-1} = \mathfrak{f}^{-1}D^{-1}.$$

Proof. The definition of the inverse implies $\mathfrak{f}D^{-1}(\mathfrak{f}D^{-1})^{-1} \subseteq \mathcal{o}D^{-1}$. Multiplying by $\mathfrak{f}^{-1}$ shows $(\mathfrak{f}D^{-1})^{-1} \subseteq \mathfrak{f}^{-1}D^{-1}$. This proves the forward inclusion. Similarly, we have $\mathfrak{f}\mathfrak{f}^{-1} \subseteq \mathcal{o}$ whence $\mathfrak{f}D^{-1}\mathfrak{f}^{-1}D^{-1} \subseteq \mathcal{o}D^{-1}$ which proves the reverse inclusion. □

Equivalently, this result says that inversion and localization are commutative. Therefore when inverting and localizing we do not need to specify the order in which these operations are performed.

When $\mathcal{o}$ is an integrally closed domain, localization respects integral closure.

Proposition 1.36. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then for any multiplicative set D of $\mathcal{o}$, $\mathcal{O}D^{-1}$ is the integral closure of $\mathcal{o}D^{-1}$ in L. In particular, $\mathcal{O}D^{-1}$ is an integrally closed domain with field of fractions L.*

Proof. To prove the first statement, we must show $\mathcal{O}D^{-1} = \overline{\mathcal{o}D^{-1}}$. For the forward inclusion, let $\frac{\eta}{\delta} \in \mathcal{O}D^{-1}$. As $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L, η is integral over $\mathcal{o}$ so that

$$\eta^n + \alpha_{n-1}\eta^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Diving by δ^n, we obtain

$$\left(\frac{\eta}{\delta}\right)^n + \frac{\alpha_{n-1}}{\delta}\left(\frac{\eta}{\delta}\right)^{n-1} + \cdots + \frac{\alpha_0}{\delta^n} = 0.$$

Thus $\frac{\eta}{\delta}$ is the root of a monic polynomial with coefficients in $\mathcal{o}D^{-1}$ whence $\frac{\eta}{\delta} \in \overline{\mathcal{o}D^{-1}}$. This proves the forward inclusion. For the reverse inclusion, suppose $\lambda \in \overline{\mathcal{o}D^{-1}}$. Then

$$\lambda^n + \frac{\alpha_{n-1}}{\delta_{n-1}}\lambda^{n-1} + \cdots + \frac{\alpha_0}{\delta_0} = 0,$$

for some positive integer n and $\frac{\alpha_i}{\delta_i} \in \mathcal{o}D^{-1}$. Then $\delta = \prod_i \delta_i$ is nonzero and upon multiplying by δ^n, we obtain

$$(\lambda\delta)^n + \frac{\alpha_{n-1}\delta}{\delta_{n-1}}(\lambda\delta)^{n-1} + \cdots + \frac{\alpha_0\delta^n}{\delta_0} = 0.$$

It follows that $\lambda\delta$ is the root of a monic polynomial with coefficients in $\mathcal{o}$. As $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L, we have $\lambda\delta \in \mathcal{O}$ and thus $\lambda \in \mathcal{O}D^{-1}$. This proves the reverse inclusion. For the second statement, observe that $\mathcal{O}D^{-1}$ is an integral domain as it is a subring of a field. Moreover, the field of fractions of $\mathcal{O}D^{-1}$ is L since the same is true for $\mathcal{O}$ by Proposition 1.4. The second statement follows. □

In view of the previous result, the localization of a Dedekind domain is again a Dedekind domain.

Proposition 1.37. *Let $\mathcal{o}$ be a Dedekind domain with field of fractions K and D be a multiplicative subset of $\mathcal{o}$. Then $\mathcal{o}D^{-1}$ is a Dedekind domain with field of fractions K.*

Proof. We see from Proposition 1.36 that $\mathcal{o}D^{-1}$ is an integrally closed domain with field of fractions K. To show it is noetherian, let $\mathfrak{A}$ be an integral ideal of $\mathcal{o}D^{-1}$ and set $\mathfrak{a} = \mathfrak{A} \cap \mathcal{o}$. Then

$$\mathfrak{A} = \mathfrak{a}D^{-1}.$$

Since $\mathcal{o}$ is noetherian, $\mathfrak{a}$ is a finitely generated $\mathcal{o}$-module and hence $\mathfrak{A}$ is a finitely generated $\mathcal{o}D^{-1}$-module. Therefore $\mathcal{o}D^{-1}$ is noetherian. It remains to show that every prime of $\mathcal{o}D^{-1}$ is maximal. By Proposition 1.33, every prime of $\mathcal{o}D^{-1}$ is of the form $\mathfrak{p}D^{-1}$ for some prime $\mathfrak{p}$ of $\mathcal{o}$. But then $\mathfrak{p}D^{-1}$ is maximal because $\mathfrak{p}$ is and the bijections in Proposition 1.33 are inclusion-preserving. ☐

An immediate consequence of this result is that the localizations $\mathcal{o}_{\mathfrak{p}}$ and $\mathcal{o}(X)$ are Dedekind domains if $\mathcal{o}$ is. In fact, from Propositions 1.32, 1.36 and 1.37 we see that $\mathcal{O}D^{-1}/\mathcal{o}D^{-1}$ is a Dedekind extension of a finite separable extension L/K if $\mathcal{O}/\mathcal{o}$ is. In this case we setup some additional notation. If $\mathfrak{p}$ is a prime of $\mathcal{o}$, we define the **localization** $\mathcal{O}_{\mathfrak{p}}$ of $\mathcal{O}$ at $\mathfrak{p}$ by

$$\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(\mathcal{o} - \mathfrak{p})^{-1}.$$

More generally, if $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{F}_{\mathfrak{p}}$ of $\mathfrak{F}$ at $\mathfrak{p}$ is defined by

$$\mathfrak{F}_{\mathfrak{p}} = \mathfrak{F}(\mathcal{o} - \mathfrak{p})^{-1}.$$

We can also consider the case of more than a single prime. Let X be a set of primes of $\mathcal{o}$. We define the **localization** $\mathcal{O}(X)$ of $\mathcal{O}$ at X by

$$\mathcal{O}(X) = \mathcal{O}\left(\mathcal{o} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

More generally, if $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{F}(X)$ of $\mathfrak{F}$ at X is defined by

$$\mathfrak{F}(X) = \mathfrak{F}\left(\mathcal{o} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

If X consists of a single prime $\mathfrak{p}$ then $\mathcal{O}(X) = \mathcal{O}_{\mathfrak{p}}$ and $\mathfrak{F}(X) = \mathfrak{F}_{\mathfrak{p}}$. Moreover, $\mathcal{O}_{\mathfrak{p}}/\mathcal{o}_{\mathfrak{p}}$ and $\mathcal{O}(X)/\mathcal{o}(X)$ are Dedekind extensions.

We will now consider a more restrictive setting. An integral domain $\mathcal{o}$ is said to be a **discrete valuation ring** if it is a principal ideal domain with a unique maximal ideal. Equivalently, $\mathcal{o}$ is a local principal ideal domain or a Dedekind domain with exactly one prime. Let $\mathfrak{p}$ be the unique maximal ideal of $\mathcal{o}$. In particular, $\mathfrak{p}$ is prime and of the form $\mathfrak{p} = \pi\mathcal{o}$ for some prime $\pi \in \mathcal{o}$. We call π a **uniformizer** of $\mathcal{o}$ and it is uniquely defined up to multiplication by a unit. On the one hand, $\mathcal{o}$ is local so every element not in $\mathfrak{p}$ is a unit.

On the other hand, $\mathcal{o}$ is a principal ideal domain so that π is the only prime of $\mathcal{o}$. Since $\mathcal{o}$ is necessarily a unique factorization domain, these facts together imply that every $\alpha \in \mathcal{o}$ is of the form

$$\alpha = \varepsilon\pi^n,$$

for some nonnegative integer n and unit ε. Moreover, α generates the integral ideal $\mathfrak{p}^n$ and every integral ideal is of this form. If K is the field of fractions of $\mathcal{o}$, it follows that every nonzero $\kappa \in K$ can be uniquely expressed as

$$\kappa = \varepsilon\pi^n,$$

for some integer n and unit ε. The most important data of a discrete valuation ring is its valuation. The **valuation** v of $\mathcal{o}$ is the function defined by

$$v : K \to \mathbb{Z} \cup \{\infty\} \qquad \kappa \mapsto v(\kappa) = \begin{cases} n & \text{if } \kappa = \varepsilon\pi^n, \\ \infty & \text{if } \kappa = 0. \end{cases}$$

We call $v(\kappa)$ the **valuation** of κ with respect to $\mathcal{o}$. Note that $v(\kappa) = 0$ if and only if κ is a unit in $\mathcal{o}$. If $\kappa \in \mathcal{o}$ then the $v(\kappa)$ is characterized by the equation

$$\kappa\mathcal{O} = \mathfrak{p}^{v(\kappa)},$$

since $\kappa = \varepsilon\pi^{v(\kappa)}$. Moreover, if $\kappa = \varepsilon\pi^n$ and $\eta = \delta\pi^m$, we have

$$\kappa\eta = \varepsilon\delta\pi^{n+m} \quad \text{and} \quad \kappa + \eta = (\varepsilon\pi^{n-\min(n,m)} + \delta\pi^{m-\min(n,m)})\pi^{\min(n,m)},$$

where $\varepsilon\pi^{n-\min(n,m)} + \delta\pi^{m-\min(n,m)}$ is a unit. These identities imply that v satisfies the properties

$$v(\kappa\eta) = v(\kappa) + v(\eta) \quad \text{and} \quad v(\kappa + \eta) \geq \min(v(\kappa), v(\eta)).$$

The first of which shows that

$$v : K^* \to \mathbb{Z} \qquad \kappa \mapsto v(\kappa),$$

is a surjective homomorphism.

A Dedekind extension $\mathcal{O}/\mathcal{o}$ is said to be **local** if $\mathcal{o}$ is a discrete valuation ring. As $\mathcal{o}$ is a principal ideal domain, $\mathcal{O}/\mathcal{o}$ admits an integral basis by Theorem 1.14 but we can say more. If $\mathfrak{p}$ is the unique prime of $\mathcal{o}$, then $\mathcal{O}$ has finitely many primes $\mathfrak{P}$ which are necessarily the prime factors of $\mathfrak{p}\mathcal{O}$ and are therefore all the primes above $\mathfrak{p}$. By Proposition 1.30, we find that $\mathcal{O}$ is also a principal ideal domain. Whence $\mathcal{O}/\mathcal{o}$ is an extension of principal ideal domains.

If $\mathcal{o}$ is a Dedekind domain then it is local if and only if it is a discrete valuation ring. Indeed, any Dedekind domain with finitely many primes is a principal ideal domain by Proposition 1.30. This will allow us to establish a connection between Dedekind domains and discrete valuation rings. In particular, a noetherian domain is a Dedekind domain if and only if its localization at any prime is a discrete valuation ring.

Theorem 1.38. *Let $\mathcal{o}$ be a noetherian domain. Then $\mathcal{o}$ is a Dedekind domain if and only if $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring for all primes $\mathfrak{p}$ of $\mathcal{o}$.*

Proof. For the forward implication, suppose $\mathcal{o}$ is a Dedekind domain and let $\mathfrak{p}$ be a prime. Then $\mathcal{o}_\mathfrak{p}$ is a local Dedekind domain and hence a discrete valuation ring as we have seen. This proves the forward implication. For the reverse implication, suppose $\mathcal{o}$ is a noetherian domain and $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring. As $\mathcal{o}_\mathfrak{p}$ is a principal ideal domain, $\mathcal{o}_\mathfrak{p}$ is an integrally closed domain by Lemma 1.3. Then Proposition 1.34 implies $\mathcal{o}$ is an integrally closed domain too. As $\mathcal{o}$ is noetherian by assumption, it remains to show that every prime $\mathfrak{q}$ of $\mathcal{o}$ is maximal. As $\mathfrak{q}$ is proper, it is contained in some maximal ideal $\mathfrak{p}$ which is necessarily prime. The inclusion-preserving bijections in Proposition 1.33 show that $\mathfrak{q}_\mathfrak{p}$ and $\mathfrak{p}_\mathfrak{p}$ are primes of $\mathcal{o}_\mathfrak{p}$. As $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring it is a Dedekind domain with exactly one prime. Whence $\mathfrak{q}_\mathfrak{p} = \mathfrak{p}_\mathfrak{p}$ and the aforementioned inclusion-preserving bijections imply that $\mathfrak{q} = \mathfrak{p}$. Hence $\mathfrak{q}$ is maximal. This proves the reverse implication. □

In view of this result, $\mathcal{O}_\mathfrak{p}/\mathcal{o}_\mathfrak{p}$ is a local Dedekind extension. Observe that $\mathfrak{P}$ is not above $\mathfrak{p}$ if and only if $\mathfrak{P}$ contains an element of $\mathcal{o} - \mathfrak{p}$. In view of Proposition 1.33, it follows the finitely many primes of $\mathcal{O}_\mathfrak{p}$ are those above the unique prime of $\mathcal{o}_\mathfrak{p}$ and they correspond to the primes of $\mathcal{O}$ above $\mathfrak{p}$. Moreover, suppose $\mathfrak{A}$ is an integral ideal of $\mathcal{O}$. By prime factorization, we may write

$$\mathfrak{A} = \prod_{\mathfrak{P}} \mathfrak{P}^{e_\mathfrak{P}},$$

where $e_\mathfrak{P}$ are nonnegative integers all but finitely many of which are zero. Then we see that

$$\mathfrak{A} = \prod_{\mathfrak{P} | \mathfrak{p}\mathcal{O}} \mathfrak{P}^{e_\mathfrak{P}}.$$

In other words, localizing an integral ideal of $\mathcal{O}$ at $\mathfrak{p}$ removes all of the prime factors that are not above $\mathfrak{p}$ and localizes the primes above $\mathfrak{p}$.

Suppose $\mathcal{o}$ is a Dedekind domain with field of fractions K and let $v_\mathfrak{p}$ denote the valuation of $\mathcal{o}_\mathfrak{p}$ in view of Theorem 1.38. We call $v_\mathfrak{p}$ the **valuation** associated to the prime $\mathfrak{p}$. These valuations are intimately connected to prime factorizations. Indeed, prime factorization implies that for any $\kappa \in K^*$, we have

$$\kappa\mathcal{O} = \prod_{\mathfrak{q}} \mathfrak{q}^{e_\mathfrak{q}},$$

for some integers $e_\mathfrak{q}$ all but finitely many of which are zero. We claim that $v_\mathfrak{p}(\kappa) = e_\mathfrak{p}$. To see this, first observe that if $\mathfrak{p}$ and $\mathfrak{q}$ are distinct primes then $\mathfrak{q}_\mathfrak{p} = \mathcal{o}_\mathfrak{p}$. Indeed, we have

$$\kappa\mathcal{O}_\mathfrak{p} = \mathfrak{p}_\mathfrak{p}^{e_\mathfrak{p}},$$

from which $v_\mathfrak{p}(\kappa) = e_\mathfrak{p}$. In particular, $v_\mathfrak{p}(\kappa) = 0$ for all but finitely many primes $\mathfrak{p}$. In view of these properties, $v_\mathfrak{p}$ is also called an **exponential valuation**.

Continue to let $\mathcal{o}$ be a Dedekind domain with field of fractions K and let X be a set of all but finitely many primes of $\mathcal{o}$. Then $\mathcal{o}(X)$ is a Dedekind domain. By Proposition 1.33, the primes $\mathfrak{p}_X$ of $\mathcal{o}(X)$ are of the form $\mathfrak{p}_X = \mathfrak{p}(X)$ for $\mathfrak{p} \in X$. Moreover, $\mathcal{o}$ and $\mathcal{o}(X)$ have the same localizations at $\mathfrak{p}$ and $\mathfrak{p}_X$ respectively because the only elements which are not inverted are those of $\mathfrak{p}$. In particular, this means

$$\mathcal{o}_\mathfrak{p} = \mathcal{o}(X)_{\mathfrak{p}_X} \quad \text{and} \quad \mathfrak{f}_\mathfrak{p} = \mathfrak{f}(X)_{\mathfrak{p}_X}. \tag{5}$$

The relationship between $\mathcal{o}$ and $\mathcal{o}(X)$ can be expressed via a long exact sequence.

Proposition 1.39. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K and let X be a set of all but finitely many primes of $\mathcal{O}$. Then there is a long exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \mathcal{O}(X)^* \longrightarrow \bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^* \longrightarrow \mathrm{Cl}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}(X)) \longrightarrow 1,$$

induced by natural inclusions and projections and where the fourth map and fifth maps are induced by $(\kappa_\mathfrak{p})_{\mathfrak{p}\notin X} \mapsto \prod_{\mathfrak{p}\notin X} \mathfrak{p}^{v_\mathfrak{p}(\kappa_\mathfrak{p})}$ and $\mathfrak{a} \mapsto \mathfrak{a}(X)$ respectively. Moreover, $K^*/\mathcal{O}_\mathfrak{p}^* \cong \mathbb{Z}$.

Proof. As the second map is injective and the fifth map is surjective, the sequence is exact at $\mathcal{O}^*$ and $\mathrm{Cl}(\mathcal{O}(X))$. So we must show exactness at $\mathcal{O}(X)^*$, $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$, and $\mathrm{Cl}(\mathcal{O})$. For exactness at $\mathcal{O}(X)^*$, $\alpha \in \mathcal{O}(X)^*$ represents the zero coset in under the third map $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$ if and only if $\alpha \in \mathcal{O}_\mathfrak{p}^*$ for $\mathfrak{p} \notin X$. But Equation (5) implies $\alpha \in \mathcal{O}_\mathfrak{p}^*$ for $\mathfrak{p} \in X$. Whence α represents the zero coset if and only if $\alpha \in \mathcal{O}^*$ by Proposition 1.34 proving exactness at $\mathcal{O}(X)^*$. For exactness at $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$, a representative $(\kappa_\mathfrak{p})_{\mathfrak{p}\notin X}$ of a coset in $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$ represents the principal class of $\mathrm{Cl}(\mathcal{O})$ under the fourth map if and only if there is a $\kappa \in K^*$ such that

$$\prod_{\mathfrak{p}\notin X} \mathfrak{p}^{v_\mathfrak{p}(\kappa_\mathfrak{p})} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_\mathfrak{p}(\kappa)}.$$

By prime factorization, $v_\mathfrak{p}(\kappa) = v_\mathfrak{p}(\kappa_\mathfrak{p})$. In particular, $v_\mathfrak{p}(\kappa) = 0$ for all $\mathfrak{p} \in X$ and $v_\mathfrak{p}(\kappa) = v_\mathfrak{p}(\kappa_\mathfrak{p})$ for $\mathfrak{p} \notin X$. As $v_\mathfrak{p}(\kappa) = 0$ for $\mathfrak{p} \in X$, we have $\kappa \in \mathcal{O}_\mathfrak{p}^*$ for these primes as well. Because the primes of $\mathcal{O}(X)$ are $\mathfrak{p}_X$ for $\mathfrak{p} \in X$, Equation (5) and Proposition 1.34 together imply $\kappa \in \mathcal{O}(X)^*$. Exactness at $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$ follows. For exactness at $\mathrm{Cl}(\mathcal{O})$, an integral ideal $\mathfrak{a}$ representing a class of $\mathrm{Cl}(\mathcal{O})$ represents the principal class under the fifth map if and only if there is a $\kappa \in K^*$ such that

$$\mathfrak{a}(X) = \kappa\mathcal{O}(X).$$

In view of Equation (5) and Proposition 1.34 again, taking the intersection with the localizations at $\mathfrak{p}_X$ for $\mathfrak{p} \notin X$ shows

$$\mathfrak{a} = \kappa\mathcal{O}.$$

Therefore $(v_\mathfrak{p}(\kappa))_{\mathfrak{p}\notin X}$ is a representative of a coset in $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_\mathfrak{p}^*$ whose image under the fourth map is $\mathfrak{a}$ proving exactness at $\mathrm{Cl}(\mathcal{O})$.

The last statement follows from the first isomorphism theorem since the valuation $v_\mathfrak{p}$ restricted to K^* is a surjective homomorphism and the kernel is exactly $\mathcal{O}_\mathfrak{p}^*$. This completes the proof. □

Number Fields

We now turn to the setting of a number field K. Let S denote a finite set of primes of $\mathcal{O}_K$ and let X denote the set of all primes that do not belong to S. The **ring of S-integers** $\mathcal{O}_{K,S}$ of K is defined by

$$\mathcal{O}_{K,S} = \mathcal{O}_K(X).$$

We call any $\alpha \in \mathcal{O}_{K,S}$ an **algebraic S-integer**. We will write $I_S(K)$ and $P_S(K)$ for the group of fractional ideals of $\mathcal{O}_{K,S}$ and the subgroup of principal fractional ideals respectively.

The S-**class group** $\mathrm{Cl}_S(K)$ of K is the ideal class group of $\mathcal{O}_{K,S}$. In particular,

$$\mathrm{Cl}_S(K) = I_S(K)/P_S(K).$$

The S-**class number** $h_{K,S}$ of K is the class number of $\mathcal{O}_{K,S}$. The S-**unit group** of K is the unit group $\mathcal{O}_{K,S}^*$ of $\mathcal{O}_{K,S}$ and we call any element of $\mathcal{O}_{K,S}^*$ an S-**unit** of K.

1.5 Orders

Recall that Dedekind domains are one-dimensional noetherian domains that are also integrally closed domains. An order will relax the condition of being integrally closed, and surprisingly, we will still be able to deduce similar information as we did for Dedekind domains. More precisely, for any integral domain $\mathcal{o}$ with field of fractions K, we define its **conductor** $\mathfrak{q}_{\mathcal{o}}$ to be

$$\mathfrak{q}_{\mathcal{o}} = \{\alpha \in \mathcal{o} : \alpha\overline{\mathcal{o}} \subseteq \mathcal{o}\}.$$

Then $\mathfrak{q}_{\mathcal{o}}$ is an ideal of $\mathcal{o}$ and it is the largest ideal which is also an ideal of $\overline{\mathcal{o}}$. Clearly $\mathcal{o}$ is an integrally closed domain if and only if the conductor is $\mathcal{o}$. In the case $\mathcal{o}$ is a noetherian domain, the conductor is an integral ideal of $\mathcal{o}$ and hence also $\overline{\mathcal{o}}$, which is to say it is nonzero, precisely when $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module.

Proposition 1.40. *Let $\mathcal{o}$ be a noetherian domain with field of fractions K. Then the conductor $\mathfrak{q}_{\mathcal{o}}$ is an integral ideal of $\mathcal{o}$ if and only if $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module.*

Proof. For the forward implication, suppose $\mathfrak{q}_{\mathcal{o}}$ is an integral ideal of $\mathcal{o}$. Then there is some nonzero $\alpha \in \mathfrak{q}_{\mathcal{o}}$ with $\alpha\overline{\mathcal{o}} \subseteq \mathcal{o}$. Whence $\alpha\overline{\mathcal{o}}$ is an ideal of $\mathcal{o}$ and hence a finitely generated $\mathcal{o}$-module as $\mathcal{o}$ is noetherian. But then $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module. For the reverse implication, let $\frac{\eta_1}{\delta_1}, \ldots, \frac{\eta_r}{\delta_r}$ be generators of $\overline{\mathcal{o}}$ as an $\mathcal{o}$-module. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\overline{\mathcal{o}} \subset \mathcal{o}$. Whence $\delta \in \mathfrak{q}_{\mathcal{o}}$. □

Accordingly, an **order** $\mathcal{o}$ is a one-dimensional noetherian domain whose conductor is an integral ideal of $\mathcal{o}$. Whence the conductor is also an integral ideal of $\overline{\mathcal{o}}$. In view of the previous result, the conductor being an integral ideal of $\mathcal{o}$ is equivalent to the normalization $\overline{\mathcal{o}}$ being a finitely generated $\mathcal{o}$-module. This means $\mathcal{o}$ is a Dedekind domain if and only if it is an integrally closed domain which happens if and only if the conductor is $\mathcal{o}$ itself. In fact, the normalization of an order is always a Dedekind domain.

Proposition 1.41. *Let $\mathcal{o}$ be an order with field of fractions K. Then $\overline{\mathcal{o}}$ is a Dedekind domain with field of fractions K.*

Proof. $\overline{\mathcal{o}}$ is an integrally closed domain with field of fractions K by definition. To see $\overline{\mathcal{o}}$ is noetherian, recall that $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module as $\mathcal{o}$ is an order. Since $\mathcal{o}$ is noetherian, every integral ideal of $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module and therefore also a finitely generated $\overline{\mathcal{o}}$-module. Whence $\overline{\mathcal{o}}$ is noetherian. It remains to show that every prime $\overline{\mathfrak{p}}$ of $\overline{\mathcal{o}}$ is maximal. This is equivalent to showing $\overline{\mathcal{o}}/\overline{\mathfrak{p}}$ is a field. So consider the homomorphism

$$\phi : \mathcal{o} \mapsto \overline{\mathcal{o}}/\overline{\mathfrak{p}} \qquad \alpha \mapsto \alpha + \overline{\mathfrak{p}},$$

induced by natural projection. Recall that $\overline{\mathfrak{p}}$ is above some prime $\mathfrak{p}$ of $\mathcal{O}$. Whence $\ker \phi = \mathfrak{p}$. By the first isomorphism theorem, ϕ induces an embedding $\mathbb{F}_{\mathfrak{p}} \to \overline{\mathcal{O}}/\overline{\mathfrak{p}}$. Therefore $\overline{\mathcal{O}}/\overline{\mathfrak{p}}$ is a finitely dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space as $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module. By primality of $\overline{\mathfrak{p}}$, $\overline{\mathcal{O}}/\overline{\mathfrak{p}}$ is also an integral domain and therefore must be a field. □

Perhaps not surprisingly, there is a generalization of the ideal class group to orders. As $\mathcal{O}$ is not a Dedekind domain, fractional ideals need not be invertible. The natural restriction is to consider all fractional ideals which are invertible. Let $I(\mathcal{O})$ denote the set of invertible fractional ideals of $\mathcal{O}$. Then it is easy to see that this is an abelian group.

Theorem 1.42. *Let $\mathcal{O}$ be an order with field of fractions K. Then $I(\mathcal{O})$ is an abelian group with identity $\mathcal{O}$.*

Proof. Associativity and commutativity of $I(\mathcal{O})$ are obvious. The identity is $\mathcal{O}$ because every invertible fractional ideal is a finitely generated $\mathcal{O}$-submodule of K. Lastly, every fractional ideal is invertible by definition of $I(\mathcal{O})$. □

Let $P(\mathcal{O})$ denote the subgroup of principal fractional ideals of $I(\mathcal{O})$. In fact, $P(\mathcal{O})$ contains every principal fraction ideal of $\mathcal{O}$ because they are all invertible. As $I(\mathcal{O})$ is abelian by the previous result, $P(\mathcal{O})$ is normal. The **Picard group** $\mathrm{Pic}(\mathcal{O})$ of $\mathcal{O}$ is defined to be the quotient

$$\mathrm{Pic}(\mathcal{O}) = I(\mathcal{O})/P(\mathcal{O}).$$

A class of $\mathrm{Pic}(\mathcal{O})$ is called an **Picard class** of $\mathcal{O}$. As every fractional ideal $\mathfrak{f}$ can be expressed as $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$, we have $\delta\mathfrak{f} = \mathfrak{a}$ and it follows that every ideal class can be represented by an integral ideal. The **class number** $h_{\mathcal{O}}$ of $\mathcal{O}$ is defined by

$$h_{\mathcal{O}} = |\,\mathrm{Pic}(\mathcal{O})|.$$

The class number is a measure of the degree of failure for every invertible ideal to be principal. Of course, in the case $\mathcal{O}$ is a Dedekind domain, the Picard group is precisely the ideal class group and the corresponding class numbers are equal. This is to say that

$$\mathrm{Pic}(\mathcal{O}) = \mathrm{Cl}(\mathcal{O}).$$

The **unit group** of $\mathcal{O}$ is defined to be $\mathcal{O}^*$. That is, the unit group is the group of units in $\mathcal{O}$. The Picard and unit groups of $\mathcal{O}$ fit into a short exact sequence analogous to Proposition 1.24.

Proposition 1.43. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is a short exact sequence*

$$1 \longrightarrow K^*/\mathcal{O}^* \longrightarrow I(\mathcal{O}) \longrightarrow \mathrm{Pic}(\mathcal{O}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the second map is induced by $\kappa \mapsto \kappa\mathcal{O}$.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $K^*/\mathcal{O}^*$ and $\mathrm{Pic}(\mathcal{O})$. So it remains to prove exactness at $I(\mathcal{O})$. This follows because the principal fractional ideals represent the identity class of $\mathrm{Pic}(\mathcal{O})$ and these are generated by a complete set of representatives for $K^*/\mathcal{O}^*$. □

It is possible to describe the Picard group in terms of principal fractional ideals for the localizations $\mathcal{O}_{\mathfrak{p}}$ at primes $\mathfrak{p}$. To prove this, we need some results specific to orders. Our first property is that localization preserves orders.

Proposition 1.44. *Let $\mathcal{O}$ be an order. Then for any multiplicative subset D of $\mathcal{O}$, $\mathcal{O}D^{-1}$ is an order.*

Proof. By Proposition 1.33, every prime of $\mathcal{O}D^{-1}$ is of the form $\mathfrak{p}D^{-1}$ for some prime $\mathfrak{p}$ of $\mathcal{O}$. But then $\mathfrak{p}D^{-1}$ is maximal because $\mathfrak{p}$ is and the bijections in Proposition 1.33 are inclusion-preserving. Therefore $\mathcal{O}D^{-1}$ is one-dimensional. To show it is noetherian, let $\mathfrak{A}$ be an integral ideal of $\mathcal{O}D^{-1}$ and set $\mathfrak{a} = \mathfrak{A} \cap \mathcal{O}$. Then

$$\mathfrak{A} = \mathfrak{a}D^{-1}.$$

As $\mathcal{O}$ is noetherian, $\mathfrak{a}$ is a finitely generated $\mathcal{O}$-module and hence $\mathfrak{A}$ is a finitely generated $\mathcal{O}D^{-1}$-module. This proves $\mathcal{O}D^{-1}$ is noetherian. It remains to show that the conductor of $\mathcal{O}D^{-1}$ is an integral ideal of $\mathcal{O}D^{-1}$. By Proposition 1.36, $\overline{\mathcal{O}}D^{-1}$ is the normalization of $\mathcal{O}D^{-1}$. So in view of Proposition 1.40, it suffices to show that $\overline{\mathcal{O}}D^{-1}$ is a finitely generated $\mathcal{O}D^{-1}$-submodule of K. But this is clear since $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-submodule of K by Proposition 1.40 again. □

In particular, this result shows that $\mathcal{O}_{\mathfrak{p}}$ is an order. Our second property is a generalization of the Chinese remainder theorem for orders.

Proposition 1.45. *Let $\mathcal{O}$ be an order. Then for any integral ideal $\mathfrak{a}$ of $\mathcal{O}$, $\mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ for all but finitely many primes $\mathfrak{p}$ those of which satisfy $\mathfrak{a} \subseteq \mathfrak{p}$. Moreover, there is an isomorphism*

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{a} \subseteq \mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}},$$

induced by natural projection which may equivalently be expressed as

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}.$$

Proof. Let $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathfrak{a}_{\mathfrak{p}} \cap \mathcal{O}$. Then $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ is an integral ideal of $\mathcal{O}$. As $\mathcal{O}$ is noetherian, Lemma 1.18 implies

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{a},$$

for some primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. Since $\mathcal{O}$ is one-dimensional, every prime is maximal whence every maximal ideal containing $\mathfrak{a}$ is one of the $\mathfrak{p}_i$. So one the one hand, $\mathfrak{a} \subsetneq \mathfrak{p}$ for all but finitely many primes $\mathfrak{p}$. Whence $\mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ and $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}$ for these primes. On the other hand, if $\mathfrak{a} \subseteq \mathfrak{p}$ then the inclusion-preserving bijections in Proposition 1.33 show that $\mathfrak{p}$ is the only prime containing $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$. As $\mathcal{O}$ is one-dimensional, this means that if $\mathfrak{p}$ and $\mathfrak{q}$ are distinct primes then

$$\widetilde{\mathfrak{a}}_{\mathfrak{p}} + \widetilde{\mathfrak{a}}_{\mathfrak{q}} = \mathcal{O},$$

which is to say $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ and $\widetilde{\mathfrak{a}}_{\mathfrak{q}}$ are comaximal. Together, these facts show that the $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ are pairwise comaximal. Now we claim

$$\mathfrak{a} = \bigcap_{\mathfrak{p}} \widetilde{\mathfrak{a}}_{\mathfrak{p}}.$$

The forward inclusion is obvious. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \widetilde{\mathfrak{a}}_{\mathfrak{p}}$ and set

$$\mathfrak{b} = \{\beta \in \mathcal{O} : \beta\eta \in \delta\mathfrak{a}\}.$$

Then $\mathfrak{b}$ is an integral ideal of $\mathcal{O}$ containing δ. As δ is not contained in any prime of $\mathcal{O}$, it follows that $\mathfrak{b}$ cannot be contained in any prime of $\mathcal{O}$. But every proper ideal is contained in a maximal ideal which is necessarily prime. Whence $\mathfrak{b}$ is not proper and so $\mathfrak{b} = \mathcal{O}$. Therefore $\eta \in \delta\mathfrak{a}$ which implies $\frac{\eta}{\delta} \in \mathfrak{a}$ proving the reverse inclusion. Altogether, the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{a} \subseteq \mathfrak{p}} \mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}},$$

induced by natural projection. As $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}$ unless $\mathfrak{a} \subseteq \mathfrak{p}$, this isomorphism can be expressed as

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}}.$$

It remains to demonstrate that natural projection induces an isomorphism $\mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}} \cong \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$. To see this, consider the homomorphism

$$\phi : \mathcal{O} \to \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} \qquad \alpha \mapsto \alpha + \mathfrak{a}_{\mathfrak{p}},$$

induced by natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker\phi = \widetilde{\mathfrak{a}}_{\mathfrak{p}}$. As $\mathcal{O}_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}_{\mathfrak{p}}$, $\mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$ has a unique maximal ideal $\mathfrak{p}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$ by the fourth isomorphism theorem. Whence

$$\mathcal{O} + \mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}},$$

proving surjectivity. Moreover, $\ker\phi = \widetilde{\mathfrak{a}}_{\mathfrak{p}}$ from the definition of $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$. □

Our third property is that a fractional ideal of $\mathcal{O}$ is invertible if and only if its localization at a prime $\mathfrak{p}$ is principal in $\mathcal{O}_{\mathfrak{p}}$.

Proposition 1.46. *Let $\mathcal{O}$ be an order with field of fractions K. Then a fractional ideal $\mathfrak{f}$ of $\mathcal{O}$ is invertible if and only if the fractional ideal $\mathfrak{f}_{\mathfrak{p}}$ of $\mathcal{O}_{\mathfrak{p}}$ is principal for every prime $\mathfrak{p}$ of $\mathcal{O}$.*

Proof. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. Then it suffices to show the claim for the integral ideal $\mathfrak{a}$. Set $\mathfrak{b} = \mathfrak{a}^{-1}$. For the forward implication, suppose $\mathfrak{a}$ is invertible so that $\mathfrak{a}\mathfrak{b} = \mathcal{O}$. Then we may find $\alpha_1, \ldots, \alpha_n \in \mathfrak{a}$ and $\beta_1, \ldots, \beta_n \in \mathfrak{b}$ such that

$$\sum_i \alpha_i\beta_i = 1.$$

Moreover, not all of the $\alpha_i\beta_i$ can lie in the unique maximal ideal $\mathfrak{p}_{\mathfrak{p}}$ of $\mathcal{O}_{\mathfrak{p}}$ for otherwise this ideal is not proper. Without loss of generality, we may assume $\alpha_1\beta_1 \notin \mathfrak{p}_{\mathfrak{p}}$ and so it is a unit in $\mathcal{O}_{\mathfrak{p}}$. As $\mathfrak{a}_{\mathfrak{p}}\mathfrak{b} = \mathcal{O}_{\mathfrak{p}}$ by the invertibility of $\mathfrak{a}$, it follows that $\mathfrak{a}_{\mathfrak{p}} = \alpha_1\mathcal{O}_{\mathfrak{p}}$. This shows $\mathfrak{a}_{\mathfrak{p}}$ is principal. For the reverse implication, suppose $\mathfrak{a}_{\mathfrak{p}}$ is principal for every prime $\mathfrak{p}$ of $\mathcal{O}$. Whence $\mathfrak{a}_{\mathfrak{p}} = \alpha_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}}$ for some nonzero $\alpha_{\mathfrak{p}} \in K$. As K is the field of fractions of $\mathcal{O}_{\mathfrak{p}}$, we may multiply by a nonzero element of $\mathcal{O}_{\mathfrak{p}}$, if necessary, to ensure that $\alpha_{\mathfrak{p}} \in \mathfrak{a}$. Now if $\mathfrak{a}$ were not invertible then there would exist a prime $\mathfrak{p}$ such that

$$\mathfrak{a}\mathfrak{a}^{-1} \subseteq \mathfrak{p},$$

since $\mathcal{O}$ is one-dimensional. Let $\alpha_1, \dots, \alpha_r$ be generators of $\mathfrak{a}$ as an $\mathcal{O}$-module. As $\alpha_i \in \mathfrak{a}_{\mathfrak{p}}$, we may write $\alpha_i = \alpha_{\mathfrak{p}}\frac{\eta_i}{\delta_i}$ with $\frac{\eta_i}{\delta_i} \in \mathcal{O}_{\mathfrak{p}}$. Then $\delta_i\alpha_i \in \alpha_{\mathfrak{p}}\mathcal{O}$ and so $\delta_i\alpha_i\alpha_{\mathfrak{p}}^{-1} \in \mathcal{O}$. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\alpha_{\mathfrak{p}}^{-1}\mathfrak{a} \subseteq \mathcal{O}$. Thus $\delta\alpha_{\mathfrak{p}}^{-1} \in \mathfrak{a}^{-1}$ from which we find $\delta \in \mathfrak{a}\mathfrak{a}^{-1}$ whence $\delta \in \mathfrak{p}$. This is a contradiction and so $\mathfrak{a}$ is invertible. □

We can now show how the Picard group is related to principal fractional ideals for the localizations $\mathcal{O}_{\mathfrak{p}}$ of an order $\mathcal{O}$ at primes $\mathfrak{p}$.

Proposition 1.47. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is an isomorphism*

$$I(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}}),$$

induced by localization. In particular, upon identifying $P(\mathcal{O})$ with its image under this isomorphism, we have

$$\mathrm{Pic}(\mathcal{O}) \cong \left(\bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}})\right)/P(\mathcal{O}).$$

Proof. It follows from Proposition 1.46 that $\mathfrak{f}_{\mathfrak{p}}$ is a principal fractional ideal of $\mathcal{O}_{\mathfrak{p}}$ for every fractional ideal $\mathfrak{f}$ of $I(\mathcal{O})$ and every prime $\mathfrak{p}$ of $\mathcal{O}$. Therefore we may consider the homomorphism

$$\phi : I(\mathcal{O}) \to \bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}}) \qquad \mathfrak{f} \mapsto (\mathfrak{f}_{\mathfrak{p}})_{\mathfrak{p}},$$

induced by localization. To prove the first statement, we must show that ϕ is an isomorphism. For surjectivity, let $(\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}})_{\mathfrak{p}} \in \bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}})$. Then $\kappa_{\mathfrak{p}} \in K$ and all but finitely satisfy $\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$. We claim that

$$\mathfrak{f} = \bigcap_{\mathfrak{p}} \kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}},$$

is an invertible fractional ideal of $\mathcal{O}$ whose image under ϕ is $(\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}})_{\mathfrak{p}}$. As $\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ for all but finitely many primes $\mathfrak{p}$, there exists a nonzero $\delta \in \mathcal{O}$ for which $\delta\kappa_{\mathfrak{p}} \in \mathcal{O}_{\mathfrak{p}}$ for all primes $\mathfrak{p}$. It follows from Proposition 1.34 that $\delta\mathfrak{f} \subseteq \mathcal{O}$. Whence $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$. To show it is invertible and that the image under ϕ is $(\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}})_{\mathfrak{p}}$, it suffices to prove

$$\mathfrak{f}_{\mathfrak{p}} = \kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}}.$$

Indeed, Proposition 1.46 will imply that $\mathfrak{f}$ is invertible and the definition of ϕ shows that $\mathfrak{f}$ will then map to $(\kappa_{\mathfrak{p}}\mathcal{O}_{\mathfrak{p}})_{\mathfrak{p}}$ under ϕ. The forward inclusion is clear by the definition of $\mathfrak{f}$. As $\delta\mathfrak{f}$ is an integral ideal, Proposition 1.45 implies that there exists an $\alpha \in \mathcal{O}$ satisfying

$$\alpha \equiv \delta\kappa_{\mathfrak{p}} \pmod{\delta\mathfrak{f}_{\mathfrak{p}}} \quad \text{and} \quad \alpha \equiv 0 \pmod{\delta\mathfrak{f}_{\mathfrak{q}}},$$

for primes $\mathfrak{q}$ with $\mathfrak{q} \neq \mathfrak{p}$. Whence $\frac{\alpha}{\delta} \in \mathfrak{f}_{\mathfrak{q}}$ for all primes $\mathfrak{q}$ and Proposition 1.34 implies $\frac{\alpha}{\delta} \in \mathfrak{f}$. But then $\kappa_{\mathfrak{p}} \in \mathfrak{f}_{\mathfrak{p}}$ and the reverse inclusion holds. Altogether, this means ϕ is surjective. To see that it is injective, suppose $\mathfrak{f} \in I(\mathcal{O})$ is such that $\mathfrak{f}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ for all primes $\mathfrak{p}$. Then Proposition 1.34 implies $\mathfrak{f} = \mathcal{O}$ which means ϕ is injective. This establishes the first isomorphism. The second isomorphism is immediate by the definition of the Picard group.□

With this formulation of the Picard group, it is possible to obtain a long exact sequence relating the Picard and ideal class groups of $\mathcal{O}$ and its normalization $\overline{\mathcal{O}}$.

Proposition 1.48. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is a long exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \overline{\mathcal{O}}^* \longrightarrow \bigoplus_{\mathfrak{p}} \overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow \operatorname{Pic}(\mathcal{O}) \longrightarrow \operatorname{Cl}(\overline{\mathcal{O}}) \longrightarrow 1,$$

induced by natural inclusions and projections, localization, as well as the maps $\kappa \mapsto \kappa\mathcal{O}$ and $\kappa \mapsto \kappa\overline{\mathcal{O}}$.

Proof. Observe that $\overline{\mathcal{O}}$ is a Dedekind domain by Proposition 1.41. The exact sequences in Propositions 1.24 and 1.43 combine to give a commutative diagram of short exact sequences given by

$$\begin{array}{ccccccccc} 1 & \longrightarrow & K^*/\mathcal{O}^* & \longrightarrow & I(\mathcal{O}) & \longrightarrow & \operatorname{Pic}(\mathcal{O}) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & K^*/\overline{\mathcal{O}}^* & \longrightarrow & I(\overline{\mathcal{O}}) & \longrightarrow & \operatorname{Cl}(\overline{\mathcal{O}}) & \longrightarrow & 1. \end{array}$$

As the first two vertical maps are surjective, exactness and commutativity together imply that the third one is surjective as well. In view of Proposition 1.47, we have isomorphisms

$$I(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}}) \quad \text{and} \quad I(\overline{\mathcal{O}}) \cong \bigoplus_{\overline{\mathfrak{p}}} P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}).$$

By Propositions 1.37 and 1.41, $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a Dedekind domain. But as $\mathcal{O}_{\mathfrak{p}}$ is local and an order by Proposition 1.44, $\mathfrak{p}_{\mathfrak{p}}$ is the unique prime which is also the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. It follows that every prime $\overline{\mathfrak{p}}_{\mathfrak{p}}$ of $\overline{\mathcal{O}}_{\mathfrak{p}}$ is above $\mathfrak{p}_{\mathfrak{p}}$ which happens if and only if $\mathfrak{p} \subseteq \overline{\mathfrak{p}}$ by Proposition 1.33. In particular, there can only be finitely many primes whence $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a principal ideal domain by Proposition 1.30. As $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$ is a discrete valuation ring by Theorem 1.38, we obtain an isomorphism

$$P(\overline{\mathcal{O}}_{\mathfrak{p}}) \cong \bigoplus_{\mathfrak{p} \subseteq \overline{\mathfrak{p}}} P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}),$$

induced by localization. Since we have isomorphisms $K^*/\mathcal{O}_{\mathfrak{p}}^* \cong P(\mathcal{O}_{\mathfrak{p}})$ and $K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^* \cong P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}})$ induced by the maps $\kappa \mapsto \kappa\mathcal{O}$ and $\kappa \mapsto \kappa\overline{\mathcal{O}}$ respectively, our previous work gives isomorphisms

$$I(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} K^*/\mathcal{O}_{\mathfrak{p}}^* \quad \text{and} \quad I(\overline{\mathcal{O}}) \cong \bigoplus_{\mathfrak{p}} K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^*,$$

and our exact commutative diagram becomes

$$\begin{array}{ccccccccc} 1 & \longrightarrow & K^*/\mathcal{O}^* & \longrightarrow & \bigoplus_{\mathfrak{p}} K^*/\mathcal{O}_{\mathfrak{p}}^* & \longrightarrow & \mathrm{Pic}(\mathcal{O}) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & K^*/\overline{\mathcal{O}}^* & \longrightarrow & \bigoplus_{\mathfrak{p}} K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^* & \longrightarrow & \mathrm{Cl}(\overline{\mathcal{O}}) & \longrightarrow & 1. \end{array}$$

As the vertical maps are surjective, the snake lemma gives a short exact sequence

$$1 \longrightarrow \overline{\mathcal{O}}^*/\mathcal{O}^* \longrightarrow \bigoplus_{\mathfrak{p}} \overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow \ker(\mathrm{Pic}(\mathcal{O}) \to \mathrm{Cl}(\overline{\mathcal{O}})) \longrightarrow 1.$$

Splicing this short exact sequence with the short exact sequences

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \overline{\mathcal{O}}^* \longrightarrow \overline{\mathcal{O}}^*/\mathcal{O}^* \longrightarrow 1,$$

and

$$1 \longrightarrow \ker(\mathrm{Pic}(\mathcal{O}) \to \mathrm{Cl}(\overline{\mathcal{O}})) \longrightarrow \mathrm{Pic}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\overline{\mathcal{O}}) \longrightarrow 1,$$

gives the desired long exact sequence. □

Recall that an order $\mathcal{O}$ is a Dedekind domain if and only if the conductor $\mathfrak{q}_{\mathcal{O}}$ is $\mathcal{O}$ itself. This suggests that the conductor should control the extent to which an order fails to be integrally closed. We now turn to a more detailed investigation of what this means. Recall from Proposition 1.44 that the localization $\mathcal{O}_{\mathfrak{p}}$ is an order for any prime $\mathfrak{p}$ of $\mathcal{O}$ and is also local. In view of Theorem 1.38, $\mathcal{O}$ is a Dedekind domain if and only if the localizations $\mathcal{O}_{\mathfrak{p}}$ are integrally closed for all primes $\mathfrak{p}$ of $\mathcal{O}$. Accordingly, $\mathfrak{p}$ is said to be **regular** if $\mathcal{O}_{\mathfrak{p}}$ is an integrally closed domain and is **irregular** otherwise. So if $\mathfrak{p}$ is regular then $\mathcal{O}_{\mathfrak{p}}$ is a Dedekind domain with exactly one prime and therefore a discrete valuation ring. This means $\mathfrak{p}$ is regular if and only if $\mathcal{O}_{\mathfrak{p}}$ is a discrete valuation ring. It turns out that the regular primes are exactly those which are do not contain the conductor.

Proposition 1.49. *Let $\mathcal{O}$ be an order. Then a prime $\mathfrak{p}$ of $\mathcal{O}$ is regular if and only if $\mathfrak{q}_{\mathcal{O}} \subsetneq \mathfrak{p}$. In the case $\mathfrak{p}$ is regular, there a unique prime $\overline{\mathfrak{p}}$ of $\overline{\mathcal{O}}$ above $\mathfrak{p}$ which is given by $\overline{\mathfrak{p}} = \mathfrak{p}_{\mathfrak{p}} \cap \overline{\mathcal{O}}$, we have $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_{\mathfrak{p}}$, and $\mathfrak{p}$ is invertible.*

Proof. For the forward implication, if $\mathfrak{p}$ is regular then $\mathcal{O}_{\mathfrak{p}}$ is an integrally closed domain. As $\overline{\mathcal{O}}$ is integral over $\mathcal{O}$ and hence also $\mathcal{O}_{\mathfrak{p}}$, it follows that $\overline{\mathcal{O}} \subseteq \mathcal{O}_{\mathfrak{p}}$. But $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module by Proposition 1.40. So we may find generators $\frac{\eta_1}{\delta_1}, \ldots, \frac{\eta_r}{\delta_r}$ for $\overline{\mathcal{O}}$ as an $\mathcal{O}$-module with $\delta_i \notin \mathfrak{p}$. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\overline{\mathcal{O}} \subseteq \mathcal{O}$ meaning $\delta \in \mathfrak{q}_{\mathcal{O}}$. But primality of $\mathfrak{p}$ forces $\delta \notin \mathfrak{p}$. Whence $\mathfrak{q}_{\mathcal{O}} \subsetneq \mathfrak{p}$. For the reverse implication, suppose $\mathfrak{q}_{\mathcal{O}} \subsetneq \mathfrak{p}$. Then we may find a nonzero $\alpha \in \mathfrak{q}_{\mathcal{O}} - \mathfrak{p}$ with $\alpha\overline{\mathcal{O}} \subseteq \mathcal{O}$. It follows that $\overline{\mathcal{O}} \subset \mathcal{O}_{\mathfrak{p}}$. As $\mathfrak{p}_{\mathfrak{p}}$ is the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$, it is necessarily a prime. In view of the previous inclusion, $\overline{\mathfrak{p}}$ is then a prime of $\overline{\mathcal{O}}$ containing $\mathfrak{p}$ and is therefore above $\mathfrak{p}$ as $\mathcal{O}$ is one-dimensional. Since $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$ is an integrally closed domain by Proposition 1.36, it is enough to show $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_{\mathfrak{p}}$. The reverse inclusion holds because $\overline{\mathfrak{p}}$ is above $\mathfrak{p}$. For the forward inclusion, Suppose $\frac{\eta}{\delta} \in \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$. Then $\alpha\eta \in \mathcal{O}$ and $\alpha\delta \notin \mathfrak{p}$ by primality of $\mathfrak{p}$. Whence $\frac{\eta}{\delta} \in \mathcal{O}_{\mathfrak{p}}$ proving the forward inclusion. This completes the proof of the first statement.

As for the second statement, we have already verified $\overline{\mathfrak{p}}$ is a prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$ and that $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_{\mathfrak{p}}$. To show it is the unique prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$, let $\overline{\mathfrak{q}}$ be another prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$. In view of the identity $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_{\mathfrak{p}}$, it follows that $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} \subseteq \overline{\mathcal{O}}_{\overline{\mathfrak{q}}}$. The inclusion-preserving bijections in Proposition 1.33 show

$$\overline{\mathfrak{p}} = \overline{\mathfrak{p}}_{\overline{\mathfrak{p}}} \cap \overline{\mathcal{O}} \quad \text{and} \quad \overline{\mathfrak{q}} = \overline{\mathfrak{q}}_{\overline{\mathfrak{q}}} \cap \overline{\mathcal{O}}.$$

These identities together with the previous inequality imply $\overline{\mathfrak{p}} \subseteq \overline{\mathfrak{q}}$ and so $\overline{\mathfrak{p}} = \overline{\mathfrak{q}}$ as primes of $\overline{\mathcal{O}}$ are maximal. This proves uniqueness. It remains to show $\mathfrak{p}$ is invertible. By Proposition 1.46, we must show that $\mathfrak{p}_{\mathfrak{q}}$ is principal for every prime $\mathfrak{q}$ of $\mathcal{O}$. If $\mathfrak{q} = \mathfrak{p}$, regularity of $\mathfrak{p}$ means $\mathcal{O}_{\mathfrak{p}}$ is a discrete valuation ring and hence a principal ideal domain. Thus $\mathfrak{p}_{\mathfrak{p}}$ is principal. If $\mathfrak{q} \neq \mathfrak{p}$ then $\mathfrak{p}_{\mathfrak{q}} = \mathcal{O}_{\mathfrak{q}}$ which is also principal. This completes the proof of the second statement. □

As $\mathcal{O}$ is noetherian, Lemma 1.18 implies

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{q}_{\mathcal{O}},$$

for some primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. Since $\mathcal{O}$ is one-dimensional, every maximal ideal containing $\mathfrak{q}_{\mathcal{O}}$ is a prime and thus one of the $\mathfrak{p}_i$. In view of the previous result, this means that only finitely many primes of $\mathcal{O}$ are irregular and they are exactly the primes which are not invertible. This is a concrete description of the failure to which $\mathcal{O}$ is not integrally closed in terms of the conductor $\mathfrak{q}_{\mathcal{O}}$.

Using the concept of regular primes, it is also possible to give a more refined description of the direct sum term in Proposition 1.48.

Proposition 1.50. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is an isomorphism*

$$(\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} \overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^*,$$

induced by natural projection.

Proof. As the conductor $\mathfrak{q}_{\mathcal{O}}$ is both an integral ideal of $\mathcal{O}$ and its normalization $\overline{\mathcal{O}}$, Proposition 1.45 gives isomorphisms

$$\mathcal{O}/\mathfrak{q}_{\mathcal{O}} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}} \quad \text{and} \quad \overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}} \cong \bigoplus_{\overline{\mathfrak{p}}} \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}/(\mathfrak{q}_{\mathcal{O}})_{\overline{\mathfrak{p}}},$$

induced by natural projection. By Propositions 1.37 and 1.41, $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a Dedekind domain. But as $\mathcal{O}_{\mathfrak{p}}$ is local and an order by Proposition 1.44, $\mathfrak{p}_{\mathfrak{p}}$ is the unique prime which is also the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. It follows that every prime $\overline{\mathfrak{p}}_{\mathfrak{p}}$ of $\overline{\mathcal{O}}_{\mathfrak{p}}$ is above $\mathfrak{p}_{\mathfrak{p}}$ which happens if and only if $\mathfrak{p} \subseteq \overline{\mathfrak{p}}$ by Proposition 1.33. By Proposition 1.45 again, we have an isomorphism

$$\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}} \cong \bigoplus_{\mathfrak{p} \subseteq \overline{\mathfrak{p}}} \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}/(\mathfrak{q}_{\mathcal{O}})_{\overline{\mathfrak{p}}},$$

induced by natural projection. Whence our previous isomorphisms together imply

$$(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \quad \text{and} \quad (\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*,$$

and we obtain an isomorphism

$$(\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*.$$

Now suppose $\mathfrak{p}$ is irregular and consider the homomorphism

$$\phi : \overline{\mathcal{O}}_{\mathfrak{p}}^* \to (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \qquad \alpha \mapsto (\alpha + (\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*,$$

induced by natural projection. Note that ϕ is surjective if the homomorphism

$$\pi : \overline{\mathcal{O}}_{\mathfrak{p}}^* \to (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \qquad \alpha \mapsto \alpha + (\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}},$$

given by natural projection is. To see that this map is surjective, recall that an element of a ring is invertible if and only if it is not contained in any maximal ideal. By the fourth isomorphism theorem, the preimages of the maximal ideals of $(\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})$ under natural projection are maximal ideals of $\overline{\mathcal{O}}_{\mathfrak{p}}$ containing $(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}}$. These preimages constitute all of the maximal ideals of $\overline{\mathcal{O}}_{\mathfrak{p}}$ since every maximal ideal of $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a prime which must be above $\mathfrak{p}_{\mathfrak{p}}$ and irregularity implies $\mathfrak{q}_{\mathcal{O}} \subseteq \mathfrak{p}$ by Proposition 1.49. Therefore π, and hence ϕ as well, is surjective. Clearly $\mathcal{O}_{\mathfrak{p}}^* \subseteq \ker\phi$. But $\ker\phi$ is also a subgroup of $\overline{\mathcal{O}}_{\mathfrak{p}}^*$ which is contained in $\mathcal{O}_{\mathfrak{p}}$. Whence $\ker\phi = \mathcal{O}_{\mathfrak{p}}^*$. The first isomorphism theorem then implies

$$\overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \cong (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*,$$

provided $\mathfrak{p}$ is irregular. If $\mathfrak{p}$ is regular, then $\mathcal{O}_{\mathfrak{p}}$ is an integrally closed domain and Proposition 1.49 implies $(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$. Therefore the isomorphism holds as well because all quotients are trivial. This completes the proof. □

Number Fields

We now turn to the setting of a number field K of degree n. An **order** $\mathcal{O}$ of K is an order satisfying

$$\mathbb{Z} \subseteq \mathcal{O} \subseteq \mathcal{O}_K,$$

and whose normalization is $\mathcal{O}_K$. Accordingly, $\mathcal{O}_K$ is called the **maximal order** of K. Clearly it is the largest order of K. As the normalization of $\mathcal{O}$ is O_K, the field of fractions of $\mathcal{O}$ is necessarily K. In fact, $\mathcal{O}$ contains a basis for $K/\mathbb{Q}$. To see this, let $\kappa_1, \ldots, \kappa_n$ be a basis for $K/\mathbb{Q}$ and write

$$k_i = \frac{\alpha_i}{\delta_i},$$

for $\alpha_i, \delta_i \in \mathcal{O}$ with δ_i nonzero. Then $\delta = \prod_i \delta_i$ is nonzero and we find that $\delta\kappa_1, \ldots, \delta\kappa_n$ is a basis for $K/\mathbb{Q}$ contained in $\mathcal{O}$.

1.6 Prime Splitting

Suppose $\mathcal{O}/\mathcal{o}$ is a Dedekind extension of a finite separable extension L/K. Let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. The **ramification index** $e_{\mathfrak{p}}(\mathfrak{P})$ of $\mathfrak{P}$ relative to $\mathfrak{p}$ is defined to be the power of $\mathfrak{P}$ appearing in the prime factorization of $\mathfrak{p}\mathcal{O}$. If $\mathfrak{p}\mathcal{O}$ has prime factors $\mathfrak{P}_1, \ldots, \mathfrak{P}_r$ then the prime factorization of $\mathfrak{p}\mathcal{O}$ is

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)}.$$

Something can also be said about the residue class fields $\mathbb{F}_{\mathfrak{P}}$ and $\mathbb{F}_{\mathfrak{p}}$. Consider the homomorphism

$$\phi : \mathcal{o} \to \mathbb{F}_{\mathfrak{P}} \qquad \alpha \mapsto \alpha + \mathfrak{P},$$

induced by natural projection. We have $\ker \phi = \mathfrak{p}$ as $\mathfrak{P}$ is above $\mathfrak{p}$. The first isomorphism theorem implies that ϕ induces an embedding $\mathbb{F}_{\mathfrak{p}} \to \mathbb{F}_{\mathfrak{P}}$. This means $\mathbb{F}_{\mathfrak{P}}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension at most n as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. In other words, $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is a field extension of degree at most n. Accordingly, we call $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ the **residue class extension** of $\mathfrak{P}$ relative to $\mathfrak{p}$. We define the **inertia degree** $f_{\mathfrak{p}}(\mathfrak{P})$ of $\mathfrak{P}$ relative to $\mathfrak{p}$ by

$$f_{\mathfrak{p}}(\mathfrak{P}) = [\mathbb{F}_{\mathfrak{P}} : \mathbb{F}_{\mathfrak{p}}].$$

That is, $f_{\mathfrak{p}}(\mathfrak{P})$ is the degree of $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ which is the dimension of the residue class field $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space. In fact, something slightly more general is true. Let $\mathfrak{A}$ and $\mathfrak{B}$ be an integral ideals of $\mathcal{O}$ with $\mathfrak{p}\mathcal{O} \subseteq \mathfrak{B} \subseteq \mathfrak{A}$. Consider the homomorphism

$$\phi : \mathfrak{A} \to \mathfrak{A}/\mathfrak{B} \qquad \alpha \mapsto \alpha + \mathfrak{B},$$

induced by natural projection. As $\mathfrak{A}$ and $\mathfrak{B}$ contain $\mathfrak{p}$, we have that $\ker \phi$ contains $\mathfrak{p}$. So ϕ induces a homomorphism $\mathbb{F}_{\mathfrak{p}} \to \mathfrak{A}/\mathfrak{B}$. Whence $\mathfrak{A}/\mathfrak{B}$ is a finite dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension at most n because $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. We obtain the residue class extension when $\mathfrak{A} = \mathcal{O}$ and $\mathfrak{B} = \mathfrak{P}$.

Suppose $\widetilde{\mathcal{O}}/\mathcal{O}/\mathcal{o}$ is a tower of Dedekind extensions for a finite separable tower $M/L/K$. Let $\widetilde{\mathfrak{P}}$, $\mathfrak{P}$, and $\mathfrak{p}$ be primes of $\widetilde{\mathcal{O}}$, $\mathcal{O}$, and $\mathcal{o}$ respectively with $\widetilde{\mathfrak{P}}$ above $\mathfrak{P}$ and $\mathfrak{P}$ above $\mathfrak{p}$. We illustrate this relationship via the diagram

$$\begin{array}{c}
\widetilde{\mathfrak{P}} \subset \widetilde{\mathcal{O}} \subseteq M \\
\Big| \\
\mathfrak{P} \subset \mathcal{O} \subseteq L \\
\Big| \\
\mathfrak{p} \subset \mathcal{o} \subseteq K.
\end{array}$$

Then $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \mid\mid \mathfrak{p}\mathcal{O}$ and $\widetilde{\mathfrak{P}}^{e_{\mathfrak{P}}(\widetilde{\mathfrak{P}})} \mid\mid \mathfrak{P}\widetilde{\mathcal{O}}$. We also have the residue extensions $\mathbb{F}_{\widetilde{\mathfrak{P}}}/\mathbb{F}_{\mathfrak{P}}$ and $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$. It follows that

$$e_{\mathfrak{p}}(\widetilde{\mathfrak{P}}) = e_{\mathfrak{P}}(\widetilde{\mathfrak{P}})e_{\mathfrak{p}}(\mathfrak{P}) \quad \text{and} \quad f_{\mathfrak{p}}(\widetilde{\mathfrak{P}}) = f_{\mathfrak{P}}(\widetilde{\mathfrak{P}})f_{\mathfrak{p}}(\mathfrak{P}). \tag{6}$$

In other words, the ramification indices and inertia degrees are multiplicative with respect to towers.

The ramification indices and inertia degrees satisfy a simple relationship to the degree of L/K that is fundamental to the study of Dedekind extensions. We first prove a useful lemma.

Lemma 1.51. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite separable extension L/K and suppose $\mathfrak{P}$ a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. Then there is an isomorphism*

$$\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \cong \bigoplus_i \mathfrak{P}^i/\mathfrak{P}^{i+1},$$

induced by natural projection of a basis adapted to the filtration

$$\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset \mathfrak{P}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset \cdots \supset \mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset 0,$$

of $\mathbb{F}_\mathfrak{P}$-vector spaces. In particular, $\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ is an $\mathbb{F}_\mathfrak{p}$-vector space of dimension $e_\mathfrak{p}(\mathfrak{P})f_\mathfrak{p}(\mathfrak{P})$.

Proof. By the third isomorphism theorem, quotients of successive layers of the filtration are isomorphic to $\mathfrak{P}^i/\mathfrak{P}^{i+1}$. Whence we obtain a short exact sequence

$$0 \longrightarrow \mathfrak{P}^{i+1}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \longrightarrow \mathfrak{P}^i/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \longrightarrow \mathfrak{P}^i/\mathfrak{P}^{i+1} \longrightarrow 0,$$

induced by natural inclusions and projections. As this is a short exact sequence of $\mathbb{F}_\mathfrak{P}$-vector spaces it splits. Choosing sections for all of these short exact sequences is equivalent to choosing a basis for $\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ adapted to the filtration and we obtain an isomorphism

$$\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \cong \bigoplus_i \mathfrak{P}^i/\mathfrak{P}^{i+1},$$

induced by natural projection of the adapted basis. This proves the first statement. The quotients $\mathfrak{P}^i/\mathfrak{P}^{i+1}$ are all isomorphic to $\mathbb{F}_\mathfrak{P}$ by Lemma 1.29. As $\mathbb{F}_\mathfrak{P}$ is an $\mathbb{F}_\mathfrak{p}$-vector spaces of dimension $f_\mathfrak{p}(\mathfrak{P})$, it follows that $\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ is an $\mathbb{F}_\mathfrak{p}$-vector space of dimension $e_\mathfrak{p}(\mathfrak{P})f_\mathfrak{p}(\mathfrak{P})$. This proves the second statement. □

We can now establish the relationship between the ramification indices, inertia degrees, and degree of L/K. The idea is that $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is of maximum dimension as an $\mathbb{F}_\mathfrak{p}$-vector space, namely n, and its dimension can be computed as a sum of products of the ramification indices and inertia degrees. This relationship is known as the **fundamental equality**.

Theorem (Fundamental equality). *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Suppose $\mathfrak{p}$ is a prime of $\mathcal{o}$ and $\mathfrak{p}\mathcal{O}$ has prime factorization*

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_\mathfrak{p}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_\mathfrak{p}(\mathfrak{P}_r)}.$$

Then

$$n = \sum_i e_\mathfrak{p}(\mathfrak{P}_i)f_\mathfrak{p}(\mathfrak{P}_i).$$

Proof. Since distinct primes are relatively prime, the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_i \mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)}.$$

As $\mathcal{O}/\mathfrak{p}\mathcal{O}$ and $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ are $\mathbb{F}_{\mathfrak{p}}$-vector spaces, it suffices to show $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is of dimension n and $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ is of dimension $e_{\mathfrak{p}}(\mathfrak{P}_i)f_{\mathfrak{p}}(\mathfrak{P}_i)$. For $\mathcal{O}/\mathfrak{p}\mathcal{O}$, we already know it is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension at most n as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. Therefore we must show that the dimension is exactly n. Let $\overline{\lambda_1}, \ldots, \overline{\lambda_m}$ be a basis for $\mathcal{O}/\mathfrak{p}\mathcal{O}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space and let $\lambda_1, \ldots, \lambda_m$ be any lift of this basis to $\mathcal{O}$. As $m \le n$, it suffices to show $\lambda_1, \ldots, \lambda_m$ spans L/K. Let M be the $\mathcal{o}$-module generated by $\lambda_1, \ldots, \lambda_m$ and set $N = \mathcal{O}/M$. Then $\mathcal{O} = M + \mathfrak{p}\mathcal{O}$ since $\lambda_1, \ldots, \lambda_m$ is a lift of a basis for $(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}$. Whence $N = \mathfrak{p}N$ and therefore N is a finitely generated $\mathcal{o}$-module of rank at most n since $\mathcal{O}$ is. Let $\omega_1, \ldots, \omega_r$ be generators for N. As $N = \mathfrak{p}N$, we have

$$\omega_i = \sum_j \alpha_{i,j}\omega_j,$$

for some $\alpha_{i,j} \in \mathfrak{p}$. These r equations are equivalent to the identity

$$\begin{pmatrix} 1-\alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & 1-\alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & 1-\alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

Let $d = \det(I - (\alpha_{i,j}))$. Then $d \neq 0$ because expanding the determinant shows $d \equiv 1 \pmod{\mathfrak{p}}$ as $\alpha_{i,j} \in \mathfrak{p}$. Cramer's rule implies that d annihilates N which is to say that $d\mathcal{O} \subseteq M$. Equivalently,

$$d\mathcal{O} \subseteq \lambda_1\mathcal{o} + \cdots + \lambda_m\mathcal{o}.$$

By Proposition 1.4, multiplication by K shows

$$L = \lambda_1 K + \cdots + \lambda_m K.$$

Hence $\lambda_1, \ldots, \lambda_m$ spans L/K whence $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension n. For $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$, the dimensionality claim follows from Lemma 1.51. □

It is useful to classify primes according to specific cases of the fundamental equality. We say $\mathfrak{p}$ is **inert** in $\mathcal{O}/\mathcal{o}$ if there is a prime $\mathfrak{P}$ above $\mathfrak{p}$ with $f_{\mathfrak{p}}(\mathfrak{P}) = n$. By the fundamental equality, this means $e_{\mathfrak{p}}(\mathfrak{P}) = 1$ and $r = 1$. Whence

$$\mathfrak{p}\mathcal{O} = \mathfrak{P},$$

which is to say $\mathfrak{p}\mathcal{O}$ is prime. More generally, $\mathfrak{p}$ is said to be **nonsplit** in $\mathcal{O}/\mathcal{o}$ if there is a prime $\mathfrak{P}$ above $\mathfrak{p}$ satisfying $e_{\mathfrak{p}}(\mathfrak{P})f_{\mathfrak{p}}(\mathfrak{P}) = n$ and is said to be **split** in $\mathcal{O}/\mathcal{o}$ otherwise. So

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \quad \text{or} \quad \mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)},$$

with r at least 2, according to if $\mathfrak{p}$ is nonsplit or split accordingly. Moreover, we say $\mathfrak{p}$ is **totally split** in $\mathcal{O}/\mathcal{o}$ if every prime $\mathfrak{P}$ above $\mathfrak{p}$ satisfies $e_{\mathfrak{p}}(\mathfrak{P}) = f_{\mathfrak{p}}(\mathfrak{P}) = 1$ and so $r = n$ by the fundamental equality. Then

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1 \cdots \mathfrak{P}_n.$$

Observe that being inert or totally split correspond to antithetical factorizations in terms of inertia degrees. In particular, the smaller the inertia degrees are the greater the tendency for $\mathfrak{p}\mathcal{O}$ to factor into distinct primes.

Now let us introduce ramification of primes. If $\mathfrak{P}$ is above $\mathfrak{p}$, we say $\mathfrak{P}$ is **unramified** in $\mathcal{O}/\mathcal{o}$ if $e_{\mathfrak{p}}(\mathfrak{P}) = 1$ and the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. Otherwise, we say $\mathfrak{P}$ is **ramified** in $\mathcal{O}/\mathcal{o}$. Moreover, $\mathfrak{P}$ is **totally ramified** in $\mathcal{O}/\mathcal{o}$ if in addition to being ramified we have $e_{\mathfrak{p}}(\mathfrak{P}) = n$ whence $f_{\mathfrak{p}}(\mathfrak{P}) = 1$. Similarly, we say that a prime $\mathfrak{p}$ of $\mathcal{o}$ is **unramified** in $\mathcal{O}/\mathcal{o}$ if every prime $\mathfrak{P}$ above it is unramified and is **ramified** in $\mathcal{O}/\mathcal{o}$ otherwise. Also, $\mathfrak{p}$ is said to be **totally ramified** in $\mathcal{O}/\mathcal{o}$ if a prime above it is totally ramified. In this case, the fundamental equality implies that there is only one prime $\mathfrak{P}$ above $\mathfrak{p}$ and

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}^n.$$

Note that being totally ramified or totally split correspond to antithetical factorizations in terms of ramification indices. In particular, the larger the maximum ramification index is the greater the tendency for $\mathfrak{p}\mathcal{O}$ to factor as a power of a single prime.

In any case, all of this suggests that it is desirable to understand how $\mathfrak{p}\mathcal{O}$ factors into primes of $\mathcal{O}$. Using orders, this is tractable for almost all primes $\mathfrak{p}$ if we know a sufficient amount of information about the minimal polynomial of a primitive element for L/K. Let θ be a primitive element for L/K so that $L = K(\theta)$. In view of Proposition 1.4 we may multiply by a nonzero element of $\mathcal{o}$, if necessary, to ensure $\theta \in \mathcal{O}$ and hence its minimal polynomial $m_\theta(x)$ over K has coefficients in $\mathcal{o}$. Then $\mathcal{o} \subseteq \mathcal{o}[\theta] \subseteq \mathcal{O}$ so that $\mathcal{o}[\theta]$ is an order. In fact, the field of fractions of $\mathcal{o}[\theta]$ is L since the field of fractions of $\mathcal{o}$ is K. As $\mathcal{o}[\theta]$ is integral over $\mathcal{o}$, it follows that the normalization of $\mathcal{o}[\theta]$ is $\mathcal{O}$. Therefore the conductor $\mathfrak{q}_{\mathcal{o}[\theta]}$ is an integral ideal of $\mathcal{o}[\theta]$ and $\mathcal{O}$. The **Dedekind-Kummer theorem** describes the prime factorization of $\mathfrak{p}\mathcal{O}$ provided it is relatively prime to $\mathfrak{q}_{\mathcal{o}[\theta]}$.

Theorem (Dedekind-Kummer theorem). *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K, θ be a primitive element for L/K contained in $\mathcal{O}$ with minimal polynomial $m_\theta(x)$ over K, and $\mathfrak{p}$ be a prime of $\mathcal{o}$ such that $\mathfrak{p}\mathcal{O}$ is relatively prime to $\mathfrak{q}_{\mathcal{o}[\theta]}$. Also let $\overline{m_\theta}(x) \in \mathbb{F}_{\mathfrak{p}}[x]$ be the image of $m_\theta(x)$ under natural projection and suppose*

$$\overline{m_\theta}(x) = \overline{m_1}(x)^{e_1} \cdots \overline{m_r}(x)^{e_r},$$

is its prime factorization. Let $m_i(x)$ be any lift of $\overline{m_i}(x)$ to $\mathcal{o}[x]$ and set

$$\mathfrak{P}_i = m_i(\theta)\mathcal{O} + \mathfrak{p}\mathcal{O}.$$

Then $\mathfrak{P}_i$ is a prime of $\mathcal{O}$ and

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r},$$

is the prime factorization of $\mathfrak{p}\mathcal{O}$. In particular,

$$e_{\mathfrak{p}}(\mathfrak{P}_i) = e_i \quad \text{and} \quad f_{\mathfrak{p}}(\mathfrak{P}_i) = \deg(\overline{m_i}(x)).$$

Proof. We will being by demonstrating a chain of isomorphisms

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{o}[\theta]/\mathfrak{p}[\theta] \cong \mathbb{F}_{\mathfrak{p}}[x]/(\overline{m_\theta}(x)).$$

For the first isomorphism, consider the homomorphism

$$\phi : \mathcal{o}[\theta] \mapsto \mathcal{O}/\mathfrak{p}\mathcal{O} \qquad \alpha \mapsto \alpha + \mathfrak{p}\mathcal{O},$$

induced by natural projection. We have $\mathfrak{q}_{\mathcal{O}[\theta]} + \mathfrak{p}\mathcal{O} = \mathcal{O}$ because $\mathfrak{p}\mathcal{O}$ is relatively prime to $\mathfrak{q}_{\mathcal{O}[\theta]}$. Whence

$$\mathcal{o}[\theta] + \mathfrak{p}\mathcal{O} = \mathcal{O},$$

proving ϕ is surjective. We claim $\ker \phi = \mathfrak{p}[\theta]$. Since $\ker \phi = \mathfrak{p}\mathcal{O} \cap \mathcal{o}[\theta]$, it suffices to show

$$\mathfrak{p}\mathcal{O} \cap \mathcal{o}[\theta] = \mathfrak{p}[\theta].$$

The reverse inclusion is clear. For the forward inclusion, we claim $(\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathcal{o}) + \mathfrak{p} = \mathcal{o}$. For if not, $(\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathcal{o}) + \mathfrak{p} \subseteq \mathfrak{q}$ for some prime $\mathfrak{q}$ of $\mathcal{o}$ as primes are maximal. Letting $\mathfrak{Q}$ be a prime of $\mathcal{O}$ above $\mathfrak{q}$, we would then have $\mathfrak{q}_{\mathcal{O}[\theta]} + \mathfrak{p}\mathcal{O} \subseteq \mathfrak{Q}$ contradicting relative primality. A short computation shows

$$((\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathcal{o}) + \mathfrak{p})(\mathcal{o}[\theta] \cap \mathfrak{p}\mathcal{O}) \subseteq \mathfrak{p}[\theta],$$

from whence the reverse inclusion follows. So $\ker \phi = \mathfrak{p}[\theta]$. By first isomorphism theorem, we obtain

$$\mathcal{o}[\theta]/\mathfrak{p}[\theta] \cong \mathcal{O}/\mathfrak{p}\mathcal{O}.$$

This establishes the first isomorphism in the chain. For the second, recall from Proposition 1.4 that $m_\theta(x)$ has coefficients in $\mathcal{o}$ whence is an element of $\mathcal{o}[x]$. Therefore $\overline{m_\theta}(x)$ is an element of $\mathbb{F}_p[x]$ and we may consider the homomorphism

$$\pi : \mathcal{o}[x] \mapsto \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)) \qquad f(x) \mapsto \overline{f}(x) + (\overline{m_\theta}(x)),$$

given by the composition of natural projections. As such, π is surjective and $\ker \pi = (\mathfrak{p}, m_\theta(x))$. Whence the first isomorphism theorem gives

$$\mathcal{o}[x]/(\mathfrak{p}, m_\theta(x)) \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

In view of the isomorphism $\mathcal{o}[\theta] \cong \mathcal{o}[x]/(m_\theta(x))$ induced by evaluation at θ and the third isomorphism theorem, the aforementioned isomorphism becomes

$$\mathcal{o}[\theta]/\mathfrak{p}[\theta] \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

This establishes the second isomorphism in the chain.

We now determine the maximal ideals of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ via this chain. As the $\overline{m}_i(x)^{e_i}$ are pairwise relatively prime, the Chinese remainder theorem gives an isomorphism

$$\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)) \cong \bigoplus_i \mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)^{e_i}),$$

induced by natural projection. Since $\overline{m_i}(x)$ is irreducible, $(\overline{m_i}(x))$ is maximal. By the fourth isomorphism theorem, $(\overline{m_i}(x))/(\overline{m_i}(x)^{e_i})$ is a maximal ideal of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)^{e_i})$. So under the aforementioned isomorphism, the maximal ideals of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ are of the form $(\overline{m_i}(x))/(\overline{m_\theta}(x))$. As the chain of isomorphisms is induced by natural projections, the maximal ideals of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ are of the form $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$.

It is now possible to show that the $\mathfrak{P}_i$ are the prime factors of $\mathfrak{p}\mathcal{O}$. To see this, consider the homomorphism

$$\pi : \mathcal{O} \to \mathcal{O}/\mathfrak{p}\mathcal{O} \qquad \alpha \mapsto \alpha + \mathfrak{p}\mathcal{O},$$

given by natural projection. As such, π is surjective. By definition of $\mathfrak{P}_i$, the image under π is $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$. As this ideal is maximal and hence prime, the preimage $\mathfrak{P}_i$ is prime too. Moreover, the $\mathfrak{P}_i$ are distinct since the $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$ are. In particular, the $\mathfrak{P}_i$ are also pairwise relatively prime. By construction, $\mathfrak{P}_i \mid \mathfrak{p}\mathcal{O}$ because to divide is to contain. Whence the $\mathfrak{P}_i$ are prime factors of $\mathfrak{p}\mathcal{O}$. They constitute all of the prime factors of $\mathfrak{p}\mathcal{O}$ because the image of any prime under π containing $\mathfrak{p}\mathcal{O}$ must be a maximal ideal of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ by the fourth isomorphism theorem and every maximal ideal is one of the $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$. Together, all of this means that $\mathfrak{p}\mathcal{O}$ admits the prime factorization

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_\mathfrak{p}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_\mathfrak{p}(\mathfrak{P}_r)},$$

for some ramification indices $e_\mathfrak{p}(\mathfrak{P}_i)$.

It remains to compute the ramification indices and inertia degrees. We will first compute the inertia degrees and then use the fundamental equality to determine the ramification indices. From our previous work, we have an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

Moreover, the primes $\mathfrak{P}_i$ of $\mathcal{O}$ correspond to the maximal ideals $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$ of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ which correspond to the maximal ideals $(\overline{m_i}(x))/(\overline{m_\theta}(x))$ of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ under this isomorphism. By the third isomorphism theorem, we obtain the isomorphism of fields

$$\mathcal{O}/\mathfrak{P}\mathcal{O} \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)).$$

Both of these fields are $\mathbb{F}_\mathfrak{p}$-vector spaces where the latter clearly has degree $\deg(\overline{m_i}(x))$. Hence $f_\mathfrak{p}(\mathfrak{P}_i) = \deg(\overline{m_i}(x))$ which is the desired formula for the inertia degrees. For the ramification indices, as the chain of isomorphisms is induced by natural projections, the ideal $(\overline{m_i}(x)^{e_i})/(\overline{m_\theta}(x))$ of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ corresponds to the ideal $m_i(\theta)^{e_i}\mathcal{O}/\mathfrak{p}\mathcal{O}$ of $\mathcal{O}/\mathfrak{p}\mathcal{O}$. Since the image of $\mathfrak{P}_i$ under π is $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$, we have that $\mathfrak{P}_i^{e_i}$ is contained in the preimage of $m_i(\theta)^{e_i}\mathcal{O}/\mathfrak{p}\mathcal{O}$ under π. As $m_\theta(\theta) \in \mathfrak{p}\mathcal{O}$, it follows that

$$\mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r} \subseteq \mathfrak{p}\mathcal{O}.$$

Prime factorization forces $e_\mathfrak{p}(\mathfrak{P}_i) \le e_i$. But then

$$\sum_i e_\mathfrak{p}(\mathfrak{P}_i) f_p(\mathfrak{P}_i) = \sum_i e_i \deg(\overline{m_i}(x)),$$

as fundamental equality says that the first sum is sum is n while the prime factorization of $m_\theta(x)$ says that the second sum is n as well. Ths only way this is possible is if $e_\mathfrak{p}(\mathfrak{P}_i) = e_i$. This is the desired formula for the ramification indices. $\square$

As we have already noted, the irregular primes of $\mathcal{o}[\theta]$ correspond to the prime factors of $\mathfrak{q}_{\mathcal{O}[\theta]}$ of which there are finitely many as $\mathcal{O}$ is a Dedekind domain. This means that the

Dedekind-Kummer theorem is valid for $\mathfrak{p}\mathcal{O}$ provided that $\mathfrak{p}$ is not below any of the prime factors of $\mathfrak{q}_{\mathcal{O}[\theta]}$. In other words, the Dedekind-Kummer theorem is valid for all but finitely many primes $\mathfrak{p}$. In fact, if $\mathfrak{q}_{\mathcal{O}[\theta]} = \mathcal{O}$ then we do not have to avoid any primes at all. This occurs when $\mathcal{O}/\mathcal{o}$ is monogenic. Indeed, Suppose $\mathcal{O} = \mathcal{o}[\theta]$ for some $\theta \in \mathcal{O}$. Then $1, \theta, \ldots, \theta^{n-1}$ is an integral basis for $\mathcal{O}/\mathcal{o}$ which is necessarily a basis for L/K. This means θ is also a primitive element for L/K which implies $\mathfrak{q}_{\mathcal{O}[\theta]} = \mathcal{O}$.

Let us now discuss how ramification indices and inertia degrees are affected under localization. It turns out that they are preserved.

Proposition 1.52. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then for the Dedekind extension $\mathcal{O}D^{-1}/\mathcal{o}D^{-1}$, $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$ and there are isomorphisms*

$$\mathbb{F}_{\mathfrak{P}D^{-1}} \cong \mathbb{F}_{\mathfrak{P}} \quad \textit{and} \quad \mathbb{F}_{\mathfrak{p}D^{-1}} \cong \mathbb{F}_{\mathfrak{p}}.$$

Lastly, we have equalities

$$f_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = f_{\mathfrak{p}}(\mathfrak{P}) \quad \textit{and} \quad e_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = e_{\mathfrak{p}}(\mathfrak{P}).$$

Proof. In view of Proposition 1.33, $\mathfrak{P}$ and $\mathfrak{p}$ are necessarily disjoint from D. In fact, the inclusion-preserving bijections show that $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$. This proves the first statement. Now consider the homomorphisms

$$\Phi : \mathcal{O} \to \mathbb{F}_{\mathfrak{P}D^{-1}} \qquad \alpha \mapsto \alpha + \mathfrak{P}D^{-1},$$

and

$$\phi : \mathcal{o} \to \mathbb{F}_{\mathfrak{p}D^{-1}} \qquad \alpha \mapsto \alpha + \mathfrak{p}D^{-1},$$

induced by natural projection. By the first isomorphism theorem, it suffices to show Φ and ϕ are surjective as well as $\ker \Phi = \mathfrak{P}$ and $\ker \phi = \mathfrak{p}$. We have

$$\mathcal{O} + \mathfrak{P}D^{-1} = \mathcal{O}D^{-1} \quad \text{and} \quad \mathcal{o} + \mathfrak{p}D^{-1} = \mathcal{o}D^{-1},$$

since $\mathfrak{P}D^{-1}$ and $\mathfrak{p}D^{-1}$ are maximal. This proves surjectivity of Φ and ϕ. Now $\ker \Phi = \mathfrak{P}D^{-1} \cap \mathcal{O}$ and $\ker \phi = \mathfrak{p}D^{-1} \cap \mathcal{o}$. The inclusion-preserving bijections in Proposition 1.33 show that

$$\mathfrak{P}D^{-1} \cap \mathcal{O} = \mathfrak{P} \quad \text{and} \quad \mathfrak{p}D^{-1} \cap \mathcal{o} = \mathfrak{p}.$$

Whence $\ker \Phi = \mathfrak{P}$ and $\ker \phi = \mathfrak{p}$ and we obtain the desired isomorphisms. In fact, these isomorphisms together imply

$$f_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = f_{\mathfrak{p}}(\mathfrak{P}).$$

As $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$, the fundamental equality forces

$$e_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = e_{\mathfrak{p}}(\mathfrak{P}). \qquad \square$$

In view of the isomorphisms from this result, the residue class extensions of $\mathcal{O}/\mathcal{o}$ and $\mathcal{O}_{\mathfrak{p}}/\mathcal{o}_{\mathfrak{p}}$ corresponding to the primes $\mathfrak{p}$ and $\mathfrak{p}_{\mathfrak{p}}$ respectively are isomorphic. Thus they are both extensions of $\mathbb{F}_\mathfrak{p}$-vector spaces. More generally, we have that if $\mathfrak{A}$ is an integral ideal of $\mathcal{O}$ containing $\mathfrak{p}$ then $\mathcal{O}/\mathfrak{A}$ and $\mathcal{O}_\mathfrak{p}/\mathfrak{A}_\mathfrak{p}$ are both a finite dimensional $\mathbb{F}_\mathfrak{p}$-vector spaces.

We are now ready to introduce the ideal norm of $\mathcal{O}/\mathcal{o}$. Suppose $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ with prime factorization

$$\mathfrak{F} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

Let $\mathfrak{p}_i$ be the prime below $\mathfrak{P}_i$. We define the **norm** $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F})$ of $\mathfrak{F}$ by

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F}) = \mathfrak{p}_1^{e_1 f_{\mathfrak{p}_1}(\mathfrak{P}_1)} \cdots \mathfrak{p}_r^{e_r f_{\mathfrak{p}_r}(\mathfrak{P}_r)}.$$

Setting $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathcal{O}) = \mathcal{o}$, prime factorization shows that we obtain a homomorphism

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}} : I(\mathcal{O}) \to I(\mathcal{o}) \qquad \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r} \mapsto \mathfrak{p}_1^{e_1 f_{\mathfrak{p}_1}(\mathfrak{P}_1)} \cdots \mathfrak{p}_r^{e_r f_{\mathfrak{p}_r}(\mathfrak{P}_r)},$$

called the **ideal norm** of $\mathcal{O}/\mathcal{o}$. This means that the ideal norm is completely multiplicative. It follows from the fundamental equality that

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{p}\mathcal{O}) = \mathfrak{p}^n,$$

and by multiplicativity of the ideal norm, any fractional ideal $\mathfrak{f}$ of $\mathcal{o}$ satisfies

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{f}\mathcal{O}) = \mathfrak{f}^n.$$

In any case, the ideal norm is aptly named because it is compatible with the field norm. To prove this, we first show that the ideal norm is compatible with localization.

Proposition 1.53. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a completely multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{N}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}}(\mathfrak{F}D^{-1}) = \mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F})D^{-1}.$$

Proof. As the ideal norm is completely multiplicative, it suffices to prove the claim for a prime of $\mathcal{O}$. So let $\mathfrak{P}$ be such a prime and let $\mathfrak{p}$ be the prime below it. If $\mathfrak{P}$ is not disjoint from D then neither is $\mathfrak{p}$, and their localizations are $\mathcal{O}D^{-1}$ and $\mathcal{o}D^{-1}$ respectively. Whence the claim is immediate as both sides are $\mathcal{o}D^{-1}$. If $\mathfrak{P}$ is disjoint from D then the claim follows from Proposition 1.52. $\square$

With this result in hand, we can shows that the ideal norm is compatible with the field norm.

Proposition 1.54. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Then*

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\lambda\mathcal{O}) = \mathrm{N}_{L/K}(\lambda)\mathcal{o},$$

for any $\lambda \in L$.

Proof. In view of Propositions 1.34 and 1.53, we may assume that $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. Moreover, as $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}$ and $\mathrm{N}_{L/K}$ are both completely multiplicative and L is the field of fractions of $\mathcal{O}$, it is enough to prove the claim when $\lambda \in \mathcal{O}$ and λ is prime. As $\mathcal{O}$ is a principal ideal domain, we have $\lambda\mathcal{O} = \mathfrak{P}$ for some prime $\mathfrak{P}$ of $\mathcal{O}$. Let $\mathfrak{p}$ be the prime of $\mathcal{o}$ below $\mathfrak{P}$ and let v be the valuation of $\mathcal{o}$. Then

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\lambda\mathcal{O}) = \mathfrak{p}^{f_{\mathfrak{p}}(\mathfrak{P})} \quad \text{and} \quad \mathrm{N}_{L/K}(\lambda)\mathcal{o} = \mathfrak{p}^{v(\mathrm{N}_{L/K}(\lambda))},$$

and we are reduced to proving

$$f_{\mathfrak{p}}(\mathfrak{P}) = v(\mathrm{N}_{L/K}(\lambda)).$$

As $f_{\mathfrak{p}}(\mathfrak{P})$ is the dimension of $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space, it suffices to show that the same is true of $v(\mathrm{N}_{L/K}(\lambda))$. To this end, $\mathrm{N}_{L/K}(\lambda) = \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\lambda)$ by Proposition 1.16. Recall that $\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\lambda)$ is the determinant of the $\mathcal{o}$-linear operator T_λ on $\mathcal{O}$. As $\mathcal{o}$ is a principal ideal domain, putting T_λ in smith normal form shows that there is a basis of $\mathcal{O}/\mathcal{o}$ for which the matrix of T_λ is

$$U \begin{pmatrix} \varepsilon_1\pi^{k_1} & & \\ & \ddots & \\ & & \varepsilon_n\pi^{k_n} \end{pmatrix} V,$$

for some invertible matrices $U, V \in \mathrm{GL}_n(\mathcal{o})$ of unit determinant, nonnegative integers k_i, units ε_i and uniformizer π for $\mathcal{o}$. Setting $\varepsilon = \prod_i \varepsilon_i$ and $k = \sum_i k_i$ shows $\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\lambda) = \varepsilon\pi^k$ whence $v(\mathrm{N}_{L/K}(\lambda)) = k$. As the image of T_λ is $\mathfrak{P}$, the smith normal form induces an isomorphism

$$\mathbb{F}_{\mathfrak{P}} \cong \bigoplus_i \mathcal{o}/\pi^{k_i}\mathcal{o}.$$

Observe that the length of $\mathcal{o}/\pi^{k_i}\mathcal{o}$ as an $\mathcal{o}$-module is k_i. Whence the length of $\mathbb{F}_{\mathfrak{P}}$ as an $\mathcal{o}$-module is k. But then k is the dimension of $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space which is also $v(\mathrm{N}_{L/K}(\lambda))$. $\square$

In light of this result, the ideal norm preserves principal ideals. This means that the ideal norm induces a homomorphism

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}} : \mathrm{Cl}(\mathcal{O}) \to \mathrm{Cl}(\mathcal{o}) \qquad \mathfrak{a}P(\mathcal{O}) \mapsto \mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{a})P(\mathcal{o}),$$

between ideal class groups.

Number Fields

We now turn to the setting where L/K is an extension of number fields. Recall that the underlying Dedekind extension is $\mathcal{O}_L/\mathcal{O}_K$. A prime $\mathfrak{p}$ of K is said to be **inert**, **nonsplit**, **split**, or **totally split** in L/K if it is in $\mathcal{O}_L/\mathcal{O}_K$. Primes $\mathfrak{P}$ and $\mathfrak{p}$ of L and K are said to be **unramified**, **ramified**, or **totally ramified** in L/K if they are in $\mathcal{O}_L/\mathcal{O}_K$. Lastly, the **ideal norm** $\mathfrak{N}_{L/K}$ of L/K is the ideal norm $\mathfrak{N}_{\mathcal{O}_L/\mathcal{O}_K}$.

In the case of a number field K, we can say more as $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis and the reside class extensions are separable extensions of finite fields. A prime p of $\mathbb{Q}$ is said to be **inert**, **nonsplit**, **split**, or **totally split** in K if it is in $\mathcal{O}_K/\mathbb{Z}$. Primes $\mathfrak{p}$ and p of K and $\mathbb{Q}$ are said to be **unramified**, **ramified**, or **totally ramified** in K if they are in $\mathcal{O}_K/\mathbb{Z}$. In particular, if $\mathfrak{p}$ is above p then $\mathfrak{p}$ is unramified in K if and only if $e_p(\mathfrak{p}) = 1$. The **ideal norm** $\mathfrak{N}_K$ of K is the ideal norm $\mathfrak{N}_{\mathcal{O}_K/\mathbb{Z}}$. As $\mathbb{Z}$ is a principal ideal domain, every fractional ideal is principal. Therefore $\mathfrak{N}_K(\mathfrak{f})$ is generated by some $r_\mathfrak{f} \in \mathbb{Q}_+$ for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}_K$. As the ideal norm is completely multiplicative, this induces a homomorphism

$$\mathrm{N}_K : I_K \to \mathbb{Q}_+ \qquad \mathfrak{f} \mapsto r_\mathfrak{f},$$

called the **norm** of K. We call $\mathrm{N}_K(\mathfrak{f})$ the **norm** of $\mathfrak{f}$. The relationship between this norm and the ideal norm is

$$\mathfrak{N}_K(\mathfrak{f}) = \mathrm{N}_K(\mathfrak{f})\mathbb{Z}.$$

In particular, if $\mathfrak{p}$ is a prime of K above p then

$$\mathfrak{N}_K(\mathfrak{p}) = p^{f_p(\mathfrak{p})}\mathbb{Z} \quad \text{and} \quad \mathrm{N}_K(\mathfrak{p}) = p^{f_p(\mathfrak{p})}.$$

For an integral ideal $\mathfrak{a}$, there is a simple relationship between the norm of $\mathfrak{a}$ and the quotient $\mathcal{O}_K/\mathfrak{a}$.

Proposition 1.55. *Let K be a number field of degree n. Then for any integral ideal $\mathfrak{a}$ of $\mathcal{O}_K$, we have*

$$\mathrm{N}_K(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|.$$

Moreover,

$$\mathrm{N}_K(\alpha\mathcal{O}_K) = |\,\mathrm{N}_K(\alpha)|,$$

for any $\alpha \in \mathcal{O}_K$.

Proof. As K admits an integral basis, every integral ideal of K is a free $\mathbb{Z}$-module of rank n by Theorem 1.14. Whence any such quotient is finite. By the Chinese remainder theorem, the right-hand side of the desired formula is multiplicative. As the norm is also multiplicative, it suffices to prove the claim when $\mathfrak{a}$ is a prime power. So fix a prime power $\mathfrak{p}^e$ and let p be the prime below $\mathfrak{p}$. On the one hand, $\mathrm{N}_K(\mathfrak{p}^e) = p^{ef_p(\mathfrak{p})}$. On the other hand, consider the filtration

$$\mathcal{O}_K/\mathfrak{p}^e \supset \mathfrak{p}/\mathfrak{p}^e \supset \cdots \supset \mathfrak{p}^{e-1}/\mathfrak{p}^e \supset 0,$$

of $\mathcal{O}_K$-modules. By the third isomorphism theorem, quotients of successive layers of the filtration are isomorphic to $\mathfrak{p}^i/\mathfrak{p}^{i+1}$. Repeated application of this isomorphism and Lagrange's theorem together give

$$|\mathcal{O}_K/\mathfrak{p}^e| = \prod_i |\mathfrak{p}^i/\mathfrak{p}^{i+1}|.$$

The quotients $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ are all isomorphic to $\mathbb{F}_\mathfrak{p}$ by Lemma 1.29. As $\mathbb{F}_\mathfrak{p}$ is an $\mathbb{F}_p$-vector space of dimension $f_p(\mathfrak{p})$, we find that $|\mathfrak{p}^i/\mathfrak{p}^{i+1}| = p^{f_p(\mathfrak{p})}$ and so $|\mathcal{O}_K/\mathfrak{p}^e| = p^{ef_p(\mathfrak{p})}$. This proves the first statement. The second statement follows from the first together with Proposition 1.54. $\square$

As a consequence of this result and Lagrange's theorem, any integral ideal $\mathfrak{a}$ contains its norm $\mathrm{N}_K(\mathfrak{a})$ and therefore the same holds for every fractional ideal $\mathfrak{f}$ as well.

1.7 Ramification

Much more can be said about the ramification of primes in a Dedekind extension $\mathcal{O}/\mathcal{o}$ when the associated extension L/K is Galois. Since L/K is assumed to be finite and separable, this amounts to further assuming L/K is normal. In this case, the Galois group is

$$\mathrm{Gal}(L/K) = \mathrm{Hom}_K(L, \overline{K}),$$

so that every K-embedding of L into $\overline{K}$ is an automorphism of L. By Proposition 1.7, the field trace and field norm of any $\lambda \in L$ are given by

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_{\sigma} \sigma(\lambda) \quad \text{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_{\sigma} \sigma(\lambda),$$

where σ runs over the elements of $\mathrm{Gal}(L/K)$. As $\mathcal{o} \subseteq K$, $\mathrm{Gal}(L/K)$ fixes $\mathcal{o}$ pointwise and hence every fractional ideal of $\mathcal{o}$ as well. In fact, for any $\alpha \in \mathcal{O}$, we see that $\mathrm{Gal}(L/K)$ permutes the roots of the monic polynomial in $\mathcal{o}[x]$ which α satisfies. Whence there is an action

$$\mathrm{Gal}(L/K) \times \mathcal{O} \to \mathcal{O} \qquad (\sigma, \alpha) \mapsto \sigma(\alpha).$$

In particular, each K-automorphism σ restricts to an automorphism of $\mathcal{O}$ and therefore $\sigma(\mathfrak{P})$ is a prime of $\mathcal{O}$ if $\mathfrak{P}$ is. Moreover, if $\mathfrak{P}$ is above $\mathfrak{p}$ then so is $\sigma(\mathfrak{P})$ as it is a prime containing $\mathfrak{p}$. Whence

$$\sigma(\mathfrak{P}) \cap \mathcal{o} = \mathfrak{p}.$$

Accordingly, we say that $\sigma(\mathfrak{P})$ is **conjugate** to $\mathfrak{P}$. It turns out that the Galois group acts transitively on the primes above a given prime.

Proposition 1.56. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K and let $\mathfrak{p}$ be a prime of $\mathcal{o}$. Then $\mathrm{Gal}(L/K)$ acts transitively on the primes above $\mathfrak{p}$. Moreover, if $\mathfrak{p}\mathcal{O}$ has prime factorization*

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)},$$

then

$$f_{\mathfrak{p}}(\mathfrak{P}_1) = \cdots = f_{\mathfrak{p}}(\mathfrak{P}_r) \quad \textit{and} \quad e_{\mathfrak{p}}(\mathfrak{P}_1) = \cdots = e_{\mathfrak{p}}(\mathfrak{P}_r).$$

Proof. Assume by contradiction that there exist distinct primes $\mathfrak{P}_i$ and $\mathfrak{P}_j$ above $\mathfrak{p}$ for which $\mathfrak{P}_i \neq \mathfrak{P}_j$ for any K-automorphism σ. As distinct primes are relatively prime, the Chinese remainder theorem implies the existence of an $\alpha \in \mathcal{O}$ such that

$$\alpha \equiv 1 \pmod{\sigma(\mathfrak{P}_i)} \quad \text{and} \quad \alpha \equiv 0 \pmod{\mathfrak{P}_j},$$

Now recall As $\mathrm{N}_{L/K}(\alpha) \in \mathcal{o}$ by Proposition 1.8. On the one hand, $\alpha \notin \sigma(\mathfrak{P}_i)$ whence $\sigma(\alpha) \notin \mathfrak{P}_i$. Then $\mathrm{N}_{L/K}(\alpha) \notin \mathfrak{P}_i$ by primality of $\mathfrak{P}_i$ and hence $\mathrm{N}_{L/K}(\alpha) \notin \mathfrak{p}$ because $\mathfrak{P}_i$ is above $\mathfrak{p}$. On the other hand, $\alpha \in \mathfrak{P}_j$ which implies $\alpha \in \sigma(\mathfrak{P}_j)$ when σ is the identity. Therefore $\mathrm{N}_{L/K}(\alpha) \in \mathfrak{P}_j$ and so $\mathrm{N}_{L/K}(\alpha) \in \mathfrak{p}$ because $\mathfrak{P}_j$ is above $\mathfrak{p}$. This gives a contradiction from which we conclude the action is transitive.

It remains to show that the ramification indices and inertia degrees of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. As the action is transitive, there exists a K-automorphism σ satisfying $\sigma(\mathfrak{P}_i) = \mathfrak{P}_j$.

Now σ is an automorphism of $\mathcal{O}$ fixing $\mathcal{o}$, and hence $\mathfrak{p}$, pointwise. Thus $\sigma(\mathfrak{p}\mathcal{O}) = \mathfrak{p}\mathcal{O}$ which implies $\mathfrak{P}_i^e \mid\mid \mathfrak{p}\mathcal{O}$ if and only if $\mathfrak{P}_j^e \mid\mid \mathfrak{p}\mathcal{O}$. Therefore the ramification indices of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. Also, being an automorphism of $\mathcal{O}$, σ induces an isomorphism

$$\mathbb{F}_{\mathfrak{P}_i} \cong \mathbb{F}_{\mathfrak{P}_j}.$$

Therefore the inertia degrees of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. $\square$

Another way to phrase this result is that the primes above $\mathfrak{p}$ are all conjugate to each other and their ramification indices and inertia degrees are all equal. We call the common ramification index e the **ramification index** of $\mathfrak{p}$ in $\mathcal{O}/\mathcal{o}$ and the common inertia degree f the **inertia degree** of $\mathfrak{p}$ in $\mathcal{O}/\mathcal{o}$. If there are r primes above $\mathfrak{p}$, the fundamental equality takes the particularly simple form

$$n = ref.$$

We define the **decomposition group** $D_{\mathfrak{p}}(\mathfrak{P})$ of $\mathfrak{P}$ over $\mathfrak{p}$ by

$$D_{\mathfrak{p}}(\mathfrak{P}) = \{\tau \in \mathrm{Gal}(L/K) : \tau(\mathfrak{P}) = \mathfrak{P}\}.$$

Equivalently, the decomposition group is the stabilizer subgroup of $\mathfrak{P}$ in the Galois group. The associated **decomposition field** $L^{D_{\mathfrak{p}}(\mathfrak{P})}$ of $\mathfrak{P}$ over $\mathfrak{p}$ is defined by

$$L^{D_{\mathfrak{p}}(\mathfrak{P})} = \{\lambda \in L : \tau(\lambda) = \lambda \text{ for } \tau \in D_{\mathfrak{p}}(\mathfrak{P})\}.$$

In other words, the decomposition field is the fixed field of L by the decomposition group and hence an intermediate field of L/K. In this case, the fundamental theorem of Galois theory says

$$\mathrm{Gal}(L/L^{D_{\mathfrak{p}}(\mathfrak{P})}) = D_{\mathfrak{p}}(\mathfrak{P}).$$

Let $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ be the integral closure of $\mathcal{o}$ in $L^{D_{\mathfrak{p}}(\mathfrak{P})}$. Then Propositions 1.5 and 1.32 together imply that $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$ is a tower of Dedekind extensions for the finite separable tower $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$. Also, write $\mathfrak{P}^D$ for the prime of $L^{D_{\mathfrak{p}}(\mathfrak{P})}$ below $\mathfrak{P}$. We illustrate this relationship via the diagram

$$\begin{array}{c} \mathfrak{P} \subset \mathcal{O} \subseteq L \\ \Big| \\ \mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})} \\ \Big| \\ \mathfrak{p} \subset \mathcal{o} \subseteq K. \end{array}$$

The decomposition group encodes how $\mathfrak{p}\mathcal{O}$ splits into distinct prime factors. Indeed, by the orbit-stabilizer theorem the number of cosets in $\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})$ is equal to the size of the orbit of $\mathfrak{P}$ under the action of $\mathrm{Gal}(L/K)$. As the action is transitive by Proposition 1.56, $\sigma(\mathfrak{P})$ runs over the primes above $\mathfrak{p}$ as σ runs over a complete set of representatives for this coset space. Whence

$$\mathfrak{p}\mathcal{O} = \left(\prod_{\sigma} \sigma(\mathfrak{P})\right)^e.$$

Then the order of $\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})$ is exactly the number of prime factors of $\mathfrak{p}\mathcal{O}$. That is,

$$|\,\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})| = r.$$

In particular, $\mathfrak{p}$ is inert in $\mathcal{O}/\mathcal{o}$ if and only if $D_{\mathfrak{p}}(\mathfrak{P}) = \mathrm{Gal}(L/K)$ which is equivalent to $L^{D_{\mathfrak{p}}(\mathfrak{P})} = K$. Antithetically, $\mathfrak{p}$ is totally split in $\mathcal{O}/\mathcal{o}$ if and only if $D_{\mathfrak{p}}(\mathfrak{P}) = \{1\}$ which is equivalent to $L^{D_{\mathfrak{p}}(\mathfrak{P})} = L$. More generally, the previous identity and fundamental equality together imply

$$|D_{\mathfrak{p}}(\mathfrak{P})| = ef. \tag{7}$$

Therefore $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef and $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ has degree r. If $\mathfrak{p}$ inert in $\mathcal{O}/\mathcal{o}$ then $n = ef$ implying $e = 1$ and $f = n$ as we have seen. If $\mathfrak{p}$ is totally split in $\mathcal{O}/\mathcal{o}$ then $1 = ef$ so that $e = f = 1$ as we have also seen. Moreover, the decomposition group of a conjugate prime is the conjugate of decomposition group. More precisely, we have

$$D_{\mathfrak{p}}(\sigma(\mathfrak{P})) = \sigma D_{\mathfrak{p}}(\mathfrak{P})\sigma^{-1}.$$

This is simply because $\tau(\mathfrak{P}) = \mathfrak{P}$ if and only if $\sigma\tau\sigma^{-1}(\sigma(\mathfrak{P})) = \sigma(\mathfrak{P})$. Also, the fundamental theorem of Galois theory implies that $D_{\mathfrak{p}}(\mathfrak{P})$ is normal if and only if $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ is Galois in which case

$$\mathrm{Gal}(L^{D_{\mathfrak{p}}(\mathfrak{P})}/K) \cong \mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P}).$$

Regardless, the decomposition field contains all of the information about the prime factors that $\mathfrak{p}\mathcal{O}$ splits into. In particular, $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$.

Proposition 1.57. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$ with inertia degree f and ramification index e, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$. Moreover,*

$$e_{\mathfrak{P}^D}(\mathfrak{P}) = e \quad \textit{and} \quad f_{\mathfrak{P}^D}(\mathfrak{P}) = f.$$

Proof. Recall $\mathrm{Gal}(L/L^{D_{\mathfrak{p}}(\mathfrak{P})}) = D_{\mathfrak{p}}(\mathfrak{P})$. It follows from Proposition 1.56 that the primes of $\mathcal{O}$ above $\mathfrak{P}^D$ are of the form $\tau(\mathfrak{P})$ for $\tau \in D_{\mathfrak{p}}(\mathfrak{P})$. These τ leave $\mathfrak{P}$ invariant which means $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$. In view of this fact and that $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef, the fundamental equality gives

$$ef = e_{\mathfrak{P}^D}(\mathfrak{P})f_{\mathfrak{P}^D}(\mathfrak{P}).$$

As $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$ is a tower of Dedekind extensions, Equation (6) implies

$$e = e_{\mathfrak{P}^D}(\mathfrak{P})e_{\mathfrak{p}}(\mathfrak{P}^D) \quad \text{and} \quad f = f_{\mathfrak{P}^D}(\mathfrak{P})f_{\mathfrak{p}}(\mathfrak{P}^D).$$

Combining all of these identities results in

$$1 = e_{\mathfrak{p}}(\mathfrak{P}^D)f_{\mathfrak{p}}(\mathfrak{P}^D).$$

Whence $e_{\mathfrak{p}}(\mathfrak{P}^D) = f_{\mathfrak{p}}(\mathfrak{P}^D) = 1$. As $\mathfrak{P}^D$ is above $\mathfrak{p}$, Proposition 1.56 implies that $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$. This proves the first statement. The second statement follows at once from the identities above. $\square$

We can view this result as a statement that $\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}/\mathcal{o}$ encodes all of the splitting information about $\mathfrak{p}\mathcal{O}$ while $\mathcal{O}/\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}$ contains all of the ramification and inertia information. We will need to do more work to unpack the latter information. As $\tau \in D_\mathfrak{p}(\mathfrak{P})$ leaves $\mathcal{O}$ and $\mathfrak{P}$ invariant, it induces an automorphism $\overline{\tau}$ of the residue class field $\mathbb{F}_\mathfrak{P}$ defined by

$$\overline{\tau} : \mathbb{F}_\mathfrak{P} \to \mathbb{F}_\mathfrak{P} \qquad \alpha \mapsto \tau(\alpha) + \mathfrak{P}.$$

We then obtain a homomorphism

$$T_\mathfrak{p}^\mathfrak{P} : D_\mathfrak{p}(\mathfrak{P}) \to \mathrm{Aut}(\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}) \qquad \tau \mapsto \overline{\tau}.$$

It turns out that the residue class extensions $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ are normal and that $T_\mathfrak{p}^\mathfrak{P}$ is surjective.

Proposition 1.58. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then the residue class extension $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ is normal. Moreover, the homomorphism $T_\mathfrak{p}^\mathfrak{P}$ is surjective.*

Proof. As $f_\mathfrak{p}(\mathfrak{P}^D) = 1$ by Proposition 1.57, we have an isomorphism $\mathbb{F}_{\mathfrak{P}^D} \cong \mathbb{F}_\mathfrak{p}$. So without loss of generality, we may assume $L^{D_\mathfrak{p}(\mathfrak{P})} = K$ which is to say $D_\mathfrak{p}(\mathfrak{P}) = \mathrm{Gal}(L/K)$. For any $\alpha \in \mathcal{O}$ let $\overline{\alpha} \in \mathbb{F}_\mathfrak{P}$ be its image under natural projection. Also let $m_\alpha(x)$ and $m_{\overline{\alpha}}(x)$ be the minimal polynomials of α and $\overline{\alpha}$ over K and $\mathbb{F}_\mathfrak{p}$ respectively. As $m_\alpha(x)$ necessarily has coefficients in $\mathcal{o}$ and hence $\mathcal{O}$, let $\overline{m_\alpha}(x) \in \mathbb{F}_\mathfrak{P}[x]$ be its image under natural projection. Note that as $\overline{\alpha}$ is a root of $\overline{m_\alpha}(x)$, $m_{\overline{\alpha}}(x)$ divides $\overline{m_\alpha}(x)$ in $\mathbb{F}_\mathfrak{P}[x]$.

For the first statement, let $\alpha \in \mathcal{O}$. Then we must show that $m_{\overline{\alpha}}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$. As $m_{\overline{\alpha}}(x)$ divides $\overline{m_\alpha}(x)$ in $\mathbb{F}_\mathfrak{P}[x]$, it suffices to show $\overline{m_\alpha}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$. As L/K is Galois and thus normal, $m_\alpha(x)$ splits into linear factors over L. These linear factors belong to $\mathcal{O}[x]$ since their roots are the Galois conjugates of α which belong to $\mathcal{O}$. It follows that $\overline{m_\alpha}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$ proving the first statement.

For the second statement, consider the maximal separable subextension of $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$. As this extension is finite, it is simple by the primitive element theorem so let $\theta \in \mathcal{O}$ be such that $\overline{\theta}$ is a primitive element. Then this subextension is $\mathbb{F}_\mathfrak{p}(\overline{\theta})/\mathbb{F}_\mathfrak{p}$. Now let $\overline{\tau} \in \mathrm{Aut}(\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p})$. Since $\overline{\tau}$ fixes $\mathbb{F}_\mathfrak{p}$ pointwise, $\overline{\tau}(\overline{\theta})$ is a root of $m_{\overline{\theta}}(x)$. As $m_{\overline{\theta}}(x)$ divides $\overline{m_\theta}(x)$ in $\mathbb{F}_\mathfrak{P}[x]$, we find that $\overline{\tau}(\overline{\theta})$ is also a root of $\overline{m_\theta}(x)$. Therefore there is a root θ' of $m_\theta(x)$ whose image under natural projection is $\overline{\tau}(\overline{\theta})$. Because L/K is Galois with Galois group $D_\mathfrak{p}(\mathfrak{P})$, we can find $\tau \in D_\mathfrak{p}(\mathfrak{P})$ such that $\tau(\theta) = \theta'$. Then the image of $\overline{\theta}$ under $T_\mathfrak{p}^\mathfrak{P}(\tau)$ and $\overline{\tau}$ are the same whence these automorphisms are identical on $\mathbb{F}_\mathfrak{p}$. As $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}(\overline{\theta})$ is purely inseparable, it has trivial automorphism group. This forces $T_\mathfrak{p}^\mathfrak{P}(\tau) = \overline{\tau}$ proving surjectivity. $\square$

As this result shows that $T_\mathfrak{p}^\mathfrak{P}$ is surjective, we define the **inertia group** $I_\mathfrak{p}(\mathfrak{P})$ of $\mathfrak{P}$ over $\mathfrak{p}$ by

$$I_\mathfrak{p}(\mathfrak{P}) = \ker T_\mathfrak{p}^\mathfrak{P}.$$

Being the kernel of a homomorphism, $I_\mathfrak{p}(\mathfrak{P})$ is a normal subgroup of $D_\mathfrak{p}(\mathfrak{P})$. The associated **inertia field** $L^{I_\mathfrak{p}(\mathfrak{P})}$ of $\mathfrak{P}$ over $\mathfrak{p}$ is defined to be

$$L^{I_\mathfrak{p}(\mathfrak{P})} = \{\lambda \in L : \tau(\lambda) = \lambda \text{ for } \tau \in I_\mathfrak{p}(\mathfrak{P})\}.$$

In other words, the inertia field of is the fixed field of L by the inertia group and hence an intermediate field of L/K. In particular, the fundamental theorem of Galois theory gives

$$\operatorname{Gal}(L/L^{I_{\mathfrak{p}}(\mathfrak{P})}) = I_{\mathfrak{p}}(\mathfrak{P}),$$

$L^{I_{\mathfrak{p}}(\mathfrak{P})}$ is an intermediate field of the Galois extension $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ as $I_{\mathfrak{p}}(\mathfrak{P})$ is a normal subgroup of $D_{\mathfrak{p}}(\mathfrak{P})$, and

$$\operatorname{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong D_{\mathfrak{p}}(\mathfrak{P})/I_{\mathfrak{p}}(\mathfrak{P}).$$

Let $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ be the integral closure of $\mathcal{o}$ in $L^{I_{\mathfrak{p}}(\mathfrak{P})}$. Then Propositions 1.5 and 1.32 together imply that $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ is a tower of Dedekind extensions for the finite separable tower $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$. We illustrate this relationship via the diagram

$$\begin{array}{c} \mathfrak{P} \subset \mathcal{O} \subseteq L \\ \Big| \\ \mathfrak{P}^I \subset \mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{I_{\mathfrak{p}}(\mathfrak{P})} \\ \Big| \\ \mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})}. \end{array}$$

The inertia field is closely related to the automorphism group of the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$. Indeed, as $T_{\mathfrak{p}}^{\mathfrak{P}}$ is surjective by Proposition 1.58, the first isomorphism theorem shows

$$\operatorname{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong \operatorname{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}).$$

In fact, the decomposition and inertia groups of $\mathfrak{P}$ over $\mathfrak{p}$ fit into a short exact sequence.

Proposition 1.59. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then there is a short exact sequence*

$$1 \longrightarrow I_{\mathfrak{p}}(\mathfrak{P}) \longrightarrow D_{\mathfrak{p}}(\mathfrak{P}) \longrightarrow \operatorname{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the third map is given by $\tau \mapsto T_{\mathfrak{p}}^{\mathfrak{P}}(\tau)$.

Proof. As the second map is injective and the third map is surjective by Proposition 1.58, the sequence is exact at $I_{\mathfrak{p}}(\mathfrak{P})$ and $\operatorname{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}})$. So it remains to prove exactness at $D_{\mathfrak{p}}(\mathfrak{P})$. This follows because $I_{\mathfrak{p}}(\mathfrak{P})$ is the kernel of $T_{\mathfrak{p}}^{\mathfrak{P}}$ by definition. □

We can say more when the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. For in this case, Proposition 1.58 implies $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is Galois. Whence

$$\operatorname{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong \operatorname{Gal}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}).$$

Then Equation (7) and this isomorphism together imply

$$|\operatorname{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})})| = f \quad \text{and} \quad |I_{\mathfrak{p}}(\mathfrak{P})| = e.$$

It follows that $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree f and $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ has degree e as $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef. In particular, $\mathfrak{P}$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$.

Proposition 1.60. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$ with inertia degree f and ramification index e, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Moreover, suppose the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. Then $\mathfrak{P}$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$.*

Proof. Recall that $\mathrm{Gal}(L/L^{I_{\mathfrak{p}}(\mathfrak{P})}) = I_{\mathfrak{p}}(\mathfrak{P})$. By Proposition 1.56, the primes of $\mathcal{O}$ above $\mathfrak{P}^I$ are of the form $\tau(\mathfrak{P})$ for $\tau \in I_{\mathfrak{p}}(\mathfrak{P})$. These τ leave $\mathfrak{P}$ invariant whence $\mathfrak{P}^I$ is nonsplit in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$. In view of this fact and that $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ has degree e, the fundamental equality implies

$$e = e_{\mathfrak{P}^I}(\mathfrak{P}) f_{\mathfrak{P}^I}(\mathfrak{P}).$$

So $\mathfrak{P}^I$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ if $f_{\mathfrak{P}^I}(\mathfrak{P}) = 1$. As $\mathfrak{P}^I$ is nonsplit in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$, we have $D_{\mathfrak{P}^I}(\mathfrak{P}) = I_{\mathfrak{p}}(\mathfrak{P})$. Whence $I_{\mathfrak{P}^I}(\mathfrak{P}) = I_{\mathfrak{p}}(\mathfrak{P})$. In short, the inertia and decomposition groups are the same. By Proposition 1.59, the only way this is possible is if $\mathrm{Gal}(\mathbb{F}_{\mathfrak{P}^I}/\mathbb{F}_{\mathfrak{P}}) = \{1\}$. This forces $f_{\mathfrak{P}^I}(\mathfrak{P}) = 1$ as desired. As $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef and $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ by Proposition 1.57, the fundamental equality implies

$$ef = e_{\mathfrak{P}^D}(\mathfrak{P}) f_{\mathfrak{P}^D}(\mathfrak{P}).$$

Since $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ is a tower of Dedekind extensions, Equation (6) gives

$$e_{\mathfrak{P}^D}(\mathfrak{P}) = e_{\mathfrak{P}^I}(\mathfrak{P}) e_{\mathfrak{P}^D}(\mathfrak{P}^I) \quad \text{and} \quad f_{\mathfrak{P}^D}(\mathfrak{P}) = f_{\mathfrak{P}^I}(\mathfrak{P}) f_{\mathfrak{P}^D}(\mathfrak{P}^I).$$

Upon combining these identities and our previous work, $f_{\mathfrak{P}^D}(\mathfrak{P}^I) = f$ which is the degree of $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$. Therefore $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$. □

In the case the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable, we fit the decomposition and inertia fields into the diagram

$$\begin{array}{c}
\mathfrak{P} \subset \mathcal{O} \subseteq L \\
{\scriptstyle e}\Big|{\scriptstyle 1} \\
\mathfrak{P}^I \subset \mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{I_{\mathfrak{p}}(\mathfrak{P})} \\
{\scriptstyle 1}\Big|{\scriptstyle f} \\
\mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})} \\
{\scriptstyle 1}\Big|{\scriptstyle 1} \\
\mathfrak{p} \subset \mathcal{o} \subseteq K.
\end{array}$$

The labels on left and right of the extensions represent the corresponding ramification indices and inertia degrees respectively which come from Propositions 1.57 and 1.60. So the extension $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ contains all of the information about the splitting, the extension $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ contains all of the information about the inertia degrees, and the extension $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ contains all of the information about the ramification indices. In particular, $\mathfrak{p}$ is unramified in $\mathcal{O}/\mathcal{o}$ if and only if $L^{I_{\mathfrak{p}}(\mathfrak{P})} = L$ which is equivalent to $I_{\mathfrak{p}}(\mathfrak{P}) = \{1\}$.

Number Fields

In the case of a Galois extension of number fields L/K and a prime $\mathfrak{p}$ of K, the **ramification index** e and **inertia degree** f of $\mathfrak{p}$ in L/K is that of $\mathcal{O}_L/\mathcal{O}_K$. Recall that all of the residue class extensions are separable. This means that the above diagram applies to L/K.

1.8 Differents and Discriminants

Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. We being by introducing the notion of lattices in this setting. We say that $\mathfrak{L}$ is an **$\mathcal{o}$-lattice** of L if it is a finitely generated $\mathcal{o}$-submodule of L. Moreover, we say that $\mathfrak{L}$ is **complete** if it spans L/K. That is, $\mathfrak{L}$ contains a basis of L/K. It is also possible to construct duals to complete $\mathcal{o}$-lattices. For this, we need a nondegenerate symmetric bilinear form. The natural choice is the trace form

$$\mathrm{Tr}_{L/K} : L \times L \to K \qquad (\lambda, \eta) \mapsto \mathrm{Tr}_{L/K}(\lambda\eta).$$

Then any basis $\lambda_1, \ldots, \lambda_n$ for L/K admits a dual basis $\lambda_1^\vee, \ldots, \lambda_n^\vee$ defined by

$$\mathrm{Tr}_{L/K}(\lambda_i \lambda_j^\vee) = \delta_{i,j},$$

If $\mathfrak{L}$ is complete, then we define the **dual** $\mathfrak{L}^\vee$ of $\mathfrak{L}$ by

$$\mathfrak{L}^\vee = \{\lambda \in L : \mathrm{Tr}_{L/K}(\lambda\mathfrak{L}) \subseteq \mathcal{o}\}.$$

We say that $\mathfrak{L}$ is **self-dual** if $\mathfrak{L}^\vee = \mathfrak{L}$. The dual $\mathfrak{L}^\vee$ is indeed a complete $\mathcal{o}$-lattice of L. To see this, let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathfrak{L}$. For any $\lambda \in L$ we may write

$$\lambda = \sum_j \kappa_j^\vee \lambda_j^\vee,$$

with $\kappa_j^\vee \in K$. As the dual basis is determined by $\mathrm{Tr}_{L/K}(\lambda_i \lambda_j^\vee) = \delta_{i,j}$, it follows that $\lambda \in \mathfrak{L}^\vee$ if and only if $\kappa_j^\vee \in \mathcal{o}$. Whence $\lambda_1^\vee, \ldots, \lambda_n^\vee$ is a basis for $\mathfrak{L}^\vee$ as an $\mathcal{o}$-submodule of L and thus $\mathfrak{L}^\vee$ is a complete $\mathcal{o}$-lattice of L. As the dual of the dual basis is the original basis, $(\mathfrak{L}^\vee)^\vee = \mathfrak{L}$. The prototypical complete $\mathcal{o}$-lattice of L is $\mathcal{O}$ whence it admits a dual $\mathcal{O}^\vee$.

Observe that an $\mathcal{o}$-lattice of L is not necessarily a fractional ideal of $\mathcal{O}$ as it need not be an $\mathcal{O}$-submodule of L. However, every fractional ideal $\mathfrak{F}$ of $\mathcal{O}$ is a complete $\mathcal{o}$-lattice of L. Indeed, $\mathfrak{F}$ is a finitely generated $\mathcal{o}$-submodule of L as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. Moreover, $\mathfrak{F}$ contains a basis of L/K. To see this, let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$ and let $\alpha \in \mathfrak{F}$ be nonzero. Then $\alpha\lambda_1, \ldots, \alpha\lambda_n$ is a basis for L/K contained in $\mathfrak{F}$. As $\mathfrak{F}$ is complete, it admits a dual $\mathfrak{F}^\vee$. The dual $\mathfrak{F}^\vee$ is also a fractional ideal of $\mathcal{O}$. To see that it is an $\mathcal{O}$-submodule of L, we have $\mathrm{Tr}_{L/K}(\mathcal{O}\mathfrak{F}) \subseteq \mathcal{o}$ by Proposition 1.8 upon writing $\mathfrak{F} = \frac{1}{\delta}\mathfrak{A}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{A}$. Whence $\mathfrak{F}^\vee$ is a finitely generated $\mathcal{O}$-submodule of L as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. In fact, there is a simple relationship between the duals $\mathfrak{F}^\vee$ and $\mathcal{O}^\vee$ given by

$$\mathfrak{F}^\vee = \mathfrak{F}^{-1}\mathcal{O}^\vee.$$

For the forward inclusion, we have $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F}) \subseteq \mathcal{o}$. Whence $\mathfrak{F}^\vee\mathfrak{F} \subseteq \mathcal{O}^\vee$ which is equivalent to $\mathfrak{F}^\vee \subseteq \mathfrak{F}^{-1}\mathcal{O}^\vee$. This proves the forward inclusion. For the reverse inclusion, observe that $\mathrm{Tr}_{L/K}(\mathcal{O}^\vee) \subseteq \mathcal{o}$. Thus $\mathfrak{F}^{-1}\mathcal{O}^\vee \subseteq \mathfrak{F}^\vee$ proving the reverse inclusion.

Also, we might hope that localization respects duals of fractional ideals in order to make their study more approachable. This turns out to be true and is a very useful property.

Proposition 1.61. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then for any fractional ideal $\mathfrak{F}$ of $\mathcal{O}$, we have*

$$(\mathfrak{F}D^{-1})^\vee = \mathfrak{F}^\vee D^{-1}.$$

Proof. As D is a subset of K, we have

$$\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee \mathfrak{F} D^{-1}) = \mathrm{Tr}_{L/K}(\mathfrak{F}^\vee \mathfrak{F})D^{-1}.$$

It follows that $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee \mathfrak{F} D^{-1}) \subseteq \mathcal{o}D^{-1}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee \mathfrak{F})D^{-1} \subseteq \mathcal{o}D^{-1}$. The forward and reverse inclusions both follows from this which proves the claim. □

In a similar spirit to Proposition 1.35, this result says that dualization and localization are commutative. So when dualizing and localizing we do not need to specify the order in which these operations are performed.

We now turn to the construction of two specific integral ideals which encode many properties of fractional ideals, ramification of primes, and $\mathcal{o}$-lattices of L. In the study of these integral ideals, it will be revealed that it is an exceptionally rare phenomena for primes of $\mathcal{O}$ and $\mathcal{o}$ to ramify. The two integral ideals we will construct, one of $\mathcal{O}$ and the other of $\mathcal{o}$, will encode the ramification of the corresponding primes. These integral ideals are the different and discriminant respectively. We being by defining the **complement** $\mathfrak{C}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ by

$$\mathfrak{C}_{\mathcal{O}/\mathcal{o}} = \mathcal{O}^\vee.$$

As we have seen, this is a fractional ideal of $\mathcal{O}$ and hence a complete $\mathcal{o}$-lattice of L. The **different** $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ is defined to be

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = (\mathcal{O}^\vee)^{-1}.$$

In other words, the different is the inverse of the complement and so

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = \{\lambda \in L : \lambda\mathcal{O}^\vee \subseteq \mathcal{O}\}.$$

As $\mathcal{O} \subseteq \mathcal{O}^\vee$ by Proposition 1.8, inverting shows $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathcal{O}$. Therefore the different $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ is an integral ideal of $\mathcal{O}$ and we have a chain of inclusions

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathcal{O} \subseteq \mathcal{O}^\vee.$$

From this chain we see that the different is a measure of how much $\mathcal{O}$ fails to be self-dual since $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = \mathcal{O}^\vee$ if and only if $\mathcal{O}$ is self-dual. In terms of the different, the dual $\mathfrak{F}^\vee$ of a fractional ideal $\mathfrak{F}$ of $\mathcal{O}$ may be expressed as

$$\mathfrak{F}^\vee = \mathfrak{F}^{-1}\mathfrak{D}_{\mathcal{O}/\mathcal{o}}^{-1}. \tag{8}$$

As for other properties of the different, it respects localization which will be useful in further developments.

Proposition 1.62. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{D}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}} = \mathfrak{D}_{\mathcal{O}/\mathcal{o}}D^{-1}.$$

Proof. By definition of the different, it is equivalent to show

$$((\mathcal{O}D^{-1})^{\vee})^{-1} = (\mathcal{O}^{\vee})^{-1}D^{-1}.$$

This identity follows from Propositions 1.35 and 1.61. □

We define the **discriminant** $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ to be the integral ideal of $\mathcal{o}$ generated by the discriminants $d_{L/K}(\lambda_1, \dots, \lambda_n)$ as $\lambda_1, \dots, \lambda_n$ run over all bases of L/K contained in $\mathcal{O}$. Recall that if $\mathcal{O}/\mathcal{o}$ admits an integral basis then $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. In this case, bases of L/K contained in $\mathcal{O}$ are precisely the integral bases of $\mathcal{O}/\mathcal{o}$. Then

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}\mathcal{o},$$

so that the discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ is generated by the discriminant $d_{\mathcal{O}/\mathcal{o}}$. In fact, this means

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}(\alpha_1, \dots, \alpha_n)\mathcal{o},$$

for any integral basis $\alpha_1, \dots, \alpha_n$ for $\mathcal{O}/\mathcal{o}$. Whence $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ is principal if $\mathcal{O}/\mathcal{o}$ admits an integral basis. In view of Theorem 1.14 this will hold if $\mathcal{o}$ is a principal ideal domain.

As for properties of the discriminant, we have the very useful fact that it respects localization just as the different does.

Proposition 1.63. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{d}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}} = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}.$$

Proof. For the forward inclusion, suppose $\frac{\lambda_1}{\delta_1}, \dots, \frac{\lambda_n}{\delta_n}$ is a basis of L/K contained in $\mathcal{O}D^{-1}$. Upon setting $\delta = \prod_i \delta_i$, we see that $\frac{\lambda_1\delta}{\delta_1}, \dots, \frac{\lambda_n\delta}{\delta_n}$ is a basis of L/K contained in $\mathcal{O}$. Then $d_{L/K}\left(\frac{\lambda_1\delta}{\delta_1}, \dots, \frac{\lambda_n\delta}{\delta_n}\right) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ whence $d_{L/K}\left(\frac{\lambda_1}{\delta_1}, \dots, \frac{\lambda_n}{\delta_n}\right) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}$. This shows the forward inclusion. For the reverse inclusion, suppose $\frac{\lambda_1}{\delta}, \dots, \frac{\lambda_n}{\delta}$ is such that $\lambda_1, \dots, \lambda_n$ is a basis of L/K contained in $\mathcal{O}$. Then $\frac{\lambda_1}{\delta}, \dots, \frac{\lambda_n}{\delta}$ is a basis for L/K contained in $\mathcal{O}D^{-1}$. So $d_{L/K}\left(\frac{\alpha_1}{\delta}, \dots, \frac{\alpha_n}{\delta}\right) \in \mathfrak{d}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}}$ whence $d_{L/K}(\lambda_1, \dots, \lambda_n) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}$. This proves the reverse inclusion. □

The different and discriminant are important integral ideals because they keep track of which primes ramify. In fact, we will show that primes of $\mathcal{O}$ and $\mathcal{o}$ ramify in $\mathcal{O}/\mathcal{o}$ if and only they divide the different $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ and discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ respectively provided the residue class extensions are separable. To this end, we will require a lemma.

Lemma 1.64. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let $\mathfrak{A}$ be an integral ideal of $\mathcal{O}$. Then $\mathfrak{A} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{A}^{-1}) \subseteq \mathcal{o}$.*

Proof. Since to divide is to contain, $\mathfrak{A} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathfrak{A}$. This equivalent to $(\mathcal{O}^\vee)^{-1} \subseteq \mathfrak{A}$ and inverting shows the further equivalence $\mathfrak{A}^{-1} \subseteq \mathcal{O}^\vee$. This last inclusion is equivalent to $\mathrm{Tr}_{L/K}(\mathfrak{A}^{-1}) \subseteq \mathcal{o}$. $\square$

With this result, we can now show that a prime of $\mathcal{O}$ is ramified in $\mathcal{O}/\mathcal{o}$ if and only if it divides the discriminant $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ provided the residue class extensions are separable. In fact, we will prove something slightly stronger.

Theorem 1.65. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and assume that all residue class extensions are separable. Let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. Then $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. Moreover, $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$. In particular, $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$.*

Proof. In view of Propositions 1.34, 1.52 and 1.63, we may assume that $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. Let e be a nonnegative integer with $e \leq e_\mathfrak{p}(\mathfrak{P})$ and write $\mathfrak{p}\mathcal{O} = \mathfrak{P}^e\mathfrak{A}$ for some integral ideal $\mathfrak{A}$ of $\mathcal{O}$. By Lemma 1.64, $\mathfrak{P}^e \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{P}^e) \subseteq \mathcal{o}$. As $\mathfrak{P}^e = \mathfrak{p}^{-1}\mathfrak{A}$, it is further equivalent to show $\mathrm{Tr}_{L/K}(\mathfrak{A}) \subseteq \mathfrak{p}$. Since $\mathcal{O}/\mathcal{o}$ admits an integral basis, we have $\mathrm{Tr}_{K/L}(\mathfrak{A}) = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\mathfrak{A})$ from Proposition 1.16. Whence it is further equivalent to show $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\mathfrak{A}) \subseteq \mathfrak{p}$ which is to say

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_\mathfrak{p}}(\overline{\alpha}) = 0,$$

for all $\alpha \in \mathfrak{A}$ where $\overline{\alpha} \in \mathfrak{A}/\mathfrak{p}\mathcal{O}$ is its image under natural projection.

To prove $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$, take $e = e_\mathfrak{p}(\mathfrak{P}) - 1$ so that $\mathfrak{P} \mid\mid \mathfrak{A}$. Then the prime factors of $\mathfrak{A}$ and $\mathfrak{p}\mathcal{O}$ are the same. Whence there is a positive integer k such that $\mathfrak{A}^k$ is divisible by $\mathfrak{p}\mathcal{O}$. So $\mathfrak{p}\mathcal{O} \mid \mathfrak{A}^k$ and therefore $\mathfrak{A}^k \subseteq \mathfrak{p}\mathcal{O}$ as to divide is to contain. It follows that the k-th power of the $\mathbb{F}_\mathfrak{p}$-linear operator $T_{\overline{\alpha}}$ on $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is the zero operator. This means $T_{\overline{\alpha}}$ is nilpotent. As nilpotent operators have trace zero, we have $\mathrm{Tr}_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_\mathfrak{p}}(\overline{\alpha}) = 0$. This proves the first statement.

To prove $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$, take $e = e_\mathfrak{p}(\mathfrak{P})$ so that $\mathfrak{P}$ and $\mathfrak{A}$ are relatively prime. The Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong (\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}) \oplus (\mathcal{O}/\mathfrak{A}),$$

induced by natural projection. So $\mathfrak{A}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ and it is further equivalent to show

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})})/\mathbb{F}_\mathfrak{p}}(\overline{\beta}) = 0,$$

for all $\beta \in \mathcal{O}$ where $\overline{\beta} \in \mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ is its image under natural projection. We will prove that this occurs if and only if $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$. To this end, consider the filtration

$$\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset \mathfrak{P}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset \cdots \supset \mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \supset 0,$$

of $\mathbb{F}_\mathfrak{p}$-vector spaces. Upon choosing a basis for $\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ adapted to the filtration, using Lemma 1.51 gives an isomorphism

$$\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P}_i)} \cong \bigoplus_i \mathfrak{P}^i/\mathfrak{P}^{i+1},$$

induced by natural projection of the adapted basis. Now let $\overline{\beta}_i \in \mathfrak{P}^i/\mathfrak{P}^{i+1}$ be the images of β under the aforementioned isomorphism. Then

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})})/\mathbb{F}_\mathfrak{p}}(\overline{\beta}) = \sum_i \mathrm{Tr}_{(\mathfrak{P}^i/\mathfrak{P}^{i+1})/\mathbb{F}_\mathfrak{p}}(\overline{\beta}_i).$$

The quotients $\mathfrak{P}^i/\mathfrak{P}^{i+1}$ are all isomorphic to $\mathbb{F}_\mathfrak{P}$ by Lemma 1.29 which is induced by multiplication and natural projection. In particular, this isomorphism is an $\mathbb{F}_\mathfrak{p}$-linear operator induced by multiplication and therefore the $\mathbb{F}_\mathfrak{p}$-linear operators $T_{\overline{\beta}_i}$ and $T_{\overline{\beta}_0}$ on $\mathfrak{P}^i/\mathfrak{P}^{i+1}$ and $\mathcal{O}/\mathfrak{P}$ respectively are conjugate via this isomorphism. As the trace is invariant under conjugation, we have

$$\mathrm{Tr}_{(\mathfrak{P}^i/\mathfrak{P}^{i+1})/\mathbb{F}_\mathfrak{p}}(\overline{\beta}_i) = \mathrm{Tr}_{\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}}(\overline{\beta}_0),$$

which implies

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})})/\mathbb{F}_\mathfrak{p}}(\overline{\beta}) = e_\mathfrak{p}(\mathfrak{P})\,\mathrm{Tr}_{\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}}(\overline{\beta}_0).$$

Since the residue class extension $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ is assumed to be separable, the trace form of $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ is nondegenerate. As both sides are elements of $\mathbb{F}_\mathfrak{p}$, the left-hand side is zero for all $\beta \in \mathcal{O}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ proving the second statement.

It remains to show $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. As all residue class extensions are separable, $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \geq 2$. But the previous two statements together imply that $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \geq 2$. Whence $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. □

In the case the reside class extensions are separable, note that this result implies $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1}$ exactly divides $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ provided $e_\mathfrak{p}(\mathfrak{P}) \not\equiv 0 \pmod{\mathfrak{p}}$. However, when $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ we only know $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})}$ divides $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. So $\mathfrak{P}$ very well might divide $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ to a higher power. The condition $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ is equivalent to the characteristic of $\mathbb{F}_\mathfrak{p}$ dividing $e_\mathfrak{p}(\mathfrak{P})$. Accordingly, if $\mathfrak{P}$ is a ramified prime of $\mathcal{O}$ then $\mathfrak{P}$ is said to be **tamely ramified** if the characteristic of $\mathbb{F}_\mathfrak{p}$ does not divide $e_\mathfrak{p}(\mathfrak{P})$ and **widely ramified** otherwise. In any case, upon factorizing $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ into a product of primes we see that only finitely many primes of $\mathcal{O}$ can ramify in $\mathcal{O}/\mathcal{o}$.

Similarly to the previous result, a prime $\mathfrak{p}$ of $\mathcal{o}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if it divides the discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ provided the residue class extensions are separable.

Theorem 1.66. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and assume that all residue class extensions are separable. Then a prime $\mathfrak{p}$ of $\mathcal{o}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{p} \mid \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$.*

Proof. In view of Propositions 1.34, 1.52 and 1.63, it suffices to assume $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. As $\mathcal{o}$ is a principal ideal domain, we have

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}\mathcal{o}.$$

Therefore $\mathfrak{p} \mid \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ if and only if $d_{\mathcal{O}/\mathcal{o}} \subset \mathfrak{p}$ which is to say

$$d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_\mathfrak{p}} = 0.$$

Now let

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_\mathfrak{p}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_\mathfrak{p}(\mathfrak{P}_r)},$$

be the prime factorization of $\mathfrak{p}\mathcal{O}$. The Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_i \mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)},$$

and Proposition 1.10 further implies

$$d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_\mathfrak{p}} = \prod_i d_{(\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)})/\mathbb{F}_\mathfrak{p}}.$$

Therefore $d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_\mathfrak{p}} = 0$ if and only if $d_{(\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)})/\mathbb{F}_\mathfrak{p}} = 0$ for some prime $\mathfrak{P}_i$. We will prove that this occurs if and only if $e_\mathfrak{p}(\mathfrak{P}_i) \geq 2$ which will complete the proof because this is exactly when $\mathfrak{p}$ ramifies since all residue class extensions are separable by assumption.

For the forward implication, we will prove the contrapositive so suppose $e_\mathfrak{p}(\mathfrak{P}_i) = 1$. As the residue class extension $\mathbb{F}_{\mathfrak{P}_i}/\mathbb{F}_\mathfrak{p}$ is separable by assumption, Proposition 1.11 implies $d_{\mathbb{F}_{\mathfrak{P}_i}/\mathbb{F}_\mathfrak{p}} \neq 0$. For the reverse implication, suppose $e_\mathfrak{p}(\mathfrak{P}_i) \geq 2$. By prime factorization, there exists $\beta_1 \in \mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)-1} - \mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)}$ and let $\overline{\beta_1} \in \mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)}$ be its image under natural projection. Then $\overline{\beta_1} \neq 0$ and $\overline{\beta_1}^2 = 0$ by construction. Since $\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)}$ is an $\mathbb{F}_\mathfrak{p}$-vector space of dimension $e_\mathfrak{p}(\mathfrak{P}_i)f_\mathfrak{p}(\mathfrak{P}_i)$ by Lemma 1.51, there exists a basis of the form $\overline{\beta_1}, \ldots, \overline{\beta_{e_\mathfrak{p}(\mathfrak{P}_i)f_\mathfrak{p}(\mathfrak{P}_i)}}$. Now $T^2_{\overline{\beta_1\beta_j}}$ is the zero operator by our choice of β_1. Whence $T_{\overline{\beta_1\beta_j}}$ is nilpotent. As nilpotent operators have trace zero, we have

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)})/\mathbb{F}_\mathfrak{p}}(\overline{\beta_1\beta_j}) = 0.$$

This means that the first row of $\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)})/\mathbb{F}_\mathfrak{p}}(\overline{\beta_1}, \ldots, \overline{\beta_{e_\mathfrak{p}(\mathfrak{P}_i)f_\mathfrak{p}(\mathfrak{P}_i)}})$ is zero. It follows that $d_{(\mathcal{O}/\mathfrak{P}_i^{e_\mathfrak{p}(\mathfrak{P}_i)})/\mathbb{F}_\mathfrak{p}} = 0$ completing the proof. □

In the case the reside class extensions are separable, factorizing $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ into a product of primes shows that only finitely many primes of $\mathcal{o}$ can ramify in $\mathcal{O}/\mathcal{o}$.

Given that the prime factors of the different and discriminant correspond to the ramification of primes, provided the residue class extensions are separable, one may suspect that they are related via the ideal norm. This turns out to be true regardless of the separability of the reside class extensions.

Proposition 1.67. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Then*

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}}) = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}.$$

Proof. In view of Propositions 1.53, 1.62 and 1.63, it suffices to assume $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. Then $\mathcal{O}/\mathcal{o}$ admits an integral basis $\alpha_1, \ldots, \alpha_n$ and

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{L/K}(\alpha_1, \ldots, \alpha_n)\mathcal{o},$$

in view of Proposition 1.16. Moreover, $\alpha_1, \ldots, \alpha_n$ is a basis for $\mathcal{O}$ as an $\mathcal{o}$-submodule of L and so the dual basis $\alpha_1^\vee, \ldots, \alpha_n^\vee$ is a basis for $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}^{-1}$ as an $\mathcal{o}$-submodule of L. Since $\mathcal{O}$ is a principal ideal domain, $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}^{-1} = \frac{1}{\delta}\mathcal{O}$ for some nonzero $\delta \in \mathcal{O}$. Whence $\frac{\alpha_1}{\delta}, \ldots, \frac{\alpha_n}{\delta}$ is also a basis for $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}^{-1}$ as an $\mathcal{o}$-submodule of L. Since these are bases for L/K, Equation (1) implies

$$d_{L/K}\left(\frac{\alpha_1}{\delta}, \ldots, \frac{\alpha_n}{\delta}\right) = \mathrm{N}_{L/K}(\delta)^{-2} d_{L/K}(\alpha_1, \ldots, \alpha_n),$$

and

$$d_{L/K}\left(\frac{\alpha_1}{\delta}, \ldots, \frac{\alpha_n}{\delta}\right) = \varepsilon^2 d_{L/K}(\alpha_1^\vee, \ldots, \alpha_n^\vee),$$

for some $\varepsilon \in \mathcal{o}^*$. In view of Propositions 1.7 and 1.12, we have

$$d_{L/K}(\alpha_1^\vee, \ldots, \alpha_n^\vee) d_{L/K}(\alpha_1, \ldots, \alpha_n) = 1.$$

Combining these identities results in

$$\varepsilon^2 \, \mathrm{N}_{L/K}(\delta)^2 = d_{L/K}(\alpha_1, \ldots, \alpha_n)^2.$$

But Proposition 1.54 implies $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}}) = \mathrm{N}_{L/K}(\delta)\mathcal{o}$. Therefore passing to integral ideals gives

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}})^2 = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}^2.$$

The claim now follows from prime factorization. $\square$

Number Fields

We now turn to the setting where L/K is an extension of number fields. Recall that the underlying Dedekind extension is $\mathcal{O}_L/\mathcal{O}_K$. The **complement** $\mathfrak{C}_{L/K}$ of L/K is the complement of $\mathcal{O}_L/\mathcal{O}_K$, the **different** $\mathfrak{D}_{L/K}$ of L/K is the different of $\mathcal{O}_L/\mathcal{O}_K$, and the **discriminant** $\mathfrak{d}_{L/K}$ of L/K is the discriminant of $\mathcal{O}_L/\mathcal{O}_K$.

For a number field K, we can say more as $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis and the residue class extensions are separable. Indeed, since $\mathbb{Z}$ is a principal ideal domain the discriminant $\mathfrak{d}_K$ is related to the discriminant Δ_K by

$$\mathfrak{d}_K = \Delta_K \mathcal{O}_K.$$

As all of the residue class extensions are separable, it follows from Theorems 1.65 and 1.66 that a prime of K ramifies in K if and only if it divides $\mathfrak{D}_K$ and a prime of $\mathbb{Q}$ ramifies in K if and only if it divides $|\Delta_K|$. In particular, finitely many primes of K and $\mathbb{Q}$ ramify in K. Also, Proposition 1.67 immediately implies

$$\mathrm{N}_K(\mathfrak{D}_K) = |\Delta_K|.$$

Then for any fractional ideal $\mathfrak{f}$ of $\mathcal{O}_K$, we find that

$$\mathrm{N}_K(\mathfrak{f}^\vee) = \frac{\mathrm{N}_K(\mathfrak{f}^{-1})}{|\Delta_K|},$$

from Equation (8).

Lastly, let $a_K(m)$ denote the number of integral ideals of $\mathcal{O}_K$ of norm m. As the norm is completely multiplicative, prime factorization implies that $a_K(m)$ is multiplicative. In fact, we can estimate the growth of $a_K(m)$.

Proposition 1.68. *Let K be a number field of degree n. Then $a_K(m) \leq \sigma_0(m)^n$.*

Proof. Let $\mathfrak{a}$ be an integral ideal of norm m. First suppose $m = p^k$ is a prime power. By the fundamental equality, there are at most n primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ above p. Let $f_p(\mathfrak{p}_1), \ldots, f_p(\mathfrak{p}_n)$ be the corresponding inertia degrees. Then

$$\mathrm{N}_K(\mathfrak{a}) = p^{e_1 f_p(\mathfrak{p}_1)} \cdots p^{e_n f_p(\mathfrak{p}_n)},$$

for integers e_i with $0 \leq e_i \leq k$. The number of such choices is equivalent to the number of solutions

$$e_1 f_p(\mathfrak{p}_1) + \cdots + e_n f_p(\mathfrak{p}_n) = k,$$

which is at most $\sigma_0(p^k)^n$. This proves the claim in the case m is a prime power. The claim follows by multiplicativity of $a_K(m)$ and the divisor function. □

As a consequence of this result, we have the slightly weaker estimate $a_K(n) \ll_\varepsilon n^\varepsilon$.

2 Geometry of Number Fields

Throughout, all rings will be understood to be commutative and with unity.

2.1 Integral Lattices

Let F be a characteristic zero field and V be an $\mathbb{F}$-vector space of dimension n with nondegenerate symmetric inner product $\langle \cdot, \cdot \rangle$. We say that a subset Λ of V is a **integral lattice** if Λ is a free $\mathbb{Z}$-module. In particular, any integral lattice Λ is of the form

$$\Lambda = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_m,$$

for some linearly independent vectors $v_1, \ldots, v_m$ of V with $m \leq n$. We say Λ is **complete** if $n = m$. This means $v_1, \ldots, v_n$ is necessarily a basis of V. If $e_1, \ldots, e_n$ is an orthonormal basis for V, write

$$v_i = \sum_i v_{i,j} e_j,$$

with $v_{i,j} \in F$, and define the associated **generator matrix** P by

$$P = \begin{pmatrix} v_{1,1} & \cdots & v_{n,1} \\ \vdots & & \vdots \\ v_{1,n} & \cdots & v_{n,n} \end{pmatrix}.$$

Then P is the base change matrix from $e_1, \ldots, e_n$ to $v_1, \ldots, v_n$. The **covolume** V_Λ of a complete integral lattice Λ is defined to be

$$V_\Lambda = |\det(P)|.$$

As the base change matrix between any two orthonormal basses is an orthogonal matrix and hence has determinant ± 1, the covolume is independent of the choice of bases. In the case where V is a real or complex vector space, we will always take $e_1, \ldots, e_n$ to be the standard basis.

If Λ is a complete integral lattice in V then the **dual** $\Lambda^\vee$ of Λ is defined by

$$\Lambda^\vee = \{v \in V : \langle \lambda, v \rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}.$$

In other words, the dual integral lattice consists of all of the vectors in V whose inner product with elements of the integral lattice are integers. Clearly the dual integral lattice is an abelian group. Less clear is that the dual integral lattice turns out to always be complete.

Proposition 2.1. *If $v_1, \ldots, v_n$ is a basis for a complete integral lattice Λ then the dual basis $v_1^\vee, \ldots, v_n^\vee$ is a basis for $\Lambda^\vee$. In particular, $\Lambda^\vee$ is a complete integral lattice.*

Proof. Let $v \in V$ and write

$$v = \sum_j a_i v_j^\vee,$$

with $v_j^\vee \in \mathbb{R}$. Since the dual basis is determined by $\langle v_i, v_j^\vee \rangle = \delta_{i,j}$, it follows that $v \in \Lambda^\vee$ if and only if $v_j^\vee \in \mathbb{Z}$. This means $v_1^\vee, \ldots, v_n^\vee$ is a basis for $\Lambda^\vee$ as a free $\mathbb{Z}$-module and thus is a complete integral lattice. □

Since the dual of the dual basis is the original basis, $(\Lambda^\vee)^\vee = \Lambda$. We say that Λ is **self-dual** if $\Lambda^\vee = \Lambda$. For example, the integral lattice $\mathbb{Z}^n$ is self-dual because the standard basis is self-dual. It also turns out that the covolume of the dual integral lattice is the inverse of the volume of the original integral lattice.

Proposition 2.2. *Let Λ be a complete integral lattice in V. Then*

$$V_{\Lambda^\vee} = \frac{1}{V_\Lambda}.$$

Proof. Let $e_1, \ldots, e_n$ be an orthonormal basis for V, let $v_1, \ldots, v_n$ be a basis for Λ, and let P be the associated generator matrix. By Proposition 2.1, $v_1^\vee, \ldots, v_n^\vee$ is a basis for $\Lambda^\vee$ and $(P^{-1})^t$ is the base change matrix from $e_1, \ldots, e_n$ to $v_1^\vee, \ldots, v_n^\vee$. As $\det((P^{-1})^t) = \det(P)^{-1}$ the claim follows. □

We now turn to the case when V is a real inner product space of dimension n. Let $d\lambda$ be the Lebesgue measure induced by the inner product with associated volume Vol given by

$$\mathrm{Vol}(M) = \int_M d\lambda,$$

for any measurable set M. Note that Λ acts on V by automorphisms given by translation. In other words, we have a group action

$$\Lambda \times V \to V \qquad (\lambda, v) \mapsto \lambda + v.$$

Moreover, Λ acts properly discontinuously on V. To see this, let $v \in V$ and let δ_v be such that $0 < \delta_v < \min_i(v - v_i)$. Taking U_v to be the ball of radius δ_v about v, the intersection

$\lambda+U_v\cap U_v$ is empty unless $\lambda=0$. As Λ is also discrete, it follows that V/Λ is also connected Hausdorff. Whence V/Λ admits a fundamental domain

$$\mathcal{M}=\{t_1v_1+\cdots+t_nv_n\in V:0\le t_i\le 1 \text{ for all } i\}.$$

Moreover, any translation of $\mathcal{M}$ by an element of Λ is also a fundamental domain. As we might expect, the covolume of Λ is equal to the volume of $\mathcal{M}$.

Proposition 2.3. *Let Λ be a complete integral lattice in a finite dimensional real inner product space V and let $\mathcal{M}$ be a fundamental domain for Λ. Then*

$$V_\Lambda=\mathrm{Vol}(\mathcal{M}).$$

Proof. Let $e_1,\ldots,e_n$ be an orthonormal basis of V, and $v_1,\ldots,v_n$ be a basis for Λ. Also let P be the associated generator matrix. The change of variables

$$x_1e_1+\cdots+x_ne_n\mapsto x_1v_1+\cdots+x_nv_n,$$

has Jacobian determinant V_Λ. Whence

$$\mathrm{Vol}(\mathcal{M})=\int_{\mathcal{M}}d\lambda=V_\Lambda\int_{[0,1]^n}d\mathbf{x}=V_\Lambda.$$

□

This result shows that the covolume of a complete integral lattice in a real inner product space is a measure of the density of the integral lattice. The smaller the covolume the smaller the fundamental domain and the more dense the integral lattice is.

It turns out that for a real inner product space, being an integral lattice is equivalent to being a discrete subgroup.

Proposition 2.4. *Let Λ be a subset of a finite dimensional real inner product space V. Then Λ is an integral lattice if and only if it is a discrete subgroup.*

Proof. It is clear that if Λ is an integral lattice then it is a discrete subgroup. So suppose Λ is a discrete subgroup. Then Λ is closed. Let V' be the subspace spanned by Λ and let its dimension be m. Choosing a basis $v_1',\ldots,v_m'$ of V', set

$$\Lambda'=\mathbb{Z}v_1'+\cdots+\mathbb{Z}v_m'.$$

Then Λ' is a complete integral lattice in V' and $\Lambda'\subseteq\Lambda\subset V$. We claim Λ/Λ' is finite. To see this, let λ be a representative of a coset in Λ/Λ' and let $\mathcal{M}'$ be a fundamental domain for Λ' in V'. As $\mathcal{M}'$ is a fundamental domain, there exists unique $w'\in\mathcal{M}'$ and $\lambda'\in\Lambda'$ such that $\lambda=w'+\lambda'$. But then $w'=\lambda-\lambda'\in\mathcal{M}'\cap\Lambda$ and since $\mathcal{M}'\cap\Lambda$ is closed, discrete, and compact, it must be finite. Hence there are finitely many w' and thus finitely many cosets in Λ/Λ'. Letting $|\Lambda/\Lambda'|=q$, we have $q\Lambda\subseteq\Lambda'$ and therefore

$$\Lambda\subseteq\Lambda'=\mathbb{Z}\frac{1}{q}v_1'+\cdots+\mathbb{Z}\frac{1}{q}v_m'.$$

In particular, Λ is a subgroup of a free $\mathbb{Z}$-module and therefore is a free $\mathbb{Z}$-module itself. This means Λ is an integral lattice. □

As for complete integral lattices in V, they are equivalent to the existence of a bounded set whose translates cover V.

Proposition 2.5. *Let Λ be an integral lattice in a finite dimensional real inner product space V. Then Λ complete if and only if there exists a bounded subset M of V whose translates by Λ cover V.*

Proof. First suppose Λ is complete. Then we may take $M = \mathcal{M}$ to be a fundamental domain of Λ which is bounded and whose translates by Λ cover V. Now suppose Λ is an integral lattice and there exists a bounded subset M of V whose translates by Λ cover V. Let W be the subspace of V spanned by Λ. Then W is closed. Moreover, Λ is complete if and only if $V = W$ and this is what we will show. So let $v \in V$. Since the translates of M by Λ cover V, for every positive integer n we may write

$$nv = w_n + \lambda_n,$$

with $v_n \in M$ and $\lambda_n \in \Lambda$. As M is bounded, $\lim_{n\to\infty} \frac{1}{n} v_n = 0$. Moreover, $\frac{1}{n}\lambda_n \in W$ and since W is closed we must have that $\lim_{n\to\infty} \frac{1}{n}\lambda_n$ exists and belongs to W. These two limits together imply

$$v = \lim_{n\to\infty} \frac{1}{n}\lambda_n,$$

and is an element of W. Thus $V = W$ which means Λ is complete. $\square$

The most important result we will require about integral lattices is **Minkowski's lattice point theorem** which states that, under some mild conditions, a set of sufficiently large volume in V contains a nonzero point of a complete integral lattice.

Theorem (Minkowski's lattice point theorem). *Suppose Λ is a complete integral lattice in a real inner product space V of dimension n. Let $X \subset V$ is a compact convex symmetric set. If*

$$\mathrm{Vol}(X) \geq 2^n V_\Lambda,$$

then there exists a nonzero $\lambda \in \Lambda \cap X$.

Proof. We will prove the claim depending on if the inequality is strict or not. First suppose $\mathrm{Vol}(X) > 2^n V_\Lambda$. Consider the linear map

$$\phi : \frac{1}{2}X \to V/\Lambda \qquad \frac{1}{2}x \mapsto \frac{1}{2}x + \Lambda.$$

If ϕ were injective then we would have

$$\mathrm{Vol}\left(\frac{1}{2}X\right) = \frac{1}{2^n}\mathrm{Vol}(X) \leq V_\Lambda,$$

so that $\mathrm{Vol}(X) \leq 2^n V_\Lambda$. This is a contradiction, so ϕ cannot be injective. Hence there exists distinct $x_1, x_2 \in \frac{1}{2}X$ such that $\phi(x_1) = \phi(x_2)$. Thus $2x_1, 2x_2 \in X$. In particular, since X is symmetric we must have $-2x_2 \in X$. Convexity of X implies

$$\left(1 - \frac{1}{2}\right)2x_1 + \frac{1}{2}(-2x_2) = x_1 - x_2,$$

is an element of X. But $x_1 - x_2 \in \Lambda$ because $\phi(x_1) = \phi(x_2)$ and ϕ is linear. Then $\lambda = x_1 - x_2$ is nonzero with $\lambda \in \Lambda \cap X$. Now suppose $\mathrm{Vol}(X) = 2^n V_\Lambda$. Then

$$\mathrm{Vol}((1+\varepsilon)X) = (1+\varepsilon)^n 2^n V_\Lambda > 2^n V_\Lambda.$$

What we have just proved shows that there exists a nonzero $\lambda_\varepsilon \in \Lambda \cap (1+\varepsilon)X$. In particular, if $\varepsilon \leq 1$ then $\lambda_\varepsilon \in 2X \cap \Lambda$. The set $2X \cap \Lambda$ is compact and discrete and therefore finite. Taking $\varepsilon = \frac{1}{n}$, the sequence $(\lambda_{\frac{1}{n}})_{n \geq 1}$ belongs to the finite set $2X \cap \Lambda$ and so must converge to a point λ. Since Λ is discrete and the $\lambda_{\frac{1}{n}}$ are nonzero so too is λ. As λ is an element of

$$\bigcap_{n \geq 1} \left(1 + \frac{1}{n}\right) X,$$

and X is closed, $\lambda \in X$ as well. Thus we have found a nonzero $\lambda \in \Lambda \cap X$. $\square$

2.2 Minkowski Space

Let K be number field of degree n. If $\sigma \in \mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ then either σ is real or complex and in the latter case it is has a paired $\mathbb{Q}$-embedding $\overline{\sigma}$ which is the complex conjugate of σ. Accordingly, let r_1 and $2r_2$ be the number of real and complex $\mathbb{Q}$-embeddings of K into $\overline{\mathbb{Q}}$ respectively. We call the pair (r_1, r_2) the **signature** of K and it satisfies the relation

$$n = r_1 + 2r_2.$$

Setting

$$K_{\mathbb{C}} = \mathbb{C}^n,$$

makes $K_{\mathbb{C}}$ into a component-wise $\mathbb{C}$-algebra. Moreover, $K_{\mathbb{C}}$ is also a complex Hilbert space with respect to the standard complex inner product which we denote by $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$. We also write $d\lambda_{K_{\mathbb{C}}}$ for the induced Lebesgue measure on $K_{\mathbb{C}}$ by this inner product. Also let $\mathrm{Tr}_{K_{\mathbb{C}}}$ and $\mathrm{N}_{K_{\mathbb{C}}}$ denote the trace and norm of $K_{\mathbb{C}}/\mathbb{C}$. Because $K_{\mathbb{C}}$ is a component-wise $\mathbb{C}$-algebra, the trace and the norm are given by

$$\mathrm{Tr}_{K_{\mathbb{C}}}(\mathbf{z}) = \sum_{\sigma} z_\sigma \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{C}}}(\mathbf{z}) = \prod_{\sigma} z_\sigma,$$

for all $\mathbf{z} \in K_{\mathbb{R}}$. We define the **canonical embedding** j of K to be the $\mathbb{Q}$-embedding

$$j : K \to K_{\mathbb{C}} \qquad \kappa \mapsto (\sigma(\kappa))_\sigma.$$

In other words, the canonical embedding is a component-wise $\mathbb{Q}$-embedding of K into $K_{\mathbb{C}}$ via the σ. In view of Proposition 1.7, we have

$$\mathrm{Tr}_{K_{\mathbb{C}}}(j(\kappa)) = \mathrm{Tr}_K(\kappa) \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{C}}}(j(\kappa)) = \mathrm{N}_K(\kappa),$$

which is to say that the compositions of the trace and norm of $K_{\mathbb{C}}/\mathbb{C}$ with the canonical embedding produces the field trace and field norm. Our primary aim is to construct an $\mathbb{R}$-subalgebra of $K_{\mathbb{C}}$ which corrects for the fact that the complex $\mathbb{Q}$-embeddings come in pairs. To this end, consider the complex conjugation map

$$F : \mathbb{C} \to \mathbb{C} \qquad z \mapsto \overline{z}.$$

This induces an automorphism of $K_{\mathbb{C}}$ given by

$$F : K_{\mathbb{C}} \to K_{\mathbb{C}} \qquad \mathbf{z} \mapsto (\overline{z_{\overline{\sigma}}})_{\sigma},$$

which is an involution. A short computation shows

$$\langle F(\mathbf{z}), F(\mathbf{w}) \rangle_{K_{\mathbb{C}}} = F(\langle \mathbf{z}, \mathbf{w} \rangle_{K_{\mathbb{C}}}),$$

for all $\mathbf{z}, \mathbf{w} \in K_{\mathbb{C}}$. In other words, inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$ is F-equivariant. We define the **Minkowski space** $K_{\mathbb{R}}$ of K by

$$K_{\mathbb{R}} = \{\mathbf{z} \in K_{\mathbb{C}} : F(\mathbf{z}) = \mathbf{z}\}.$$

In other words, $K_{\mathbb{R}}$ consists of all of the F-invariant elements of $K_{\mathbb{C}}$. Note that $K_{\mathbb{R}}$ is an $\mathbb{R}$-subalgebra of $K_{\mathbb{C}}$. We denote the restriction of the inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$ to $K_{\mathbb{R}}$ by $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$. Moreover, the inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$ turns $K_{\mathbb{R}}$ into a real Hilbert space. Indeed, the conjugate symmetry and F-equivariance of the inner product together give

$$\overline{\langle \mathbf{z}, \mathbf{w} \rangle} = \langle \mathbf{z}, \mathbf{w} \rangle,$$

for any $\mathbf{z}, \mathbf{w} \in K_{\mathbb{R}}$. Accordingly, we call $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$ the **Minkowski inner product** of K. We denote the restriction of the Lebesgue measure $d\lambda_{\mathbb{C}}$ to $K_{\mathbb{R}}$ by $d\lambda_{K_{\mathbb{R}}}$ which is also the Lebesgue measure associated to $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$. We call $d\lambda_{K_{\mathbb{R}}}$ the **Minkowski measure**. Lastly, we denote the restrictions of $\mathrm{Tr}_{K_{\mathbb{C}}}$ and $\mathrm{N}_{K_{\mathbb{C}}}$ to $K_{\mathbb{R}}$ by $\mathrm{Tr}_{K_{\mathbb{R}}}$ and $\mathrm{N}_{K_{\mathbb{R}}}$ respectively which are the norm and trace of $K_{\mathbb{R}}/\mathbb{R}$. We call $\mathrm{Tr}_{K_{\mathbb{R}}}$ and $\mathrm{N}_{K_{\mathbb{R}}}$ the **Minkowski trace** and **Minkowski norm** of K respectively. As the complex $\mathbb{Q}$-embeddings of K into $\overline{\mathbb{Q}}$ come in conjugate pairs, $j(K) \subset K_{\mathbb{R}}$ so that the canonical embedding is a $\mathbb{Q}$-embedding of K into Minkowski space. Whence

$$\mathrm{Tr}_{K_{\mathbb{R}}}(j(\kappa)) = \mathrm{Tr}_K(\kappa) \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{R}}}(j(\kappa)) = \mathrm{N}_K(\kappa).$$

So the compositions of the Minkowski trace and Minkowski norm with the canonical embedding produces the field trace and field norm.

To give a more explicit description of Minkowski space, let $N_\sigma = 1, 2$ according to if σ is real or complex respectively. Also let ρ and τ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Note that $N_\rho = 1$ and $N_\tau = 2$. Then

$$K_{\mathbb{R}} = \{\mathbf{z} \in K_{\mathbb{C}} : \mathrm{Im}(z_\rho) = 0 \text{ and } z_{\overline{\tau}} = \overline{z_\tau} \text{ for all } \rho \text{ and } \tau\}.$$

In fact, this gives an explicit isomorphism between Minkowski space and $\mathbb{R}^n$.

Proposition 2.6. *Let K be a number field of degree n and signature (r_1, r_2). Then the map*

$$z_\sigma \mapsto x_\sigma = \begin{cases} z_\sigma & \text{if } \sigma = \rho, \\ \mathrm{Re}(z_\sigma) & \text{if } \sigma = \tau, \\ \mathrm{Im}(z_\sigma) & \text{if } \sigma = \overline{\tau}, \end{cases}$$

induces a component-wise isomorphism between $K_{\mathbb{R}}$ and $\mathbb{R}^n$ making $K_{\mathbb{R}}$ a real vector space of dimension n. Moreover, the inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$ induced by the Minkowski inner product is given by

$$\langle \mathbf{x}, \mathbf{x}' \rangle = \sum_{\sigma} N_\sigma x_\sigma x'_\sigma,$$

for any $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$.

Proof. This map has inverse

$$x_\sigma \mapsto z_\sigma = \begin{cases} x_\sigma & \text{if } \sigma = \rho, \\ x_\sigma + i x_{\overline{\sigma}} & \text{if } \sigma = \tau, \\ x_{\overline{\sigma}} + i x_\sigma & \text{if } \sigma = \overline{\tau}. \end{cases}$$

As both maps are $\mathbb{R}$-linear, it follows that the original map induces a component-wise isomorphism between $K_{\mathbb{R}}$ and $\mathbb{R}^n$ proving the first statement. For the second statement, let $\mathbf{z}, \mathbf{z}' \in K_{\mathbb{R}}$ and let $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$ be the corresponding elements in $\mathbb{R}^n$ under this isomorphism respectively. If $\sigma = \rho$ then

$$x_\rho = z_\rho \quad \text{and} \quad x'_\rho = z'_\rho,$$

and thus

$$x_\rho x'_\rho = z_\rho \overline{z'_\rho}.$$

If $\sigma = \tau$ then

$$x_\tau = \mathrm{Re}(z_\tau), \quad x_{\overline{\tau}} = \mathrm{Im}(z_\tau), \quad x'_\tau = \mathrm{Re}(z'_\tau), \quad \text{and} \quad x'_{\overline{\tau}} = \mathrm{Im}(z'_\tau).$$

Whence a short computation shows

$$2(x_\tau x'_\tau + x_{\overline{\tau}} x'_{\overline{\tau}}) = z_\tau \overline{z'_\tau} + z_{\overline{\tau}} \overline{z'_{\overline{\tau}}}.$$

The claim about the inner product follows. □

In view of this result, we define the **Minkowski embedding** σ_K of K by

$$\sigma_K : K \to \mathbb{R}^n \qquad \kappa \mapsto ((\rho(\kappa))_\rho, (\mathrm{Re}(\tau(\kappa)), \mathrm{Im}(\tau(\kappa)))_\tau).$$

Note that the Minkowski embedding is a $\mathbb{Q}$-embedding of K into $\mathbb{R}^n$ as it is the composition of the canonical embedding and the isomorphism established in the previous result. Also, the Minkowski embedding is independent of the representatives τ because they occur in conjugate pairs.

We have now constructed Minkowski space and determined its dimension as a real vector space. Our goal now is to pass from $\mathbb{Z}$-lattices of K to integral lattices of Minkowski space. Being an inner product space, this will permit us to measure fractional ideals of K geometrically via their images inside Minkowski space. As j is a $\mathbb{Q}$-embedding and any fractional ideal $\mathfrak{f}$ of K is a complete $\mathbb{Z}$-lattice of K, $j(\mathfrak{f})$ is a complete integral lattice of $K_{\mathbb{R}}$. The covolume $V_{j(\mathfrak{f})}$ of $j(\mathfrak{f})$ is easily determined.

Proposition 2.7. *Let K be a number field with signature (r_1, r_2) and let $\mathfrak{f}$ be a fractional ideal of K. Then $V_{j(\mathfrak{f})}$ is the absolute value of the determinant of any embedding matrix of a basis for $K/\mathbb{Q}$ contained in $\mathfrak{f}$. In particular,*

$$V_{j(\mathfrak{f})} = \mathrm{N}_K(\mathfrak{f})\sqrt{|\Delta_K|},$$

and

$$V_{j(\mathcal{O}_K)} = \sqrt{|\Delta_K|}.$$

Proof. Upon writing $\mathfrak{f} = \frac{1}{d}\mathfrak{a}$ for some nonzero $d \in \mathbb{Z}$ and integral ideal $\mathfrak{a}$ and Proposition 1.55, it suffices to prove the statements for the integral ideal $\mathfrak{a}$. Let $\kappa_1, \ldots, \kappa_n$ be a basis for $K/\mathbb{Q}$ contained in $\mathfrak{a}$. Then the associated generator matrix for $j(\mathfrak{a})$ is the embedding matrix $M(\kappa_1, \ldots, \kappa_n)$. Hence

$$V_{j(\mathfrak{a})} = |\det(M(\kappa_1, \ldots, \kappa_n))|,$$

proving the first statement. To prove the first identity of the second statement, in view of Equation (3) it suffices to show

$$|\det(M(\kappa_1, \ldots, \kappa_n))| = \mathrm{N}_K(\mathfrak{a})|\det(M(\alpha_1, \ldots, \alpha_n))|,$$

for any integral basis $\alpha_1, \ldots, \alpha_n$ for $K/\mathbb{Q}$. Note that $\kappa_1, \ldots, \kappa_n$ is necessarily a basis for $\mathfrak{a}/\mathbb{Z}$ by Theorem 1.14 and write

$$\kappa_i = \sum_j a_{i,j}\alpha_j,$$

with $a_{i,j} \in \mathbb{Z}$. Then $(a_{i,j})$ is the base change matrix from $\alpha_1, \ldots, \alpha_n$ to $\kappa_1, \ldots, \kappa_n$. Whence the image of $\mathcal{O}_K$ under this matrix is $\mathfrak{a}$. As $\mathbb{Z}$ is a principal ideal domain, the smith normal form of this matrix implies that

$$\mathrm{N}_K(\mathfrak{a}) = |\det((a_{i,j}))|.$$

This fact together with Equation (1) and Proposition 1.12 gives

$$|\det(M(\kappa_1, \ldots, \kappa_n))| = \mathrm{N}_K(\mathfrak{a})|\det(M(\alpha_1, \ldots, \alpha_n))|,$$

as claimed. The second identity follows from the first upon taking $\mathfrak{f} = \mathcal{O}_K$. □

As σ_K is also a $\mathbb{Q}$-embedding, $\sigma_K(\mathfrak{f})$ is a complete integral lattice of $\mathbb{R}^n$. As a corollary of the previous result, we can determine the covolume of $V_{\sigma_K(\mathfrak{f})}$ of $\sigma_K(\mathfrak{f})$ as well.

Corollary 2.8. *Let K be a number field with signature (r_1, r_2) and let $\mathfrak{f}$ be a fractional ideal of K. Then*

$$V_{\sigma_K(\mathfrak{f})} = \mathrm{N}_K(\mathfrak{f})\frac{\sqrt{|\Delta_K|}}{2^{r_2}},$$

and

$$V_{\sigma_K(\mathcal{O}_K)} = \frac{\sqrt{|\Delta_K|}}{2^{r_2}}.$$

Proof. Let $\kappa_1, \ldots, \kappa_n$ be a basis for $K/\mathbb{Q}$ contained in $\mathfrak{f}$. Also let $\rho_1, \ldots, \rho_{r_1}$ and $\tau_1, \ldots, \tau_{r_2}$ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Then the associated generator matrix for $\sigma_K(\mathfrak{f})$ is

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \mathrm{Re}(\tau_1(\kappa_1)) & \mathrm{Im}(\tau_1(\kappa_1)) & \cdots & \mathrm{Re}(\tau_{r_2}(\kappa_1)) & \mathrm{Im}(\tau_{r_2}(\kappa_1)) \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \mathrm{Re}(\tau_1(\kappa_n)) & \mathrm{Im}(\tau_1(\kappa_n)) & \cdots & \mathrm{Re}(\tau_{r_2}(\kappa_n)) & \mathrm{Im}(\tau_{r_2}(\kappa_n)) \end{pmatrix}^t.$$

By Proposition 2.7 we are done if the absolute value of the determinant of this matrix is $\frac{1}{2^{r_2}}$ times the absolute value of the determinate of the embedding matrix for $\kappa_1, \ldots, \kappa_n$. To show this, first add an i multiple of the imaginary columns to their corresponding real columns and then apply the identity $\mathrm{Im}(z) = \frac{z - \overline{z}}{2i}$ to the imaginary columns. We obtain the matrix

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \tau_1(\kappa_1) & \frac{\tau_1(\kappa_1) - \overline{\tau_1}(\kappa_1)}{2i} & \cdots & \tau_{r_2}(\kappa_1) & \frac{\tau_{r_2}(\kappa_1) - \overline{\tau_{r_2}}(\kappa_1)}{2i} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \tau_1(\kappa_n) & \frac{\tau_1(\kappa_n) - \overline{\tau_1}(\kappa_n)}{2i} & \cdots & \tau_{r_2}(\kappa_n) & \frac{\tau_{r_2}(\kappa_n) - \overline{\tau_{r_2}}(\kappa_n)}{2i} \end{pmatrix}^t,$$

which has the same determinant as the generator matrix. Now multiply the imaginary columns by $-2i$ and then adding the preceding columns to annihilate the negative terms. This results in the matrix

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \tau_1(\kappa_1) & \overline{\tau_1}(\kappa_1) & \cdots & \tau_{r_2}(\kappa_1) & \overline{\tau_{r_2}}(\kappa_1) \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \tau_1(\kappa_n) & \overline{\tau_1}(\kappa_n) & \cdots & \tau_{r_2}(\kappa_n) & \overline{\tau_{r_2}}(\kappa_n) \end{pmatrix}^t,$$

whose determinant is $(-2i)^{-r_2}$ that of the generator matrix. But this latter matrix is the embedding $M(\kappa_1, \ldots, \kappa_n)$. Whence

$$V_{\sigma_K(\mathfrak{f})} = \frac{|\det(M(\kappa_1, \ldots, \kappa_n))|}{2^{r_2}}.$$

□

2.3 Finiteness of the Class Number

Recall that the class number of a Dedekind domain is a measure of how much the Dedekind domain fails to be a principal ideal domain and even need not be finite from Remark 1.23. However, we will show that the class number h_K for a number field K is always finite and thus $\mathcal{O}_K$ only has finite failure from being a principal ideal domain. In doing so, we will also show that every ideal class is represented by an integral ideal of bounded norm.

Theorem 2.9. *Let K be a number field of degree n and signature (r_1, r_2). Also, let $X \subseteq \mathbb{R}^n$ be a compact convex symmetric set and set $M = \max_{\mathbf{x} \in X} \left(\prod_i |x_i|\right)$. Then every ideal class of* $\mathrm{Cl}(K)$ *is represented by an integral ideal $\mathfrak{a}$ satisfying*

$$\mathrm{N}_K(\mathfrak{a}) \leq \frac{2^{r_1 + r_2} M}{\mathrm{Vol}(X)} \sqrt{|\Delta_K|}.$$

Moreover, the ideal class group $\mathrm{Cl}(K)$ *is finite which is to say that the class number h_K is.*

Proof. Let $\mathfrak{f}$ be a fractional ideal, n be a positive integer, and set

$$\lambda^n = 2^n \frac{V_{\sigma_K(\mathfrak{f}^{-1})}}{\mathrm{Vol}(X)}.$$

Then by construction, we have

$$\mathrm{Vol}(\lambda X) = 2^n V_{\sigma_K(\mathfrak{f}^{-1})}.$$

By Minkowski's lattice point theorem, there exists a nonzero element of $\sigma_K(\mathfrak{f}^{-1}) \cap \lambda X$. Whence there is a nonzero $\alpha \in \mathfrak{f}^{-1}$ such that $\sigma_K(\alpha) \in \lambda X$. Moreover, $\alpha\mathfrak{f} \subseteq \mathcal{O}_K$ so that $\mathfrak{a} = \alpha\mathfrak{f}$ is an integral ideal in the same class as $\mathfrak{f}$. Using Propositions 1.7 and 1.55, a short computation shows

$$\mathrm{N}_K(\mathfrak{a}) \leq \lambda^n M \,\mathrm{N}_K(\mathfrak{f}).$$

In view of our choice of λ^n and Corollary 2.8, this inequality becomes

$$\mathrm{N}_K(\mathfrak{a}) \leq \frac{2^{r_1+r_2} M}{\mathrm{Vol}(X)} \sqrt{|\Delta_K|},$$

which proves the first statement.

We now prove that the class group is finite. By the first statement, we can find a complete set of representatives for $\mathrm{Cl}(K)$ consisting of integral ideals of bounded norm. Since the norm is multiplicative, the prime factors of these representatives have bounded norm as well. Moreover, the norm of a prime $\mathfrak{p}$ is $p^{f_p(\mathfrak{p})}$ where p is the prime below $\mathfrak{p}$. Whence there are only finitely many possible primes p. As there are finitely many primes $\mathfrak{p}$ above any prime p, it follows that these representatives have finitely many prime factors. Altogether this means that there are finitely many representatives. Hence $\mathrm{Cl}(K)$ is finite and so the class number h_K is too. $\square$

Observe that the bound in the previous result is not explicit. To obtain an explicit bound, we need to make a choice for the compact convex symmetric set X. In order for such a bound to be sharp, we need to ensure that the volume of X is large while the constant M is small. The following lemma dictates our choice of X and computes its volume:

Lemma 2.10. *Suppose n is a positive integer and write $n = r_1 + 2r_2$ for some nonnegative integers r_1 and r_2. Let $X \subset \mathbb{R}^n$ be the compact convex symmetric set given by*

$$X = \left\{ (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^n : \sum_i |x_{i,1}| + 2 \sum_j \sqrt{x_{2j-1,2}^2 + x_{2j,2}^2} \leq n \right\}.$$

Then

$$\mathrm{Vol}(X) = \frac{n^n}{n!} 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2}.$$

Proof. Making the change of variables $x_{2j-1,2} \mapsto r_j \sin(\theta_j)$ and $x_{2j,2} \mapsto r_j \cos(\theta_j)$ gives

$$\mathrm{Vol}(X) = \int_{X'} r_1 \cdots r_{r_2} \, d\mathbf{x}_1 \, d\mathbf{r} \, d\boldsymbol{\theta},$$

because the Jacobian determinant is $r_1 \cdots r_{r_2}$, and where

$$X' = \left\{ (\mathbf{x}_1, \mathbf{r}, \boldsymbol{\theta}) \in [-n,n]^{r_1} \oplus [0,n]^{r_2} \oplus [0,2\pi]^{r_2} : \sum_i |x_i| + 2\sum_j r_j \le n \right\}.$$

Making the change of variables $r_j \mapsto \frac{r_j}{2}$ and using the facts that the integrand is symmetric in the x_i and independent of θ_j gives

$$\mathrm{Vol}(X) = 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2} \int_{X''} r_1 \cdots r_{r_2} \, d\mathbf{x}_1 \, d\mathbf{r},$$

where

$$X'' = \left\{ (\mathbf{x}_1, \mathbf{r}) \in [0,n]^{r_1+r_2} : \sum_i x_i + 2\sum_j r_j \le n \right\}.$$

To compute the remaining integral, for nonnegative integers ℓ and k, let

$$X''_{\ell,k}(t) = \left\{ (\mathbf{x}_1, \mathbf{r}) \in [0,t]^{\ell+k} : \sum_i x_i + 2\sum_j r_j \le t \right\},$$

for $t > 0$ and define

$$I_{\ell,k}(t) = \int_{X''_{\ell,k}(t)} r_1 \cdots r_\ell \, d\mathbf{x}_1 \, d\mathbf{r}.$$

In this notation, we have

$$\mathrm{Vol}(X) = 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2} I_{r_1,r_2}(n).$$

To compute the remaining integral, the change of variables $x_i \mapsto tx_i$ and $r_j \mapsto tr_j$ gives

$$I_{\ell,k}(t) = t^{\ell+2k} I_{\ell,k}(1).$$

Whence

$$I_{\ell,k}(1) = \int_0^1 (1-x_\ell)^{\ell-1+2k} I_{\ell-1,k}(1) \, dx_\ell = \frac{1}{\ell+2k} I_{\ell-1,k}(1).$$

Upon repeating this procedure $\ell - 1$ times results in

$$I_{\ell,k}(1) = \frac{1}{(\ell+2k)\cdots(2k+1)} I_{0,k}(1).$$

Similarly, we have

$$I_{0,k}(1) = \int_0^1 r_k (1-r_k)^{2k-2} I_{0,k-1}(1) \, dr_k = \frac{1}{2k} I_{0,k-1}(1),$$

because the latter integral is the beta function $B(1, 2k-1)$. Upon repeating this procedure $k-1$ times results in

$$I_{0,k}(1) = \frac{1}{k!}.$$

Altogether, this means

$$I_{\ell,k}(t) = \frac{t^{\ell+2k}}{(\ell+2k)!},$$

and therefore

$$\mathrm{Vol}(X) = \frac{n^n}{n!} 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2}. \qquad \square$$

Observe that the compact convex symmetric set X consists of those elements in $\mathbb{R}^n$ whose norm corresponding to the inner product induced by the Minkowski inner product from Proposition 2.6 at most n. We can now obtain an explicit bound known as the **Minkowski bound**.

Theorem (Minkowski bound). *Let K be a number field of degree n and signature (r_1, r_2). Then every ideal class of $\mathrm{Cl}(K)$ is represented by an integral ideal $\mathfrak{a}$ satisfying*

$$\mathrm{N}_K(\mathfrak{a}) \leq \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|\Delta_K|}.$$

Proof. Theorem 2.9 and Lemma 2.10 together show that every ideal class of $\mathrm{Cl}(K)$ is represented by an integral ideal $\mathfrak{a}$ satisfying

$$\mathrm{N}_K(\mathfrak{a}) \leq M \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|\Delta_K|},$$

where $M = \max_{\mathbf{x} \in X} \left(\prod_i |x|\right)$. But for all $\mathbf{x} \in X$, the arithmetic-geometric mean inequality implies

$$\left(\prod_i |x_i|\right)^{\frac{1}{n}} \leq 1,$$

Whence $M \leq 1$ completing the proof. $\square$

As a corollary we can obtain a lower bound for the discriminant of a number field and show that every number field K, other than $\mathbb{Q}$, has at least one ramified prime.

Corollary 2.11. *Let K be a number field of degree n. Then*

$$|\Delta_K| \geq \left(\frac{\pi}{4}\right)^{\frac{n}{2}} \frac{n^n}{n!}.$$

In particular, there is at least one ramified prime of K provided the degree of K is at least 2.

Proof. The inequality follows from the Minkowski bound since the norm of every integral ideal is at least 1 and r_2 is at most n. This proves the first statement. For the second statement, suppose $n \geq 2$. In the case $n = 2$, the lower bound is larger than 1 so that $|\Delta_K|$ is at least 2. Whence $|\Delta_K|$ has a prime divisor and so some prime above this divisor is a ramified prime of K. Now suppose $n \geq 3$. As $n^n \geq n!$, $\left(\frac{\pi}{4}\right)^{\frac{n}{2}} \frac{n^n}{n!}$ is an increasing function in n. Therefore $|\Delta_K| \geq 2$ and, as before, there is at least one ramified prime of K. $\square$

Generally speaking, the class number of K is one of the most difficult pieces of arithmetic data about K to compute. For example, it is still unknown if there are infinitely many number fields of class number 1. That is to say, it is unknown if there are infinitely many number fields such that their ring of integers are principal ideal domains.

Let S be a finite set of primes of K. The S-class number $h_{K,S}$ is also finite. In this setting, the long exact sequence in Proposition 1.39 takes the form

$$1 \longrightarrow \mathcal{O}_K^* \longrightarrow \mathcal{O}_{K,S}^* \longrightarrow \mathbb{Z}^{|S|} \longrightarrow \mathrm{Cl}(K) \longrightarrow \mathrm{Cl}_S(K) \longrightarrow 1,$$

implying

$$\mathrm{Cl}_S(K) \cong \mathrm{Cl}(K)/\langle \mathfrak{p}P(K) : \mathfrak{p} \in S \rangle.$$

Whence

$$h_{K,S} = \frac{h_K}{|\langle \mathfrak{p}P(K) : \mathfrak{p} \in S \rangle|},$$

which shows that the S-class number of K is finite and generally smaller than the class number of K. In fact, finiteness implies that it is possible to choose S large enough (consisting of all the prime factors for a complete set of integral ideal representatives) such that $h_{K,S} = 1$. This means we can always find a finite set of primes S such that the ring of S-integers of K is a principal ideal domain.

Let $\mathcal{o}$ be an order of K. Then the class number $h_{\mathcal{o}}$ is finite as well. In this setting and using Proposition 1.50, the long exact sequence in Proposition 1.48 takes the form

$$1 \longrightarrow \mathcal{o}^* \longrightarrow \mathcal{O}_K^* \longrightarrow (\mathcal{O}_K/\mathfrak{q}_{\mathcal{o}})^*/(\mathcal{o}/\mathfrak{q}_{\mathcal{o}})^* \longrightarrow \mathrm{Pic}(\mathcal{o}) \longrightarrow \mathrm{Cl}(K) \longrightarrow 1.$$

By Proposition 1.55, the quotient $\mathcal{O}_K/\mathfrak{q}_{\mathcal{o}}$ is finite whence $(\mathcal{O}_K/\mathfrak{q}_{\mathcal{o}})^*/(\mathcal{o}/\mathfrak{q}_{\mathcal{o}})^*$ is as well. Exactness then implies $\mathcal{O}_K^*/\mathcal{o}^*$ is also finite and that

$$h_{\mathcal{o}} = \frac{h_K}{|\mathcal{O}_K^*/\mathcal{o}^*|} \frac{|(\mathcal{O}_K/\mathfrak{q}_{\mathcal{o}})^*|}{|(\mathcal{o}/\mathfrak{q}_{\mathcal{o}})^*|},$$

so $h_{\mathcal{o}}$ is finite too. This identity also shows that the class number of an order of K is generally larger than the class number of K.

2.4 Dirichlet's Unit Theorem

Let K be a number field of degree n and signature (r_1, r_2). We define the **rank** r_K of K by

$$r_K = r_1 + r_2 - 1.$$

The rank of K is intimately related to the structure of the unit group $\mathcal{O}_K^*$. Observe that we can express the unit group as

$$\mathcal{O}_K^* = \{\alpha \in \mathcal{O}_K : \mathrm{N}_K(\alpha) = \pm 1\}.$$

Our main goal will be to describe the group structure of $\mathcal{O}_K^*$ completely. Let $\mu(K)$ be the subgroup of K^* consisting of all the roots of unity in K. As any m-th root of unity is a root of $x^m - 1$, it follows that roots of unity are algebraic integers. Whence $\mu(K)$ is a subgroup of $\mathcal{O}_K^*$. Clearly $\{\pm 1\} \subseteq \mu(K)$. In fact, $\mu(K)$ always is finite. To see this, let $\omega \in \mu(K)$ and recall that any root of unity is a primitive root of unity of some, potentially smaller, order. So we may assume ω is a primitive m-th root of unity. The minimal polynomial of ω over $\mathbb{Q}$ is the m-th cyclotomic polynomial $\Phi_m(x)$ which has degree $\varphi(m)$. Whence $\varphi(m) \le n$. Since $\varphi(m) \to \infty$ as $m \to \infty$, there can only be finitely many such ω. This means $\mu(K)$ is finite. Accordingly, set

$$w_K = |\mu(K)|.$$

Our primary aim will be to show that $\mathcal{O}_K^*$ is a direct product of $\mu(K)$ and a free Z-module of rank r_K. Determining that the rank of the free Z-module is exactly r_K will be the most difficult part of the argument. To accomplish this, we will require a map on K^* that transitions between the Minkowski trace and Minkowski norm. To define this map, let ρ and τ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Consider the homomorphism

$$\ell : K_{\mathbb{R}}^* \to \mathbb{R}^{r_K+1} \qquad (z_\sigma)_\sigma \mapsto ((\log|z_\rho|)_\rho, (2\log|z_\tau|)_\tau).$$

This map is independent of the choice of representatives τ by definition of Minkowski space. Then the Minkowski trace and Minkowski norm and related via the identity

$$\mathrm{Tr}_{K_{\mathbb{R}}}(\ell(\mathbf{z})) = \log|\,\mathrm{N}_{K_{\mathbb{R}}}(\mathbf{z})|,$$

for all $\mathbf{z} \in K_{\mathbb{R}}$. We define the **logarithmic embedding** $\log_K$ of K to be the homomorphism given by

$$\log_K : K^* \to \mathbb{R}^{r_K+1} \qquad \kappa \mapsto ((\log|\rho(\kappa)|)_\rho, (2\log|\rho(\tau)|)_\tau).$$

Then $\log_K$ is just the restriction of j to K^* composed with ℓ. As such, it is also independent of the representatives τ. We distinguish the subsets of $\mathbb{R}^{r_K+1}$ given by

$$S = \{\mathbf{x} \in \mathbb{R}_+^{r_K+1} : \mathrm{N}_{\mathbb{R}^{r_K+1}/\mathbb{R}}(\mathbf{x}) = 1\} \quad \text{and} \quad H = \{\mathbf{x} \in \mathbb{R}^{r_K+1} : \mathrm{Tr}_{\mathbb{R}^{r_K+1}/\mathbb{R}}(\mathbf{x}) = 0\}.$$

We call S and H the **norm-one hypersurface** and **trace-zero hyperplane** of $\mathbb{R}^{r_K+1}$ respectively. Note that H is a subspace of $\mathbb{R}^{r_K+1}$ of dimension r_K. We will also make use of the subset

$$U = \{\mathbf{z} \in K_{\mathbb{R}}^* : \mathrm{N}_{K_{\mathbb{R}}}(\mathbf{z}) = \pm 1\}.$$

Observe j maps $\mathcal{O}_K^*$ into U and ℓ maps U into H. Let λ denote the restriction of $\log_K$ to $\mathcal{O}_K^*$ and set

$$\Lambda_K = \log_K(\mathcal{O}_K^*),$$

so that Λ_K is the image of λ. In particular, $\Lambda_K \subset H$. We call Λ_K the **unit lattice** of K. It is not immediately obvious that Λ_K is an integral lattice, but we will show this and more. We first show that the unit group and unit lattice of K fit into a short exact sequence.

Proposition 2.12. *Let K be a number field. Then there is a short exact sequence*

$$1 \longrightarrow \mu(K) \longrightarrow \mathcal{O}_K^* \longrightarrow \Lambda_K \longrightarrow 1,$$

induced by natural inclusions and where the third map is given by λ.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $\mu(K)$ and Λ_K. So it remains to prove exactness at $\mathcal{O}_K^*$. For this we must show $\ker\lambda = \mu(K)$. The reverse inclusion holds since the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ preserve roots of unity. For the forward inclusion, the elements of $j(\ker\lambda)$ are such that each component is absolutely bounded. So one the one hand, this set belongs to a compact subset of $K_{\mathbb{R}}$. On the other hand, this set is also a subset of the integral lattice $j(\mathcal{O}_K)$ which is a discrete subset of $K_{\mathbb{R}}$. Together, this means $j(\ker\lambda)$ is a subset of a compact and discrete set which is necessarily finite. As j is a $\mathbb{Q}$-embedding, it follows that the subgroup $\ker\lambda$ of $\mathcal{O}_K^*$ contains finitely many elements and hence only roots of unity since these are the only elements of $\mathbb{C}$ with finite order. This proves the forward inclusion. □

Our aim now is to show that the unit lattice Λ_K is a a free Z-module of rank r_K. For this, we will require a lemma.

Lemma 2.13. *Let* K *be a number field. There are finitely many elements in* $\mathcal{O}_K$ *of a given field norm up to multiplication by units.*

Proof. The claim is true for algebraic integers of norm ± 1 as these are exactly the units of K. Now suppose $m \geq 2$. By Proposition 1.55, $\mathcal{O}_K/m\mathcal{O}_K$ is finite with $\mathrm{N}_K(m)$ many elements. Therefore, it suffices to show that in each coset there is at most finitely many element of norm n up to multiplication by units. In fact, we will show that there is at most one such element. So let α and β be two representatives in the same coset which are of field norm n. Writing $\alpha = \beta + n\gamma$ for some $\gamma \in \mathcal{O}_K$, we may further write

$$\frac{\alpha}{\beta} = 1 + \frac{\mathrm{N}_K(\beta)}{\beta}\gamma.$$

As any integral ideal contains its norm, Proposition 1.55 implies $\frac{\mathrm{N}_K(\beta)}{\beta} \in \mathcal{O}_K$. Whence $\frac{\alpha}{\beta} \in \mathcal{O}_K$ and interchanging the roles of α and β shows that $\frac{\beta}{\alpha} \in \mathcal{O}_K$ too. But then $\frac{\alpha}{\beta}$ is a unit in $\mathcal{O}_K$ and thus α and β differ up to multiplication by a unit. □

With this lemma, we can show Λ_K is a complete integral lattice of H and compute its rank as a free Z-module.

Theorem 2.14. *Let* K *be a number field of degree* n *and signature* (r_1, r_2). *Then* Λ_K *is a complete integral lattice of* H. *In particular,* Λ_K *is a free* Z*-module of rank* r_K.

Proof. We will begin by showing that Λ_K is a lattice of H. Since λ is a homomorphism, Λ_K is a subgroup of H. Therefore Λ_K is a lattice if and only if it is discrete. To prove this, we will show that for any positive constant c, the compact subset

$$X = \{\mathbf{x} \in \mathbb{R}^{r_K+1} : |x_\rho| \leq c \text{ and } |x_\tau| \leq 2c \text{ for all } \rho \text{ and } \tau\},$$

contains only finitely many elements of Λ_K. The preimage of X under ℓ is the compact subset

$$\ell^{-1}(X) = \{\mathbf{z} \in K_{\mathbb{R}} : e^{-c} \leq |z_\sigma| \leq e^c \text{ for all } \sigma\},$$

and hence it contains finitely many elements of $j(\mathcal{O}_K^*)$ which is a subset of the complete integral lattice $j(\mathcal{O}_K)$ of $K_{\mathbb{R}}$. Whence $\ell^{-1}(X) \cap j(\mathcal{O}_K^*)$ is finite. The image of this intersection under ℓ is exactly $X \cap \Lambda_K$. Whence $X \cap \Lambda_K$ is finite too. Therefore Λ_K is discrete and thus a lattice of H.

We now show that Λ_K is a complete integral lattice of H and since H is a real vector space of dimension r_K, this will prove the claim about the rank of Λ_K as well. It is enough to show that there is a bounded subset M of H whose translates by Λ_K cover H. Since ℓ is surjective it suffices to construct a bounded subset T of U such that

$$U = \bigcup_{\varepsilon} j(\varepsilon) T.$$

Indeed, if such a T exists then any $\mathbf{z} \in T$ satisfies $|\operatorname{N}_{K_{\mathbb{R}}}(\mathbf{z})| = 1$ which forces every component of $\mathbf{z}$ is absolutely bounded above and away from zero because T is bounded. Setting $M = \ell(T)$, it follows that M is also bounded and

$$H = \bigcup_{\lambda} (M + \lambda).$$

To construct T, fix constants $c_\sigma > 0$ satisfying $c_\sigma = c_{\overline{\sigma}}$ and

$$\prod_{\sigma} c_\sigma > \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|\Delta_K|}.$$

Also set $C = \prod_\sigma c_\sigma$. Now consider the bounded subset

$$Z = \{\mathbf{z} \in K_{\mathbb{R}} : |z_\sigma| < c_\sigma \text{ for all } \sigma\}.$$

By construction, the absolute value of the norm of any element of Z is at most C. For any $\mathbf{w} \in K_{\mathbb{R}}^*$, we can form the bounded subset

$$\mathbf{w}Z = \{\mathbf{z} \in K_{\mathbb{R}} : |z_\sigma| < |w_\sigma| c_\sigma \text{ for all } \sigma\}.$$

Letting $\mathbf{u} \in U$, the absolute value of the norm of any element of $\mathbf{u}Z$ is at most C. By Proposition 2.6, the volume of $\mathbf{u}Z$ is then 2^{r_2} times the volume of

$$X = \{\mathbf{x} \in \mathbb{R}^n : x_\rho^2 < c_\rho^2 \text{ and } x_\tau^2 + x_{\overline{\tau}}^2 < c_\tau^2 \text{ for all } \rho \text{ and } \tau\},$$

which is $C 2^{r_1} \pi^{r_2}$ because X is the product of r_1 many intervals each of length $2c_\rho$ and r_2 many disks each of of radius c_τ. Thus $\operatorname{Vol}(\mathbf{u}Z) = C 2^{r_K+1} \pi^{r_2}$. By Proposition 2.7, $V_{j(\mathcal{O}_K)} = \sqrt{|\Delta_K|}$ and our choice of C gives

$$\operatorname{Vol}(\mathbf{u}Z) > 2^n V_{j(\mathcal{O}_K)}.$$

By Minkowski's lattice point theorem, there exists a nonzero element of $j(\mathcal{O}_K) \cap \mathbf{u}Z$. Now from Lemma 2.13, let $\alpha_1, \dots, \alpha_m$ be a complete set of representatives of $\mathcal{O}_K$ whose field norm is at most C up to multiplication by a unit. Set

$$T = U \cap \left(\bigcup_i j(\alpha_i)^{-1} Z \right).$$

Then T is a bounded subset of U since the $j(\alpha_i)^{-1}Z$ are. We now claim that

$$U = \bigcup_{\varepsilon} j(\varepsilon)T.$$

The reverse inclusion holds because each $j(\varepsilon)T$ is a subset of U. For the forward inclusion, let $\mathbf{u} \in U$. Then $\mathbf{u}^{-1} \in U$ whence there exists a nonzero element of $j(\mathcal{O}_K) \cap \mathbf{u}^{-1}Z$. This means $j(\alpha) = \mathbf{u}^{-1}\mathbf{z}$ for some $\alpha \in \mathcal{O}_K$ and $\mathbf{z} \in Z$ with

$$|\operatorname{N}_K(\alpha)| < C.$$

But then there exists an α_i and $\varepsilon \in \mathcal{O}_K^*$ such that $\alpha_i = \alpha\varepsilon$. Writing $\mathbf{u} = j(\alpha)^{-1}\mathbf{z}$, a short computation shows

$$\mathbf{u} = j(\varepsilon)j(\alpha_i)^{-1}\mathbf{z},$$

As $\mathbf{u}, j(\varepsilon) \in U$, we see that $j(\alpha_i)^{-1}\mathbf{z} \in U$ and thus $j(\alpha_i)^{-1}\mathbf{z} \in T$. But then $\mathbf{u} \in j(\varepsilon)T$ proving the forward inclusion. □

In view of this result, there exist units $\varepsilon_1, \ldots, \varepsilon_{r_K}$ of K such that $\lambda(\varepsilon_1), \ldots, \lambda(\varepsilon_{r_K})$ is a basis for the unit lattice Λ_K. Indeed, such elements are obtained by taking the preimage under j of any basis for $\Lambda_K/\mathbb{Z}$. Accordingly, we say that $\varepsilon_1, \ldots, \varepsilon_{r_K}$ is a **system of fundamental units** for K and we call any such element a **fundamental unit** for K. **Dirichlet's unit theorem** determines the group structure of the units of K and says that $\mathcal{O}_K^*$ is a direct product of the root of unity in K and products of powers of fundamental units.

Theorem (Dirichlet's unit theorem). *Let K be a number field of degree n and signature (r_1, r_2). Then*

$$\mathcal{O}_K^* \cong \mu(K) \times \mathbb{Z}^{r_K}.$$

In particular, if $\varepsilon_1, \ldots, \varepsilon_{r_K}$ is a system of fundamental units for K then any unit ε of K is of the form

$$\varepsilon = \omega\varepsilon_1^{\nu_1} \cdots \varepsilon_{r_K}^{\nu_{r_K}},$$

for some $\omega \in \mu(K)$ and integers ν_i.

Proof. As Λ_K is a free $\mathbb{Z}$-module by Theorem 2.14, the short exact sequence in Proposition 2.12 splits. Whence

$$\mathcal{O}_K^* \cong \mu(K) \times \mathbb{Z}^{r_K},$$

because the rank of Λ_K is r_K. This proves the first statement. For the second statement, let $\varepsilon_1, \ldots, \varepsilon_{r_K}$ be a system of fundamental units for K and let E be the subgroup of $\mathcal{O}_K^*$ generated by them. Then Proposition 2.12 and the first isomorphism theorem together show that λ induces an isomorphism $E \cong \mathbb{Z}^{r_K}$. Then

$$\mathcal{O}_K^* \cong \mu(K) \times E,$$

from which the second statement follows. □

Recall from Theorem 2.14 that Λ_K is a complete integral lattice of H. The determination of its covolume V_{Λ_K} it important as it measures how dense the units of K are. To this end, let $\varepsilon_1, \dots, \varepsilon_{r_K}$ be a system of fundamental units for K so that $\lambda(\varepsilon_1), \dots, \lambda(\varepsilon_{r_K})$ is a basis for $\Lambda_K/\mathbb{Z}$. Setting

$$\lambda_0 = \frac{\mathbf{1}}{\sqrt{r_K+1}},$$

we see that λ_0 is a unit vector in $\mathbb{R}^{r_K+1}$ orthogonal to H. Whence λ_0 is orthogonal to Λ and it follows that $\lambda_0, \lambda(\varepsilon_1), \dots, \lambda(\varepsilon_{r_K})$ is a basis for the complete integral lattice $\Lambda'_K = \mathbb{Z}\lambda_0 + \Lambda_K$ of $\mathbb{R}^{r_K+1}$. Since λ_0 is a unit vector, the volume of a fundamental domain for Λ'_K in $\mathbb{R}^{r_K+1}$ is equal to the volume of a fundamental domain for Λ_K in H. This means that the corresponding covolumes are equal which is to say $V_{\Lambda_K} = V_{\Lambda'_K}$. So it suffices to compute $V_{\Lambda'_K}$. The associated generator matrix for Λ'_K is given by

$$\begin{pmatrix} \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & \vdots & & \vdots \\ \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix}.$$

Adding all of the rows to an arbitrary fixed row results in

$$\begin{pmatrix} \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & \vdots & & \vdots \\ \sqrt{r_K+1} & 0 & & 0 \\ \vdots & \vdots & & \vdots \\ \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix},$$

which has the same determinant as that of the generator matrix. Cofactor expanding along the row with all zeros except the first entry shows

$$V_{\Lambda_K} = \sqrt{r_K+1}R_K,$$

where R_K is the absolute value of the determinant of any rank r_K minor of

$$\begin{pmatrix} \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & & \vdots \\ \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix}.$$

Indeed, we may choose any rank r_K minor since the cofactor expansion was along an arbitrary row. We call R_K the **regulator** of K. Since the covolume V_{Λ_K} of Λ_K is independent of the choice of basis for $\Lambda_K/\mathbb{Z}$, the regulator R_K is independent of the choice of a system of fundamental units for K. The regulator is a rough measure of the density of the units for K in the sense that smaller the regulator is the more dense the units are.

Let S be a finite set of primes of K. Then it is a simple matter to determine the structure of the S-unit group of $\mathcal{O}_{K,S}$. As the ideal class group is finite, the long exact sequence Proposition 1.39 in this setting contracts to the short exact sequence

$$1 \longrightarrow \mathcal{O}_K^* \longrightarrow \mathcal{O}_{K,S}^* \longrightarrow \mathbb{Z}^{|S|} \longrightarrow 1,$$

which splits because $\mathbb{Z}^{|S|}$ is a free $\mathbb{Z}$-module. This fact and Dedekind's unit theorem together imply

$$\mathcal{O}_{K,S}^* \cong \mu(K) \times \mathbb{Z}^{r_K+|S|}.$$

So the rank of the S-unit group of K is generally larger than the rank of the unit group of K.

Let $\mathcal{O}$ be an order of K. Then we can also describe the structure of the unit group of $\mathcal{O}$. Indeed, we have already shown $\mathcal{O}_K^*/\mathcal{O}^*$ is finite. By the structure theorem for finitely generated modules over principal ideal domains, $\mathcal{O}^*$ must have the same rank as $\mathcal{O}_K^*$. Then Dirichlet's unit theorem implies

$$\mathcal{O}^* \cong \mu(\mathcal{O}) \times \mathbb{Z}^{r_K},$$

for some subgroup $\mu(\mathcal{O})$ of $\mu(K)$. Therefore the unit group of $\mathcal{O}$ is generally smaller than the unit group of K.